

MATH EXCELLENCE SERIES

SAT MATH

Practice Problems

Algebra • Advanced Math • Data Analysis • Geometry

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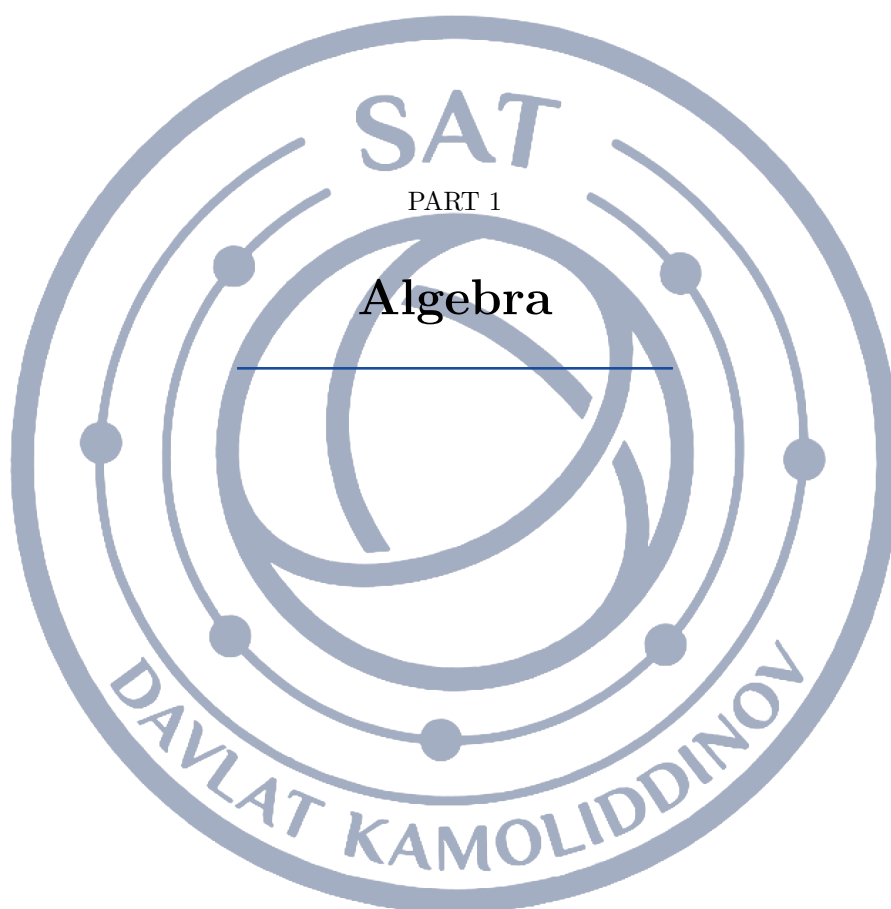
Math Instructor & Problem Setter



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Lesson 1: Linear Equations in One Variable

Module 1

1If $3x + 12 = 30$, what is the value of x ?

- A) 6
- B) 10
- C) 14
- D) 18

2If $7x = 4x + 21$, what is the value of x ?**3**If $\frac{x}{4} + 3 = 11$, what is the value of x ?

- A) 8
- B) 32
- C) 44
- D) 56

4If $3(x - 4) = 18$, what is the value of x ?

- A) 2
- B) 6
- C) 10
- D) 22

5

A gym charges a one-time membership fee of \$30 plus \$15 for each month of use. If a member paid a total of \$180, for how many months did the member use the gym?

- A) 6
- B) 10
- C) 12
- D) 14

6If $4x - 9 = 23$, what is the value of $x + 2$?**7**If $-2x + 9 = 1$, what is the value of x ?

- A) -5
- B) -4
- C) 4
- D) 5

8If $5x - 8 = 3x + 14$, what is the value of x ?

- A) -3
- B) 3
- C) 11
- D) 22

9If $\frac{5}{2}x - 3 = 7$, what is the value of x ?

- A) 4
- B) 10
- C) 20
- D) 25

10If $4(x + 3) = 2(x + 9)$, what is the value of x ?**11**If $3x + 2y = 12$, which expression gives x in terms of y ?

- A) $\frac{12 - 2y}{3}$
- B) $\frac{12 + 2y}{3}$
- C) $12 - \frac{2y}{3}$
- D) $\frac{12 - 2y}{2}$



12

At noon the temperature in a town was 12°F , and it fell at a constant rate of 3°F per hour. After how many hours was the temperature -9°F ?

- A) -7
- B) 1
- C) 3
- D) 7

13

In the equation $4(x - a) = 4x - 12$, a is a constant. If the equation is true for all values of x , what is the value of a ?

- A) -3
- B) 3
- C) 12
- D) 48

14

The equation $6x + 12 = 3(2x + k)$, where k is a constant, has infinitely many solutions. What is the value of k ?

15

If $0.4x + 1.2 = 3.6$, what is the value of x ?

- A) 0.6
- B) 6
- C) 9
- D) 12

16

If $\frac{x+5}{3} - \frac{x-1}{2} = 1$, what is the value of x ?

- A) -7
- B) 1
- C) 7
- D) 12

17

The formula $C = \frac{5}{9}(F - 32)$ converts a Fahrenheit temperature F to a Celsius temperature C . Which of the following gives F in terms of C ?

- A) $\frac{5}{9}C + 32$
- B) $\frac{9}{5}(C + 32)$
- C) $\frac{9}{5}C - 32$
- D) $\frac{9}{5}C + 32$

18

If $2(3x - 5) - 4(x - 7) = 3x + 6$, what is the value of x ?

19

Tank A holds 240 liters of water and is draining at a constant rate of 8 liters per minute. Tank B holds 120 liters and is filling at a constant rate of 4 liters per minute. After how many minutes will the two tanks hold the same amount of water?

- A) 10
- B) 15
- C) 20
- D) 30

20

In the equation $ax + 15 = 4(3x + 5) - 8$, a is a constant. If the equation has no solution, what is the value of a ?

- A) 3
- B) 4
- C) 5
- D) 12

21

If $\frac{2x-7}{5} = \frac{x+4}{3}$, what is the value of x ?



22

The perimeter of a rectangle is 48 centimeters. The length of the rectangle is 4 centimeters more than its width. What is the length, in centimeters, of the rectangle?

- A) 10
- B) 14
- C) 20
- D) 24





Module 2

1

If $7 - 2x = 19$, what is the value of x ?

- A) -12
- B) -6
- C) 6
- D) 13

2

If $\frac{x}{3} - \frac{x}{5} = 4$, what is the value of x ?

3

If $5(2x - 5) + 4 = 39$, what is the value of x ?

- A) 1
- B) 4
- C) 6
- D) 60

4

A phone plan costs \$25 per month plus \$0.10 for each text message sent. In one month the bill was \$43. How many text messages were sent that month?

- A) 680
- B) 430
- C) 180
- D) 18

5

If $5m - 2n = 20$, which expression gives n in terms of m ?

- A) $\frac{20 - 5m}{2}$
- B) $5m - 10$
- C) $\frac{5m + 20}{2}$
- D) $\frac{5m - 20}{2}$

6

If $2(x + 6) = 5x - 3$, what is the value of x ?

- A) -5
- B) 3
- C) 5
- D) 15

7

If $5(x - 2) = 3x + 4$, what is the value of x ?

8

If $2(3x - 1) = 4(x + 5) + 2$, what is the value of x ?

- A) -12
- B) 10
- C) 12
- D) 24

9

If $\frac{2}{3}(x + 6) = 10$, what is the value of x ?

- A) 6
- B) 9
- C) 15
- D) 21

10

Ana bought 4 identical notebooks and one pen that cost \$8. The total cost of the items, before tax, was \$32. What was the cost, in dollars, of one notebook?

- A) 6
- B) 8
- C) 10
- D) 24

11

If $\frac{x + 2}{4} = \frac{x - 6}{2}$, what is the value of x ?

12

If $y = \frac{2x - 5}{3}$, which expression gives x in terms of y ?

- A) $\frac{3y + 5}{2}$
- B) $\frac{3y - 5}{2}$
- C) $\frac{2y + 5}{3}$
- D) $\frac{3y}{2} + 5$



13

The equation $5(2x + 3) = ax + 15$, where a is a constant, has infinitely many solutions. What is the value of a ?

- A) 2
- B) 3
- C) 5
- D) 10

14

In the equation $3(2x - 4) + 5 = 6x + c$, c is a constant. For which value of c does the equation have infinitely many solutions?

- A) -12
- B) -7
- C) 5
- D) 7

15

The length of a rectangle is 3 meters more than twice its width. The perimeter of the rectangle is 36 meters. What is the width, in meters, of the rectangle?

- A) 5
- B) 6
- C) 11
- D) 13

16

The equation $\frac{2}{3}(6x - 9) = 4x + k - 10$, where k is a constant, has infinitely many solutions. What is the value of k ?

17

A studio charges a one-time registration fee of \$60 plus \$10 for each class. A second studio charges \$25 for each class and no registration fee. For how many classes is the total cost at the two studios the same?

- A) 100
- B) 6
- C) 4
- D) -4

18

If $3(x - a) = 2(x + a)$, where a is a nonzero constant, what is x in terms of a ?

- A) $-a$
- B) $\frac{a}{5}$
- C) a
- D) $5a$

19

The sum of three consecutive even integers is 96. What is the least of the three integers?

- A) 30
- B) 32
- C) 34
- D) 36

20

A jar contains only nickels and dimes and has twice as many dimes as nickels. The total value of the coins is \$2.50. How many nickels are in the jar?

- A) 30
- B) 25
- C) 20
- D) 10

21

If $\frac{3}{4}x + \frac{1}{2} = \frac{5}{8}x + 3$, what is the value of x ?

22

In the equation $2(kx - 3) = 6x + 5 - k$, k is a constant. If the equation has no solution, what is the value of k ?

- A) -6
- B) 3
- C) 6
- D) 11



Lesson 2: Linear Functions

Module 1

1

What is the slope of the line that passes through the points $(2, 3)$ and $(6, 11)$?

- A) -2
- B) $\frac{1}{2}$
- C) 2
- D) 4

2

The function f is defined by $f(x) = 3x - 5$. What is the value of $f(4)$?

- A) -17
- B) 2
- C) 7
- D) 12

3

The function f is defined by $f(x) = 7x + 2$. If $f(x) = 30$, what is the value of x ?

4

What is the slope of the graph of $y = -3x + 8$ in the xy -plane?

- A) -8
- B) -3
- C) 3
- D) 8

5

A plumber charges a fixed call-out fee plus an hourly rate. The total charge, in dollars, for a job lasting h hours is given by $C(h) = 45h + 60$. What is the best interpretation of the number 60 in this context?

- A) The number of hours the plumber works.
- B) The plumber's hourly rate, in dollars per hour.
- C) The total cost, in dollars, of a one-hour job.
- D) The fixed fee, in dollars, charged before any hours are worked.

6

x	1	2	3	4
$f(x)$	5	9	13	17

The table shows several values of the linear function f . What is the value of $f(0)$?

7

A linear function f has a slope of 5, and $f(0) = -2$. Which of the following defines $f(x)$?

- A) $5x - 2$
- B) $5x + 2$
- C) $\frac{1}{5}x - 2$
- D) $-2x + 5$

8

A line in the xy -plane passes through the points $(-4, 9)$ and $(6, 4)$. What is the slope of this line?

- A) -2
- B) $-\frac{1}{2}$
- C) $\frac{1}{2}$
- D) 2

9

For the linear function f , $f(2) = 5$ and $f(6) = 17$. What is the value of $f(10)$?

10

A storage tank is being drained. The volume of water in the tank, in gallons, after t minutes of draining is given by $V(t) = 180 - 12t$. Which statement is the best interpretation of the number 12 in this context?

- A) The volume of water decreases by 12 gallons each minute.
- B) The volume of water increases by 12 gallons each minute.
- C) The tank holds 12 gallons when draining begins.
- D) The tank is completely empty after 12 minutes.

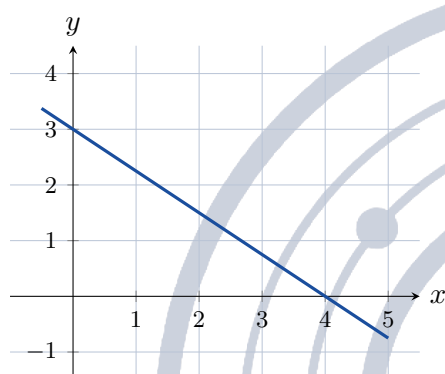


11

A gym charges a one-time membership fee of \$25 plus \$4 for each fitness class attended. Which expression gives the total cost, in dollars, for a member who attends n classes?

- A) $4n + 25$
- B) $25n + 4$
- C) $29n$
- D) $4n - 25$

12



The graph of a linear function is shown in the xy -plane. What is the slope of the graph?

- A) $-\frac{4}{3}$
- B) $-\frac{3}{4}$
- C) $\frac{3}{4}$
- D) $\frac{4}{3}$

13

x	0	3	6	9
$f(x)$	14	8	2	-4

The table gives values of the linear function f . For what value of x does $f(x) = 0$?

14

The function f is defined by $f(x) = \frac{3}{4}x + 6$. For what value of x does $f(x) = 0$?

- A) -8
- B) $-\frac{9}{2}$
- C) -6
- D) 8

15

At 2:00 p.m. the temperature outside was 68°F , and at 8:00 p.m. it was 56°F . If the temperature changed at a constant rate, what was that rate, in degrees Fahrenheit per hour?

- A) -12
- B) -6
- C) -2
- D) 2

16

The function f is defined by $f(x) = ax + b$, where a and b are constants. If $f(1) = 4$ and $f(5) = 16$, what is the value of $f(0)$?

- A) -3
- B) 1
- C) 3
- D) 4

17

In the xy -plane, a line with slope 3 passes through the points $(2, k)$ and $(6, 17)$. What is the value of k ?

18

The number of subscribers to a streaming service is modeled by $S(m) = 250m + 3200$, where m is the number of months after January 2020. Which statement is the best interpretation of the number 250 in this model?

- A) There were 250 subscribers in January 2020.
- B) The service reached 250 subscribers after 3200 months.
- C) The number of subscribers increases by 250 each year.
- D) The number of subscribers increases by 250 each month.

19

What is the slope of the line given by the equation $4x - 6y = 18$ in the xy -plane?

- A) $-\frac{3}{2}$
- B) $-\frac{2}{3}$
- C) $\frac{2}{3}$
- D) $\frac{3}{2}$



20

For the linear function g , $g(3) = 10$ and $g(7) = 2$. What is the value of $g(0)$?

21

The function f is defined by $f(x) = 5x - 3$. If $f(c) = 2c + 9$ for some constant c , what is the value of c ?

- A) -4
- B) 2
- C) 4
- D) 12

22

A candle burns at a constant rate. Its height, in centimeters, after t hours of burning is given by $h(t) = 18 - 1.5t$. Which of the following is the best interpretation of the solution to the equation $h(t) = 0$?

- A) The height, in centimeters, of the candle before it is lit.
- B) The number of centimeters the candle burns in one hour.
- C) The height, in centimeters, of the candle after one hour.
- D) The number of hours it takes for the candle to burn down completely.





Module 2

1

The function f is defined by $f(x) = -2x + 11$. What is the value of $f(-3)$?

- A) -17
- B) 5
- C) 6
- D) 17

2

A line in the xy -plane passes through the points $(-2, 5)$ and $(4, -7)$. What is the slope of this line?

3

x	2	5	8	11
$g(x)$	3	12	21	30

The table shows several values of the linear function g . What is the rate of change of g with respect to x ?

- A) $\frac{1}{3}$
- B) 3
- C) $\frac{9}{2}$
- D) 9

4

A car travels at a constant speed along a highway. Its distance from a rest stop, in miles, after t hours of driving is given by $d(t) = 45t + 30$. What is the best interpretation of the number 30 in this model?

- A) The car travels 30 miles each hour.
- B) The car drives for a total of 30 hours.
- C) The car travels a total distance of 30 miles.
- D) The car was 30 miles from the rest stop when it started driving.

5

The function f is defined by $f(x) = \frac{1}{2}x - 7$. If $f(x) = -3$, what is the value of x ?

- A) -20
- B) -8
- C) 2
- D) 8

6

The graph of $2x + 5y = 20$ is drawn in the xy -plane. What is the y -coordinate of the point where the graph crosses the y -axis?

- A) -4
- B) 4
- C) 10
- D) 20

7

A linear function f has a slope of 4, and its graph passes through the point $(3, 5)$. Which of the following defines $f(x)$?

- A) $4x - 7$
- B) $4x + 5$
- C) $4x + 17$
- D) $\frac{1}{4}x - 7$

8

The function f is defined by $f(x) = ax + 7$, where a is a constant. If $f(4) = -5$, what is the value of a ?

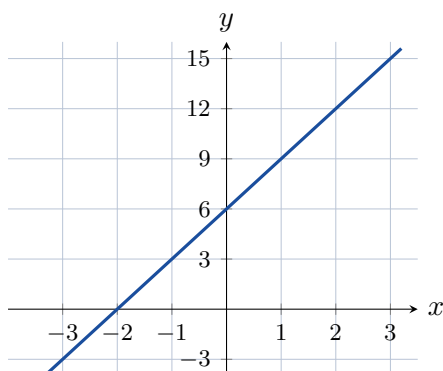
- A) -12
- B) -3
- C) $-\frac{1}{3}$
- D) 3

9

For the linear function f , $f(-2) = 13$ and $f(4) = -5$. What is the value of $f(1)$?



10



The graph of the linear function f is shown in the xy -plane. What is the value of $f(3)$?

- A) 9
- B) 12
- C) 15
- D) 18

11

Water is pumped into a pool that already holds some water. The amount of water in the pool, in liters, after t minutes of pumping is given by $W(t) = 25t + 150$. Which statement about the pool must be true?

- A) The pool gains 25 liters of water each minute.
- B) The pool gains 150 liters of water each minute.
- C) The pool is empty when pumping begins.
- D) The pool is full after 25 minutes.

12

In the xy -plane, a line with slope $-\frac{3}{2}$ passes through the points $(0, 9)$ and $(t, 0)$. What is the value of t ?

13

A candle burns at a constant rate. After 2 hours its height is 21 centimeters, and after 5 hours its height is 15 centimeters. What was the height, in centimeters, of the candle before it was lit?

- A) 19
- B) 23
- C) 25
- D) 27

14

The equation $3x - 4y = 24$ defines a line in the xy -plane. What is the value of y when $x = 12$?

- A) -3
- B) 3
- C) 9
- D) 12

15

The function f is defined by $f(x) = 2x - 5$. Which of the following is equivalent to $f(x + 3)$?

- A) $2x - 15$
- B) $2x - 2$
- C) $2x + 1$
- D) $2x + 6$

16

For the linear function f , $f(3) = 8$ and $f(x + 1) - f(x) = -4$ for every value of x . What is the value of $f(6)$?

- A) -4
- B) 4
- C) 12
- D) 20

17

A linear function f satisfies $f(-3) = -1$ and $f(6) = 5$. If $f(x) = mx + b$, what is the value of b ?

18

The population of a bird colony is modeled by $P(t) = 1500 - 45t$, where t is the number of years after 2010. Which of the following best describes the meanings of 1500 and 45 in this model?

- A) The colony had 1500 birds in 2010, and it gains 45 birds each year.
- B) The colony had 45 birds in 2010, and it loses 1500 birds each year.
- C) The colony had 1500 birds in 2010, and it disappears after 45 years.
- D) The colony had 1500 birds in 2010, and it loses 45 birds each year.



19

The function f is defined by $f(x) = 3x + c$, where c is a constant. If $f(4) = 2f(1)$, what is the value of c ?

- A) 2
- B) 3
- C) 6
- D) 9

20

The total cost, in dollars, of a gym membership for m months is given by $C(m) = 35m + 80$. After how many months will the total cost be \$1200?

21

x	-4	1	7
$f(x)$	23	8	-10

The table gives three values of the linear function f . What is the value of $f(0)$?

- A) 5
- B) 8
- C) 11
- D) 23

22

For the linear function f , $f(0) = k$ and $f(4) = 3k$, where k is a constant. If $f(2) = 10$, what is the value of k ?

- A) $\frac{5}{2}$
- B) 5
- C) 10
- D) 20



Lesson 3: Linear Equations in Two Variables

Module 1

1

The equation of line ℓ in the xy -plane is $y = -3x + 7$. What is the slope of line ℓ ?

- A) -3
- B) $\frac{1}{3}$
- C) 3
- D) 7

2

Line m is given by $y = \frac{1}{2}x - 6$. What is the y -coordinate of the point where line m crosses the y -axis?

- A) $\frac{1}{2}$
- B) -6
- C) 6
- D) 12

3

A line in the xy -plane passes through $(0, 4)$ and has slope 3 . What is the value of y when $x = 5$?

4

Which equation is equivalent to $2x + y = 8$?

- A) $y = 2x + 8$
- B) $y = 2x - 8$
- C) $y = -2x + 8$
- D) $y = -2x - 8$

5

What is the x -coordinate of the x -intercept of the graph of $3x + 4y = 12$?

- A) -4
- B) 3
- C) $\frac{4}{3}$
- D) 4

6

What is the slope of the line that passes through the points $(-1, 4)$ and $(3, 16)$?

7

Which equation represents a line parallel to the line $y = 4x - 1$ in the xy -plane?

- A) $y = -4x + 5$
- B) $y = 4x + 5$
- C) $y = -\frac{1}{4}x + 5$
- D) $y = \frac{1}{4}x + 5$

8

What is the slope of the line $3x - 5y = 15$ in the xy -plane?

- A) $\frac{3}{5}$
- B) $-\frac{3}{5}$
- C) $\frac{5}{3}$
- D) 3

9

A line in the xy -plane has slope $\frac{5}{2}$ and passes through the point $(0, -2)$. Which equation represents the line?

- A) $y = \frac{5}{2}x + 2$
- B) $y = -\frac{5}{2}x - 2$
- C) $y = \frac{5}{2}x - 2$
- D) $y = \frac{2}{5}x - 2$

10

What is the slope of a line that is perpendicular to the line $2x + 3y = 9$?

- A) $-\frac{2}{3}$
- B) $\frac{2}{3}$
- C) $\frac{3}{2}$
- D) $-\frac{3}{2}$

11

Which equation represents the line that passes through the points $(1, 4)$ and $(3, 10)$?

- A) $y = 3x + 1$
- B) $y = 3x - 1$
- C) $y = \frac{1}{3}x + 1$
- D) $y = 3x + 4$



12

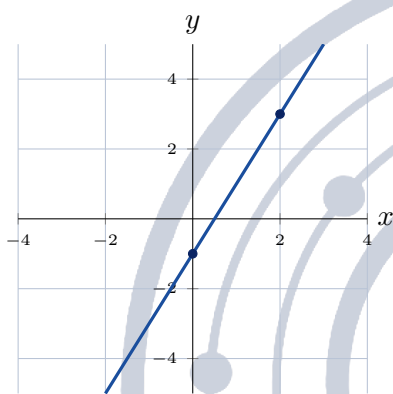
What is the x -coordinate of the x -intercept of the graph of $y = \frac{2}{3}x - 8$?

13

Which equation is equivalent to $y = -\frac{3}{4}x + 2$?

- A) $3x - 4y = 8$
- B) $4x + 3y = 8$
- C) $-3x + 4y = 8$
- D) $3x + 4y = 8$

14



The graph of a line in the xy -plane is shown above. Which equation represents this line?

- A) $y = -2x - 1$
- B) $y = 2x - 1$
- C) $y = 2x + 1$
- D) $y = \frac{1}{2}x - 1$

15

Line k is parallel to the line $y = -2x + 1$ and passes through the point $(4, 7)$. What is the y -coordinate of the y -intercept of line k ?

16

Line n passes through the point $(-3, 5)$ and is perpendicular to the line $y = \frac{3}{4}x + 2$. Which equation represents line n ?

- A) $y = \frac{3}{4}x + 1$
- B) $y = -\frac{4}{3}x + 9$
- C) $y = \frac{4}{3}x + 9$
- D) $y = -\frac{4}{3}x + 1$

17

In the xy -plane, the line $kx + 6y = 12$ is perpendicular to the line $2x - 3y = 5$. What is the value of k ?

- A) 9
- B) -9
- C) 4
- D) -4

18

The graph of $4x + ky = 20$ in the xy -plane has the same slope as the graph of $y = -2x + 7$. What is the value of k ?

19

A line in the xy -plane has x -intercept 6 and y -intercept -4 . Which equation represents this line?

- A) $2x + 3y = 12$
- B) $3x - 2y = 12$
- C) $2x - 3y = 12$
- D) $2x - 3y = -12$

20

Line ℓ passes through the point $(2, -1)$ and is perpendicular to the line that passes through $(0, 3)$ and $(4, 1)$. What is the x -coordinate of the x -intercept of line ℓ ?

- A) -5
- B) $\frac{5}{2}$
- C) 5
- D) $-\frac{2}{5}$

21

In the xy -plane, the line $y = mx + b$ passes through the points $(2, 7)$ and $(5, 16)$. What is the value of $m + b$?

- A) 4
- B) $\frac{20}{3}$
- C) 2
- D) 10

22

Line ℓ has equation $y = \frac{2}{5}x - 3$. Line m is perpendicular to line ℓ and passes through the point $(4, 1)$. What is the y -coordinate of the point where line m crosses the y -axis?



Module 2

1

What is the slope of the line that passes through the points $(-1, 4)$ and $(3, -4)$?

- A) 2
- B) -2
- C) $-\frac{1}{2}$
- D) $\frac{1}{2}$

2

What is the y -coordinate of the y -intercept of the graph of $2x - 5y = 20$ in the xy -plane?

3

Which equation represents a line perpendicular to the line $y = -\frac{1}{2}x + 3$?

- A) $y = -2x + 1$
- B) $y = -\frac{1}{2}x - 1$
- C) $y = 2x - 1$
- D) $y = \frac{1}{2}x + 3$

4

Which equation is equivalent to $6x - 2y = 18$?

- A) $y = 3x + 9$
- B) $y = -3x + 9$
- C) $y = -3x - 9$
- D) $y = 3x - 9$

5

A line in the xy -plane passes through the point $(0, 5)$ and has slope $-\frac{3}{4}$. Which equation represents the line?

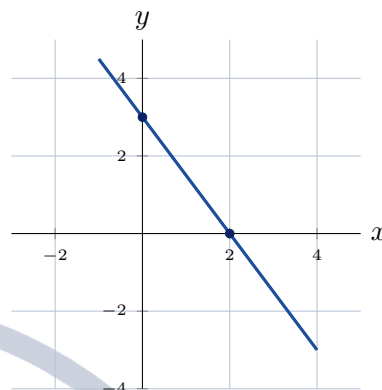
- A) $y = -\frac{3}{4}x + 5$
- B) $y = \frac{3}{4}x + 5$
- C) $y = -\frac{3}{4}x - 5$
- D) $y = \frac{4}{3}x + 5$

6

The graph of $y = 4x + c$, where c is a constant, passes through the point $(2, 3)$. What is the value of c ?

- A) 11
- B) -5
- C) 5
- D) -11

7



A line in the xy -plane is graphed above. Which of the following is an equation of that line?

- A) $y = \frac{3}{2}x + 3$
- B) $y = -\frac{2}{3}x + 3$
- C) $y = -\frac{3}{2}x + 3$
- D) $y = -\frac{3}{2}x - 3$

8

A line in the xy -plane has x -intercept 8 and passes through the point $(2, 9)$. What is the y -coordinate of its y -intercept?

- A) 8
- B) -12
- C) $\frac{3}{2}$
- D) 12

9

In the xy -plane, the graphs of $2x + 3y = 12$ and $y = mx - 4$ are parallel, where m is a constant. What is the value of m ?

10

Line ℓ passes through the point $(-4, 2)$ and has slope $\frac{1}{2}$. What is the x -coordinate of the x -intercept of line ℓ ?

- A) -8
- B) 8
- C) 4
- D) -4



11

The total cost C , in dollars, of a gym membership for m months is given by $C = 25m + 60$, where 60 dollars is a one-time joining fee. Which equation gives m in terms of C ?

- A) $m = \frac{C+60}{25}$
- B) $m = 25C - 60$
- C) $m = \frac{C-60}{25}$
- D) $m = \frac{C}{25} - 60$

12

The line that passes through the points $(1, -3)$ and $(k, 9)$ has slope 4, where k is a constant. What is the value of k ?

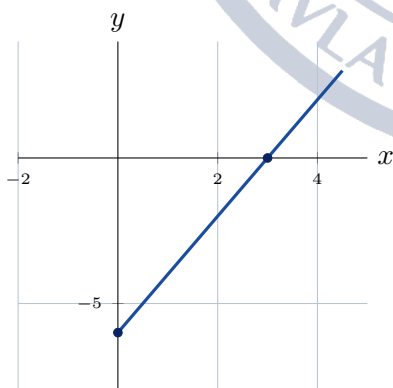
- A) -2
- B) 4
- C) 49
- D) 2

13

Which equation represents a line perpendicular to the line $4x - 6y = 15$?

- A) $y = \frac{3}{2}x + 1$
- B) $y = -\frac{2}{3}x + 1$
- C) $y = \frac{2}{3}x + 1$
- D) $y = -\frac{3}{2}x + 1$

14



The line shown in the xy -plane above passes through the two marked points. Which equation represents this line in the form $Ax + By = C$?

- A) $2x + y = 6$
- B) $x - 2y = 6$
- C) $2x - y = 6$
- D) $2x - y = -6$

15

What is the x -coordinate of the x -intercept of the graph of $y = -\frac{5}{3}x + 10$?

16

In the xy -plane, the graph of $ax - 4y = 12$ is parallel to the line that passes through the points $(1, 2)$ and $(5, 8)$, where a is a constant. What is the value of a ?

- A) 6
- B) -6
- C) $\frac{3}{2}$
- D) 8

17

Line m is perpendicular to the line $y = \frac{3}{5}x - 2$ and passes through the point $(3, 4)$. What is the value of y on line m when $x = 6$?

18

Line ℓ passes through the points $(-2, 7)$ and $(4, -5)$. Line m is perpendicular to line ℓ and passes through the midpoint of the segment joining those two points. Which equation represents line m ?

- A) $y = -2x + 3$
- B) $y = \frac{1}{2}x - \frac{1}{2}$
- C) $y = 2x - 1$
- D) $y = \frac{1}{2}x + \frac{1}{2}$

19

In the xy -plane, the graph of $y = mx + 4$ passes through the point at which the graph of $2x + y = 16$ crosses the x -axis. What is the value of m ?

- A) 2
- B) $-\frac{1}{2}$
- C) $\frac{1}{2}$
- D) -2

20

The graph of a line in the xy -plane has slope -3 and passes through the points $(a, 0)$ and $(0, 12)$, where a is a constant. What is the value of a ?

- A) -4
- B) 36
- C) 4
- D) -36



21

In the xy -plane, the graph of $3x + ky = 12$ is perpendicular to the graph of $y = \frac{3}{4}x - 1$, where k is a constant. What is the value of k ?

22

A technician charges a one-time service fee plus a constant hourly rate. The total charge is 145 dollars for a 2-hour job and 280 dollars for a 5-hour job. If the total charge C , in dollars, is a linear function of the number of hours h , what is the one-time service fee, in dollars?

- A) 45
- B) 35
- C) 100
- D) 55





Lesson 4: Systems of Two Linear Equations in Two Variables

Module 1

1

In the system shown, what is the value of x ?

$$\begin{cases} y = 2x \\ 3x + y = 20 \end{cases}$$

- A) 4
- B) 5
- C) 8
- D) 20

2

What is the value of x in the system below?

$$\begin{cases} x + y = 10 \\ x - y = 4 \end{cases}$$

- A) 3
- B) 5
- C) 7
- D) 14

3

If $x + y = 12$ and $2x + y = 17$, what is the value of y ?

4

The system below has exactly one solution (x, y) . What is the value of y ?

$$\begin{cases} y = x + 3 \\ y = 2x - 1 \end{cases}$$

- A) 4
- B) 5
- C) 7
- D) 11

5

For the solution (x, y) of the system shown, what is the value of $x + y$?

$$\begin{cases} 4x + y = 17 \\ y = x + 2 \end{cases}$$

- A) 2
- B) 3
- C) 5
- D) 8

6

What is the value of y in the system below?

$$\begin{cases} 2x + y = 7 \\ 3x - y = 8 \end{cases}$$

- A) -1
- B) 1
- C) 2
- D) 3

7

A theater sells adult tickets for \$12 each and child tickets for \$8 each. On one evening the theater sold 24 tickets and collected \$232. How many adult tickets were sold?

8

What is the value of y in the system below?

$$\begin{cases} 3x + 2y = 16 \\ 5x - 4y = -10 \end{cases}$$

- A) 2
- B) 5
- C) 7
- D) 11



9

If $2x - 3y = 5$ and $4x + 3y = 19$, what is the value of $x + y$?

10

The ordered pair (x, y) is the solution to the system shown. What is the value of $2x - y$?

$$\begin{cases} 3x + 2y = 19 \\ x - 2y = -7 \end{cases}$$

- A) 1
- B) 3
- C) 5
- D) 11

11

For what value of k does the system below have infinitely many solutions?

$$\begin{cases} kx + 6y = 10 \\ 2x + 3y = 5 \end{cases}$$

- A) 2
- B) 3
- C) 4
- D) 6

12

The system below has no solution. What is the value of k ?

$$\begin{cases} 6x + ky = 9 \\ 2x + 5y = 4 \end{cases}$$

- A) $\frac{5}{3}$
- B) 3
- C) 10
- D) 15

13

At a school store a pen costs \$2 and a notebook costs \$5. Maria buys 12 items in total and pays \$39. How many notebooks does she buy?

- A) 3
- B) 5
- C) 7
- D) 12

14

Two cars start at the same time from points 300 miles apart and drive toward each other along a straight road. They meet after 3 hours. One car travels 20 miles per hour faster than the other. What is the speed, in miles per hour, of the faster car?

- A) 40
- B) 50
- C) 60
- D) 80

15

If $5x + 2y = 2$ and $3x - 2y = 14$, what is the value of $x - y$?

16

What is the value of y in the system below?

$$\begin{cases} \frac{x}{2} + \frac{y}{3} = 4 \\ x - y = 3 \end{cases}$$

- A) 3
- B) 6
- C) 9
- D) 12

17

If $4x + 7y = 17$ and $7x + 4y = 27$, what is the value of $x + y$?

- A) 2
- B) 4
- C) 11
- D) 44

18

The system $ax + 4y = 10$ and $3x + 6y = 12$ has no solution. What is the value of a ?

19

A chemist mixes a 10% acid solution with a 30% acid solution to make 20 liters of a 25% acid solution. How many liters of the 30% solution are used?

- A) 5
- B) 10
- C) 12
- D) 15



20

Which of the following statements about the system $2x - 5y = 8$ and $-4x + 10y = -16$ is true?

- A) The system has no solution.
- B) The system has exactly one solution.
- C) The system has exactly two solutions.
- D) The system has infinitely many solutions.

21

If $5x + 3y = 29$ and $3x + 5y = 27$, what is the value of $x - y$?

22

For what value of c does the system below have infinitely many solutions?

$$\begin{cases} 4x - 6y = 10 \\ -2x + 3y = c \end{cases}$$

- A) -10
- B) -5
- C) 5
- D) 10





Module 2

1

The system shown has solution (x, y) . What is the value of $x + y$?

$$\begin{cases} y = 4x - 1 \\ y = 2x + 7 \end{cases}$$

- A) 4
- B) 15
- C) 19
- D) 23

2

What is the value of y in the system below?

$$\begin{cases} 2x + 5y = 19 \\ 4x - 5y = -7 \end{cases}$$

- A) 2
- B) 3
- C) 5
- D) 19

3

A concert sold student tickets for \$9 each and adult tickets for \$15 each. A total of 40 tickets were sold for \$456. How many adult tickets were sold?

4

If (x, y) satisfies $7x + 3y = 27$ and $2x + 3y = 12$, what is the value of $2x - y$?

- A) 2
- B) 3
- C) 4
- D) 8

5

The system $8x + ky = 12$ and $2x + 5y = 7$ has no solution. What is the value of k ?

- A) -20
- B) -5
- C) 5
- D) 20

6

The graphs of $y = -2x + 9$ and $y = x - 3$ intersect at the point (x, y) . What is the value of $x + 2y$?

- A) 4
- B) 5
- C) 6
- D) 9

7

A bakery sells muffins for \$3 each and scones for \$4 each. One morning it sold 25 of these items for a total of \$87. How many muffins did it sell?

- A) 10
- B) 11
- C) 12
- D) 13

8

What is the value of y in the system below?

$$\begin{cases} 0.5x + 0.2y = 2.6 \\ x - y = 1 \end{cases}$$

- A) 1
- B) 3
- C) 4
- D) 7

9

If $5x + 4y = 6$ and $3x + 2y = 4$, what is the value of $x - y$?

10

In the system $2x + 3y = 7$ and $6x + 9y = c$, c is a constant. For what value of c does the system have infinitely many solutions?

- A) 7
- B) 14
- C) 21
- D) 63



11

A museum charges \$8 for an adult ticket and \$5 for a child ticket. On Saturday it sold 90 tickets and collected \$600. How many child tickets were sold?

- A) 20
- B) 30
- C) 40
- D) 50

12

If $4x + y = 13$ and $2x + 3y = 9$, what is the value of $3x + 2y$?

13

Gym A charges a \$40 registration fee plus \$15 per month. Gym B charges a \$10 registration fee plus \$20 per month. After how many months will the two gyms have cost the same total amount?

- A) 2
- B) 5
- C) 6
- D) 10

14

If $3(x + y) = 15$ and $2x - y = 4$, what is the value of $2y - x$?

- A) 1
- B) 3
- C) 5
- D) 7

15

A boat travels 24 miles downstream in 2 hours and the same 24 miles upstream in 3 hours. What is the speed of the current, in miles per hour?

16

The system $ax + 5y = 12$ and $4x + 10y = 7$ has no solution, where a is a constant. What is the value of a ?

- A) 2
- B) $\frac{5}{2}$
- C) 8
- D) 20

17

The system $3x + ky = 8$ and $6x + 10y = 16$ has infinitely many solutions, where k is a constant. What is the value of k ?

- A) 2
- B) 5
- C) 10
- D) 20

18

The sum of two numbers is 34. Three times the larger number exceeds twice the smaller number by 62. What is the larger number?

- A) 8
- B) 18
- C) 26
- D) 34

19

If $7x + 5y = 31$ and $5x + 7y = 17$, what is the value of $x - y$?

- A) 4
- B) 7
- C) 12
- D) 14

20

The system $y = 5x + 2$ and $y = (k - 1)x - 3$ has no solution, where k is a constant. What is the value of k ?

21

In the system $2x - y = 4$ and $6x - 3y = c$, c is a constant. For what value of c does the system have at least one solution?

- A) 2
- B) 4
- C) 8
- D) 12



22

At a shop, 3 shirts and 2 hats cost \$85, and 2 shirts and 5 hats cost \$75. What is the total cost, in dollars, of one shirt and one hat?

- A) 5
- B) 20
- C) 25
- D) 30





Lesson 5: Linear Inequalities in One or Two Variables

Module 1

1

Which of the following is equivalent to the inequality $3x + 5 \leq 20$?

- A) $x \leq 15$
- B) $x \leq 5$
- C) $x \geq 5$
- D) $x \leq \frac{25}{3}$

2

A delivery van can carry a total package weight of at most 1200 pounds. If w represents the total weight, in pounds, of the packages in the van, which inequality represents this situation?

- A) $w \geq 1200$
- B) $w > 1200$
- C) $w \leq 1200$
- D) $w < 1200$

3

What is the least integer value of x that satisfies $2x - 7 > 5$?

4

Which of the following ordered pairs (x, y) is a solution to the inequality $y > 2x - 1$?

- A) (0, 0)
- B) (1, 1)
- C) (2, 3)
- D) (3, 4)

5

A student buys x notebooks that cost \$3 each and y pens that cost \$2 each. The student spends no more than \$30. Which inequality represents this situation?

- A) $2x + 3y \leq 30$
- B) $3x + 2y \geq 30$
- C) $3x + 2y < 30$
- D) $3x + 2y \leq 30$

6

What is the greatest value of x that satisfies $4x + 3 \leq 27$?

- A) 5
- B) 6
- C) 7
- D) 24

7

If $5 - x \geq 2$, which of the following must be true?

- A) $x \geq 3$
- B) $x \leq 7$
- C) $x \leq 3$
- D) $x \geq -3$

8

Which of the following values of x satisfies $-3 < 2x + 1 \leq 7$?

- A) -3
- B) 3
- C) -2
- D) 4

9

What is the least integer value of n that satisfies $\frac{n}{3} + 4 > 10$?

10

Movie tickets cost \$9 each, and an online order also carries a one-time service fee of \$6. Maria has \$60 to spend. What is the greatest number of tickets she can order?

- A) 4
- B) 5
- C) 6
- D) 7



11

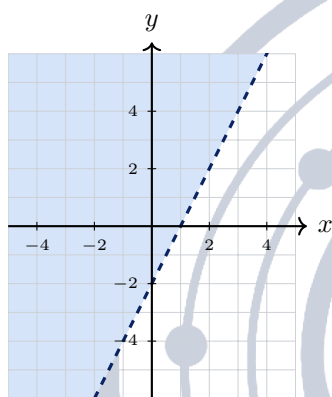
Which ordered pair (x, y) is a solution to the system $y \geq 2x - 3$ and $x + y < 5$?

- A) (2, 4)
- B) (3, 3)
- C) (4, 1)
- D) (0, 0)

12

The solution to the inequality $12 - 5x \geq -3$ is $x \leq k$. What is the value of k ?

13



The shaded region in the xy -plane above is the graph of a linear inequality whose boundary line passes through $(0, -2)$ and $(1, 0)$. Which inequality is graphed?

- A) $y \geq 2x - 2$
- B) $y > 2x - 2$
- C) $y < 2x - 2$
- D) $y \leq 2x - 2$

14

A student's grade is the average (arithmetic mean) of four test scores. The student scored 76, 85, and 82 on the first three tests and wants an average of at least 80. Which inequality gives all possible scores s on the fourth test?

- A) $s \geq 77$
- B) $s > 77$
- C) $s \geq 80$
- D) $s \leq 77$

15

What is the greatest integer value of x that satisfies $-4x + 9 > -11$?

16

If $-2 \leq 3 - 2x < 7$, which of the following describes all possible values of x ?

- A) $-2 \leq x < \frac{5}{2}$
- B) $-\frac{5}{2} \leq x < 2$
- C) $-2 < x \leq \frac{5}{2}$
- D) $-\frac{1}{2} < x \leq 2$

17

A caterer needs at least 40 pounds of fruit and will buy a pounds of apples at \$2 per pound and p pounds of pears at \$3 per pound. The caterer can spend at most \$90. Which system of inequalities represents this situation?

- A) $a + p \leq 40$ and $2a + 3p \geq 90$
- B) $a + p \geq 40$ and $2a + 3p \leq 90$
- C) $a + p \geq 40$ and $3a + 2p \leq 90$
- D) $a + p \geq 40$ and $2a + 3p \geq 90$

18

A phone plan costs \$25 per month plus \$0.10 for each text message sent. A customer wants the monthly bill to be at most \$40. What is the greatest number of text messages the customer can send in a month?

- A) 15
- B) 150
- C) 250
- D) 400

19

How many ordered pairs (x, y) satisfy both $y \leq -x + 4$ and $y \geq -x + 7$?

- A) 0
- B) 1
- C) 2
- D) Infinitely many

20

In the xy -plane, the boundary line of the inequality $y \geq mx + 1$ passes through the point $(4, 9)$. What is the value of m ?



21

Which of the following describes all values of x that satisfy $\frac{2x-1}{3} \geq \frac{x+4}{2}$?

- A) $x \geq \frac{13}{7}$
- B) $x \geq 10$
- C) $x \geq 14$
- D) $x \leq 14$

22

Which of the following describes all values of x that satisfy $-\frac{1}{2}x + 3 > x - 6$?

- A) $x < 6$
- B) $x > 6$
- C) $x < \frac{9}{2}$
- D) $x < -6$





Module 2

1

Which of the following is equivalent to $-3x \geq 12$?

- A) $x \geq -4$
- B) $x \leq -4$
- C) $x \leq 4$
- D) $x \geq 4$

2

What is the greatest integer value of x that satisfies $7 - 2x \geq 1$?

3

An elevator can hold at most 8 people. There are already 3 people inside. If p represents the number of additional people who can enter, which inequality represents all possible values of p ?

- A) $p \geq 5$
- B) $p \leq 8$
- C) $p < 5$
- D) $p \leq 5$

4

Which of the following ordered pairs (x, y) is NOT a solution to $2x + 3y \leq 12$?

- A) (0, 0)
- B) (1, 1)
- C) (3, 2)
- D) (6, 1)

5

Which of the following describes all values of x that satisfy $5(x - 2) < 3x$?

- A) $x < 5$
- B) $x > 5$
- C) $x < \frac{5}{2}$
- D) $x < -5$

6

What is the least integer value of x that satisfies $6x + 5 \geq 41$?

7

To qualify for a tournament, a player needs a total of at least 240 points over three games. After two games the player has 152 points. Which inequality gives all possible scores p in the third game that qualify the player?

- A) $p \geq 88$
- B) $p > 88$
- C) $p \leq 88$
- D) $p \geq 392$

8

If $4 \leq 3x - 5 \leq 16$, what is the greatest possible value of x ?

- A) 3
- B) 6
- C) 7
- D) 9

9

An empty truck weighs 4200 pounds. It is loaded with boxes that weigh 65 pounds each. A bridge allows a total weight of no more than 7000 pounds. What is the greatest number of boxes the truck can carry across the bridge?

10

A club buys x shirts at \$4 each and y hats at \$7 each. The club needs at least 20 items in all and can spend at most \$100. Which system represents this situation?

- A) $x + y \leq 20$ and $4x + 7y \geq 100$
- B) $x + y \geq 20$ and $7x + 4y \leq 100$
- C) $x + y \geq 20$ and $4x + 7y \leq 100$
- D) $x + y > 20$ and $4x + 7y < 100$

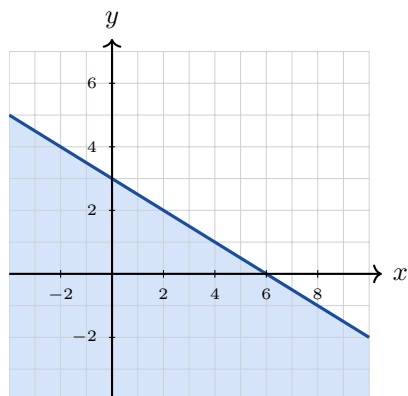
11

Which of the following describes all values of x that satisfy $\frac{3}{4}x - 2 \leq 7$?

- A) $x \leq \frac{20}{3}$
- B) $x \leq 9$
- C) $x \geq 12$
- D) $x \leq 12$



12



The shaded region above is the graph of a linear inequality whose boundary line passes through $(0, 3)$ and $(6, 0)$. Which inequality is graphed?

- A) $y \geq -\frac{1}{2}x + 3$
- B) $y \leq -\frac{1}{2}x + 3$
- C) $y < -\frac{1}{2}x + 3$
- D) $y \leq -2x + 3$

13

Which of the following describes all values of x that satisfy $-\frac{x}{4} + 1 > 3$?

- A) $x < -8$
- B) $x > -8$
- C) $x < 8$
- D) $x > 8$

14

How many integer values of x satisfy $-7 < 2x + 1 \leq 9$?

- A) 5
- B) 7
- C) 8
- D) 9

15

The point $(3, k)$ is a solution to the inequality $2x + y \geq 11$. What is the least possible value of k ?

16

The solution to the inequality $3 - kx \geq 12$ is $x \leq -3$, where k is a constant. What is the value of k ?

- A) -3
- B) $-\frac{1}{3}$
- C) $\frac{1}{3}$
- D) 3

17

A student has at most 12 hours this weekend to study for a math exam and a science exam, and must spend at least 4 hours on science. If m is the number of hours spent on math and s is the number of hours spent on science, which system represents this situation?

- A) $m + s \leq 12$ and $s \geq 4$
- B) $m + s \geq 12$ and $s \geq 4$
- C) $m + s \leq 12$ and $s \leq 4$
- D) $m + s \leq 12$ and $m \geq 4$

18

What is the greatest integer value of x that satisfies $\frac{2x+1}{5} < \frac{x+4}{3}$?

19

Which of the following has the same solution set as $2(3 - x) \leq 4x - 6$?

- A) $x \leq 2$
- B) $x \geq \frac{1}{2}$
- C) $x \geq 2$
- D) $x \leq \frac{2}{3}$

20

At a school fair a vendor sells hot dogs for \$3 each and drinks for \$2 each, and must take in at least \$180 in total. The vendor has already sold 40 drinks. What is the least number of hot dogs the vendor must sell to reach the goal?

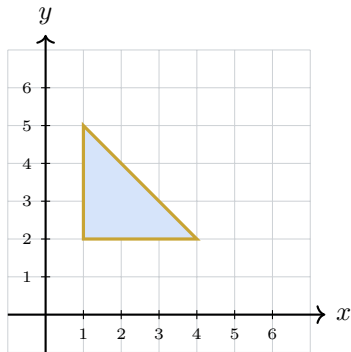
21

Which ordered pair (x, y) is a solution to the system $3x - y > 6$ and $x + 2y \leq 4$?

- A) $(0, 2)$
- B) $(2, 0)$
- C) $(3, -2)$
- D) $(4, 1)$

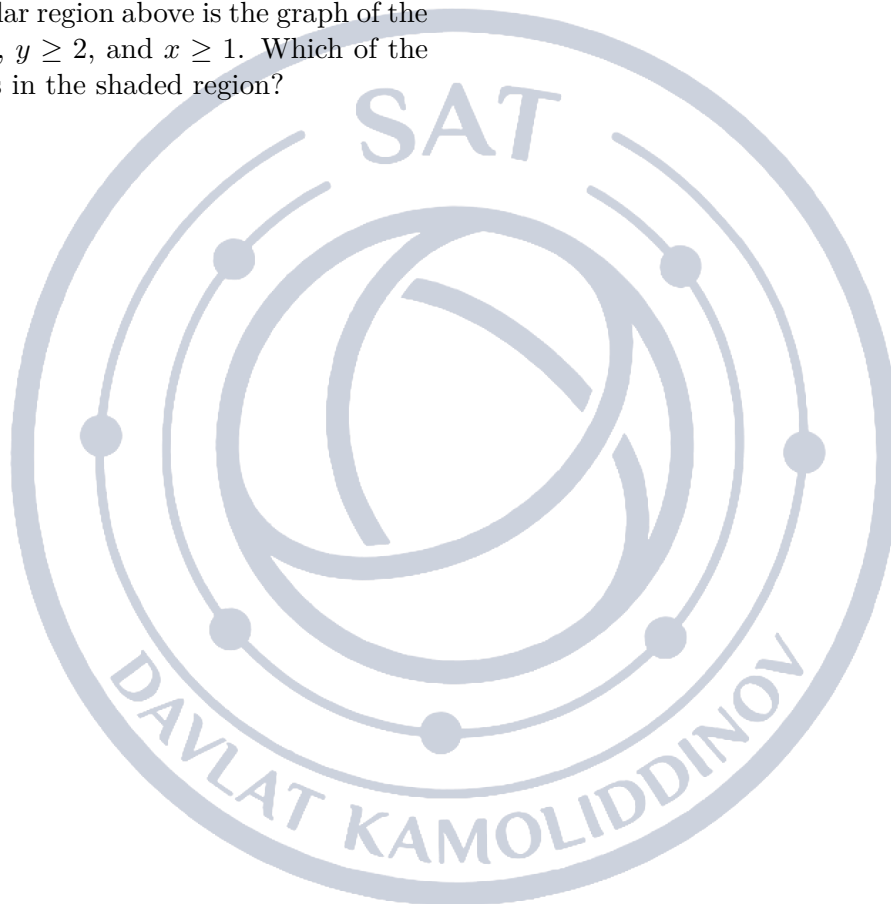


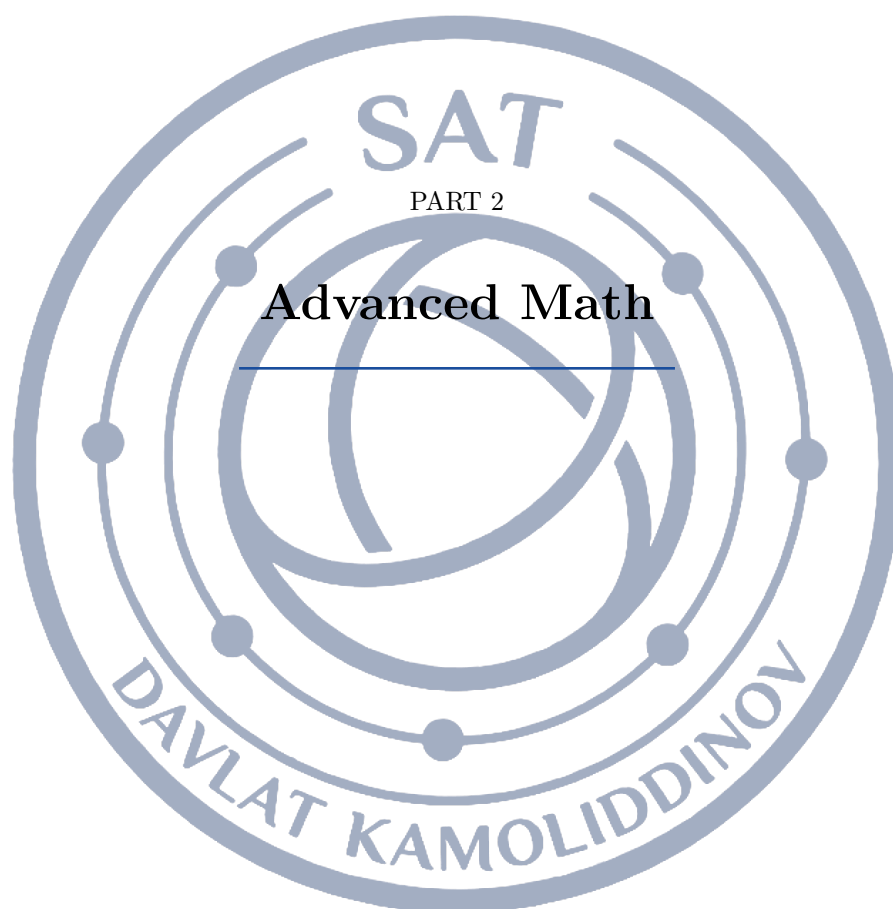
22



The shaded triangular region above is the graph of the system $y \leq -x + 6$, $y \geq 2$, and $x \geq 1$. Which of the following points lies in the shaded region?

- A) (2, 3)
- B) (0, 3)
- C) (2, 1)
- D) (3, 4)







Lesson 6: Equivalent Expressions

Module 1

1

Which expression is equivalent to $(x + 3)(x + 5)$?

- A) $x^2 + 8x + 15$
- B) $x^2 + 15$
- C) $x^2 + 8x + 8$
- D) $x^2 + 15x + 8$

2

Which expression is equivalent to $3x(2x^2 - 4x + 5)$?

- A) $6x^3 - 12x^2 + 15$
- B) $6x^3 - 12x^2 + 15x$
- C) $6x^3 - 4x + 5$
- D) $5x^3 - x^2 + 8x$

3

The expression $(2x + 3)(x - 1)$ can be written in the form $2x^2 + bx - 3$, where b is a constant. What is the value of b ?

4

Which of the following is equivalent to $x^2 - 9$?

- A) $(x - 3)^2$
- B) $(x + 9)(x - 1)$
- C) $x(x - 9)$
- D) $(x + 3)(x - 3)$

5

Which expression is equivalent to $\frac{x^5 \cdot x^3}{x^2}$ for $x \neq 0$?

- A) x^6
- B) x^8
- C) x^{10}
- D) x^{13}

6

Which expression is equivalent to $\frac{12x^5}{3x^2}$ for $x \neq 0$?

- A) $9x^3$
- B) $4x^7$
- C) $4x^3$
- D) $36x^3$

7

Which expression is equivalent to $6x^2 + 9x$?

- A) $3x(2x + 9)$
- B) $6x(x + 3)$
- C) $3x(3x + 2)$
- D) $3x(2x + 3)$

8

Which of the following is equivalent to $x^2 + 7x + 12$?

- A) $(x + 2)(x + 6)$
- B) $(x + 1)(x + 12)$
- C) $(x - 3)(x - 4)$
- D) $(x + 3)(x + 4)$

9

Which expression is equivalent to $(2x - 5)^2$?

- A) $4x^2 + 25$
- B) $4x^2 - 25$
- C) $4x^2 - 20x + 25$
- D) $4x^2 - 10x + 25$

10

For $x \neq 4$, the expression $\frac{x^2 - 16}{x - 4}$ is equivalent to $x + c$, where c is a constant. What is the value of c ?



11

For $x \neq -2$, which expression is equivalent to $\frac{x^2 + 5x + 6}{x + 2}$?

- A) $x + 6$
- B) $\frac{5x + 6}{2}$
- C) $x + 4$
- D) $x + 3$

12

For $x \neq 0$, which expression is equivalent to $(3x^{-2})^{-1}$?

- A) $\frac{x^2}{3}$
- B) $3x^2$
- C) $\frac{1}{3x^2}$
- D) $-3x^2$

13

What is the value of $16^{3/4}$?

14

For $x > 0$, which expression is equivalent to $\sqrt{72x^2}$?

- A) $36x\sqrt{2}$
- B) $6x^2\sqrt{2}$
- C) $6x\sqrt{2}$
- D) $8x\sqrt{3}$

15

For $x \neq 0$, which expression is equivalent to $\frac{1}{x} + \frac{1}{2x}$?

- A) $\frac{1}{3x}$
- B) $\frac{2}{3x}$
- C) $\frac{3}{2x}$
- D) $\frac{1}{2x^2}$

16

Which of the following is equivalent to $2x^2 + 7x + 3$?

- A) $(2x + 1)(x + 3)$
- B) $(2x + 3)(x + 1)$
- C) $(2x - 1)(x - 3)$
- D) $(x + 1)(x + 3)$

17

For $x \neq 2$ and $x \neq -2$, which expression is equivalent to $\frac{x + 2}{x^2 - 4}$?

- A) $\frac{1}{x^2 - 2}$
- B) $\frac{1}{x + 2}$
- C) $x - 2$
- D) $\frac{1}{x - 2}$

18

For $x \neq 2$ and $x \neq -3$, the expression $\frac{x^2 - 9}{x^2 + x - 6}$ is equivalent to $\frac{x - 3}{x - c}$, where c is a constant. What is the value of c ?

19

For $x \neq 1$ and $x \neq -1$, which expression is equivalent to $\frac{1}{x - 1} - \frac{1}{x + 1}$?

- A) 0
- B) $\frac{-2}{x^2 - 1}$
- C) $\frac{2}{x^2 - 1}$
- D) $\frac{2x}{x^2 - 1}$

20

For $x > 0$, which expression is equivalent to $(8x^6)^{2/3}$?

- A) $4x^4$
- B) $4x^9$
- C) $16x^4$
- D) $64x^4$

21

The equation $(3x + a)(2x + b) = 6x^2 + 19x + 15$ is true for all x , where a and b are positive integers. What is the value of $a + b$?

22

For $x \neq -5$, which expression is equivalent to $\frac{2x^2 + 7x - 15}{x + 5}$?

- A) $2x + 3$
- B) $2x - 3$
- C) $2x - 15$
- D) $x - 3$



Module 2

1

Which expression is equivalent to $(x - 4)(x + 4)$?

- A) $x^2 - 16$
- B) $x^2 + 16$
- C) $x^2 - 8x - 16$
- D) $x^2 - 16x$

2

Which expression is equivalent to $2x(3x - 7) + 5x$?

- A) $6x^2 - 2x$
- B) $6x^2 + 5x - 7$
- C) $6x^2 - 9x$
- D) $6x^2 - 19x$

3

The expression $x^2 + 10x + 25$ is equivalent to $(x + k)^2$, where k is a positive constant. What is the value of k ?

4

Which of the following is equivalent to $4x^2 - 25$?

- A) $(2x - 5)^2$
- B) $(4x + 5)(x - 5)$
- C) $(2x + 5)(2x - 5)$
- D) $(2x + 5)^2$

5

For $x \neq 0$ and $y \neq 0$, which expression is equivalent to $\frac{x^4 y^3}{xy^5}$?

- A) $x^3 y^2$
- B) $\frac{x^4}{y^2}$
- C) $\frac{x^3}{y^8}$
- D) $\frac{x^3}{y^2}$

6

What is the value of $27^{2/3}$?

7

Which expression shows $3x^2 - 12$ factored completely?

- A) $3(x^2 - 4)$
- B) $3(x + 2)(x - 2)$
- C) $(x + 2)(x - 2)$
- D) $3(x - 2)^2$

8

Which expression is equivalent to $(x + 3)(x^2 - 2x + 4)$?

- A) $x^3 + x^2 + 2x + 12$
- B) $x^3 + x^2 - 2x + 12$
- C) $x^3 - 5x^2 + 10x + 12$
- D) $x^3 + 27$

9

The equation $(2x - 3)(3x + k) = 6x^2 + x - 15$ is true for all values of x , where k is a constant. What is the value of k ?

10

For $x \neq -3$, which expression is equivalent to $\frac{3x^2 - 27}{x + 3}$?

- A) $3x - 9$
- B) $3x + 9$
- C) $3x - 27$
- D) $x - 9$

11

For $x \neq 0$ and $x \neq -1$, which expression is equivalent to $\frac{2}{x+1} + \frac{3}{x}$?

- A) $\frac{5}{x^2 + x}$
- B) $\frac{5x + 3}{x^2 + x}$
- C) $\frac{5}{2x + 1}$
- D) $\frac{5x + 2}{x^2 + x}$

12

For $x > 0$, the expression $x^{1/2} \cdot x^{3/2}$ is equivalent to x^k , where k is a constant. What is the value of k ?



13

Which expression is equivalent to $\sqrt{48} - \sqrt{12}$?

- A) 6
- B) $2\sqrt{3}$
- C) $6\sqrt{3}$
- D) $2\sqrt{6}$

14

For $x \neq 0$ and $y \neq 0$, which expression is equivalent to $(2x^3y^{-2})^3$?

- A) $\frac{8x^9}{y^6}$
- B) $\frac{6x^9}{y^6}$
- C) $\frac{8x^6}{y^5}$
- D) $8x^9y^6$

15

For $x \neq 3$, which expression is equivalent to $\frac{x^2 - 5x + 6}{x - 3}$?

- A) $x + 2$
- B) $x - 6$
- C) $x - 2$
- D) $x - 3$

16

Which of the following is equivalent to $6x^2 - x - 15$?

- A) $(2x - 3)(3x + 5)$
- B) $(2x + 3)(3x - 5)$
- C) $(6x + 5)(x - 3)$
- D) $(2x + 5)(3x - 3)$

17

For $x \neq 1$ and $x \neq -3$, which expression is equivalent to $\frac{x^2 - 1}{x + 3} \cdot \frac{x^2 + 3x}{x - 1}$?

- A) $x^2 - x$
- B) $x + 1$
- C) $\frac{x^2 + x}{x + 3}$
- D) $x^2 + x$

18

For $x \neq \frac{1}{2}$, the expression $\frac{2x^2 + 5x - 3}{2x - 1}$ is equivalent to $x + c$, where c is a constant. What is the value of c ?

19

For nonzero values of x and y , which expression is equivalent to $\frac{1}{x^{-1} + y^{-1}}$?

- A) $x + y$
- B) $\frac{xy}{x + y}$
- C) $\frac{x + y}{xy}$
- D) $\frac{1}{x + y}$

20

The expression $(2x + 3)^2 - (2x - 3)^2$ is equivalent to kx , where k is a constant. What is the value of k ?

21

For $x \neq 2$ and $x \neq -2$, which expression is equivalent to $\frac{x^2 - x - 6}{x^2 - 4}$?

- A) $\frac{x + 3}{x + 2}$
- B) $\frac{x + 6}{4}$
- C) $\frac{x - 3}{x + 2}$
- D) $\frac{x - 3}{x - 2}$

22

For $x \neq 1$ and $x \neq -1$, which expression is equivalent to $\frac{x}{x - 1} - \frac{1}{x + 1}$?

- A) $\frac{x - 1}{x^2 - 1}$
- B) $\frac{x^2 + x - 1}{x^2 - 1}$
- C) $\frac{x^2 + 1}{x^2 - 1}$
- D) $\frac{x^2 - x + 1}{x^2 - 1}$



Lesson 7: Quadratic Equations

Module 1

1

What is the larger of the two solutions to $x^2 - 7x + 10 = 0$?

- A) 2
- B) 5
- C) 7
- D) 10

2

If $(x - 3)^2 = 49$, what is the greatest possible value of x ?

- A) -4
- B) 7
- C) 10
- D) 52

3

If $2x^2 = 50$, what is the positive value of x ?

4

What is the sum of the solutions to $x^2 - 9x + 14 = 0$?

- A) -9
- B) 2
- C) 7
- D) 9

5

What is the product of the solutions to $(x + 4)(x - 6) = 0$?

- A) -24
- B) -2
- C) 2
- D) 24

6

What is the value of the discriminant of the equation $x^2 + 6x + 5 = 0$?

- A) 4
- B) 16
- C) 36
- D) 56

7

Which of the following is a solution to $x^2 + x - 12 = 0$?

- A) 3
- B) 4
- C) 6
- D) 12

8

The solutions to $x^2 - 6x + 4 = 0$ can be written in the form $3 \pm \sqrt{k}$, where k is a positive constant. What is the value of k ?

9

The equation $x^2 + 8x - 3 = 0$ can be rewritten in the form $(x + 4)^2 = b$, where b is a constant. What is the value of b ?

- A) -19
- B) 3
- C) 13
- D) 19

10

What is the sum of the solutions to $2x^2 - 10x + 3 = 0$?

- A) -5
- B) $\frac{3}{2}$
- C) 5
- D) 10

11

A ball is kicked from ground level, and its height in feet after t seconds is given by $h = -16t^2 + 48t$. After how many seconds does the ball return to the ground?



12

In the equation $x^2 + kx + 9 = 0$, k is a positive constant and the equation has exactly one real solution. What is the value of k ?

- A) 3
- B) 6
- C) 9
- D) 12

13

What is the smaller of the two solutions to $2x^2 - 7x + 3 = 0$?

- A) $\frac{1}{2}$
- B) 1
- C) $\frac{3}{2}$
- D) 3

14

If $(2x - 1)^2 = 25$, what is the sum of the possible values of x ?

- A) -2
- B) 1
- C) 3
- D) 5

15

The equation $3x^2 + bx + 12 = 0$ has exactly one real solution, and b is positive. What is the value of b ?

16

The equation $2x^2 + 12x + 5 = 0$ can be written in the form $2(x + 3)^2 = k$, where k is a constant. What is the value of k ?

- A) -13
- B) 5
- C) 13
- D) 23

17

What is the product of the solutions to $3x^2 - 5x - 2 = 0$?

- A) $-\frac{2}{3}$
- B) $-\frac{1}{3}$
- C) $\frac{2}{3}$
- D) $\frac{5}{3}$

18

In the xy -plane, the graph of $y = x^2 + 6x + c$ intersects the x -axis at exactly one point. What is the value of c ?

19

A rectangular garden is 3 meters longer than it is wide, and its area is 88 square meters. What is the width of the garden, in meters?

- A) 5
- B) 8
- C) 11
- D) 16

20

The solutions to $x^2 - 4x - 8 = 0$ can be written as $a \pm b\sqrt{3}$, where a and b are positive integers. What is the value of $a + b$?

- A) 2
- B) 3
- C) 4
- D) 6

21

The solutions to $x^2 + px + 40 = 0$ are $x = 4$ and $x = r$, where p and r are constants. What is the value of $p + r$?

- A) -14
- B) -4
- C) 4
- D) 14

22

The equation $2x^2 + 8x + k = 0$ has no real solutions. Which of the following could be the value of k ?

- A) 2
- B) 4
- C) 8
- D) 10



Module 2

1

What is the positive solution to $x^2 - 2x - 15 = 0$?

- A) -5
- B) -3
- C) 3
- D) 5

2

If $3x^2 - 27 = 0$, what is the positive value of x ?

3

How many distinct real solutions does the equation $x^2 + 4x + 7 = 0$ have?

- A) 0
- B) 1
- C) 2
- D) 3

4

The equation $x^2 + 10x + 18 = 0$ can be rewritten as $(x + 5)^2 = c$, where c is a constant. What is the value of c ?

- A) -18
- B) 7
- C) 18
- D) 43

5

What is the product of the solutions to $x^2 + 7x - 18 = 0$?

- A) -18
- B) -7
- C) 7
- D) 18

6

A stone is thrown from the top of a cliff, and its height in feet after t seconds is $h = -16t^2 + 64t + 80$. After how many seconds does the stone hit the ground?

7

What are the solutions to $x^2 - 4x + 1 = 0$?

- A) $2 \pm \sqrt{3}$
- B) $2 \pm \sqrt{5}$
- C) $-2 \pm \sqrt{3}$
- D) $4 \pm 2\sqrt{3}$

8

The length of a rectangular rug is 5 feet greater than its width, and the rug covers 84 square feet. What is the width of the rug, in feet?

- A) -12
- B) 6
- C) 7
- D) 12

9

The solutions to $x^2 + bx + c = 0$ are -3 and 5 , where b and c are constants. What is the value of c ?

10

The equation $3x^2 - 12x + 5 = 0$ can be written in the form $3(x - 2)^2 = k$, where k is a constant. What is the value of k ?

- A) -7
- B) 5
- C) 7
- D) 17

11

In the equation $kx^2 + 12x + 9 = 0$, k is a nonzero constant and the equation has exactly one real solution. What is the value of k ?

- A) 3
- B) 4
- C) 6
- D) 9

12

A rocket's height in feet after t seconds is $h = -16t^2 + 96t$. What is the smaller of the two values of t at which the height is 128 feet?

**13**

What is the positive difference between the two solutions to $x^2 - 6x + 2 = 0$?

- A) $\sqrt{7}$
- B) $2\sqrt{7}$
- C) 6
- D) 7

14

For which of the following values of k does the equation $x^2 + kx + 4 = 0$ have two distinct real solutions?

- A) 0
- B) 2
- C) 4
- D) 5

15

What is the sum of the solutions to $4x^2 - 20x + 9 = 0$?

- A) $\frac{9}{4}$
- B) $\frac{5}{2}$
- C) 5
- D) 20

16

If $x^2 + 6x = 16$, what is the sum of all possible values of x ?

- A) -16
- B) -8
- C) -6
- D) 16

17

If $\frac{12}{x} = x - 1$ and $x \neq 0$, what is the greatest possible value of x ?

- A) -3
- B) 3
- C) 4
- D) 12

18

In the xy -plane, the line $y = 2x + 3$ intersects the parabola $y = x^2 + k$ at exactly one point, where k is a constant. What is the value of k ?

19

The equation $x^2 - 14x + k = 0$ has two positive integer solutions that differ by 4. What is the value of k ?

20

A ball is launched from a platform, and its height in feet after t seconds is $h = -16t^2 + 40t + 24$. After how many seconds does the ball hit the ground?

- A) $\frac{3}{2}$
- B) 2
- C) 3
- D) 5

21

The solutions to $2x^2 - 8x + 3 = 0$ can be written as $\frac{a \pm \sqrt{b}}{2}$, where a and b are positive integers. What is the value of $a + b$?

- A) 14
- B) 18
- C) 44
- D) 48

22

The equation $x^2 + kx + k + 3 = 0$ has exactly one real solution. What is the sum of all possible values of k ?

- A) -4
- B) 2
- C) 4
- D) 6



Lesson 8: Quadratic and Polynomial Functions

Module 1

1

The function f is defined by $f(x) = (x - 4)^2 + 7$. The vertex of the graph of $y = f(x)$ in the xy -plane is the point (h, k) . What is the value of h ?

- A) -7
- B) -4
- C) 4
- D) 7

2

The function f is defined by $f(x) = x^2 - 6x + 11$. The axis of symmetry of the graph of $y = f(x)$ in the xy -plane is the line $x = c$. What is the value of c ?

3

The function g is defined by $g(x) = (x - 2)(x + 5)$. What are the x -intercepts of the graph of $y = g(x)$ in the xy -plane?

- A) $x = -2$ and $x = 5$
- B) $x = 2$ and $x = 5$
- C) $x = -2$ and $x = -5$
- D) $x = 2$ and $x = -5$

4

What is the y -coordinate of the y -intercept of the graph of $y = (x - 3)(x + 8)$ in the xy -plane?

- A) -24
- B) -11
- C) 5
- D) 24

5

The function f is defined by $f(x) = -(x - 1)^2 + 6$. What is the maximum value of $f(x)$?

- A) -6
- B) -1
- C) 1
- D) 6

6

The remainder when the polynomial $p(x) = x^2 + 3x - 5$ is divided by $x - 2$ is r . What is the value of r ?

7

In the xy -plane, which of the following equations defines a parabola with vertex $(5, -2)$?

- A) $y = (x + 5)^2 - 2$
- B) $y = (x - 5)^2 + 2$
- C) $y = (x - 5)^2 - 2$
- D) $y = (x - 2)^2 + 5$

8

Which of the following expressions is equivalent to $x^2 + 8x + 3$ and displays the minimum value of the function $f(x) = x^2 + 8x + 3$ as a constant or coefficient?

- A) $(x + 4)^2 + 3$
- B) $(x - 4)^2 - 13$
- C) $(x + 8)^2 - 13$
- D) $(x + 4)^2 - 13$

9

The function f is defined by $f(x) = 3x^2 - 12x + 7$. What is the x -coordinate of the vertex of the graph of $y = f(x)$ in the xy -plane?

10

A ball is thrown upward from a platform. Its height above the ground, in feet, t seconds after it is thrown is given by $h(t) = -16t^2 + 64t + 5$. What is the maximum height, in feet, reached by the ball?

- A) 2
- B) 5
- C) 64
- D) 69



11

Which of the following expressions is equivalent to $x^2 - 2x - 15$ and displays the x -intercepts of the graph of $y = x^2 - 2x - 15$ as constants or coefficients?

- A) $(x + 5)(x - 3)$
- B) $(x - 5)(x + 3)$
- C) $(x - 5)(x - 3)$
- D) $(x - 15)(x + 1)$

12

The polynomial function p is defined by $p(x) = x(x - 3)^2(x + 4)$. Which zero of p has multiplicity 2?

- A) -4
- B) 0
- C) 2
- D) 3

13

The polynomial function p is defined by $p(x) = -2x^3 + 5x - 1$. Which of the following describes the end behaviour of the graph of $y = p(x)$ in the xy -plane?

- A) As x increases without bound, $p(x)$ increases without bound; as x decreases without bound, $p(x)$ decreases without bound.
- B) As x increases without bound, $p(x)$ decreases without bound; as x decreases without bound, $p(x)$ increases without bound.
- C) As x increases or decreases without bound, $p(x)$ increases without bound.
- D) As x increases or decreases without bound, $p(x)$ decreases without bound.

14

Which of the following is a factor of the polynomial $p(x) = x^3 - 7x + 6$?

- A) $x + 1$
- B) $x + 2$
- C) $x - 1$
- D) $x - 3$

15

In the xy -plane, the graph of the quadratic function g crosses the x -axis at $(1, 0)$ and at $(7, 0)$. What is the x -coordinate of the vertex of the graph of g ?

- A) 3
- B) 4
- C) 6
- D) 8

16

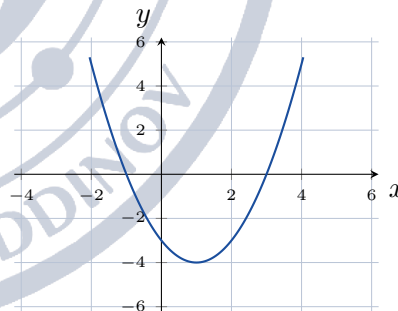
In the xy -plane, the graph of the quadratic function f has vertex $(2, -3)$ and passes through the point $(4, 5)$. The function can be written as $f(x) = a(x - 2)^2 - 3$, where a is a constant. What is the value of a ?

- A) $\frac{1}{2}$
- B) 2
- C) 4
- D) 8

17

The polynomial p is defined by $p(x) = 2x^3 + kx - 6$, where k is a constant. When $p(x)$ is divided by $x - 2$, the remainder is 4. What is the value of k ?

18



The graph of the quadratic function f is shown in the xy -plane above. Which equation could define f ?

- A) $f(x) = (x + 1)^2 - 4$
- B) $f(x) = (x - 1)^2 + 4$
- C) $f(x) = -(x - 1)^2 - 4$
- D) $f(x) = (x - 1)^2 - 4$



19

Which of the following expressions is equivalent to $(x - 3)^2 - 16$ and displays the y -coordinate of the y -intercept of the graph of $y = (x - 3)^2 - 16$ as a constant or coefficient?

- A) $x^2 - 6x + 9$
- B) $x^2 + 6x - 7$
- C) $x^2 - 6x - 7$
- D) $x^2 - 9x - 7$

20

The polynomial p is defined by $p(x) = x^3 + ax^2 - 9x - 18$, where a is a constant. If $x + 2$ is a factor of $p(x)$, what is the value of a ?

21

A store's daily revenue, in dollars, from selling an item at a price of p dollars is modelled by $R(p) = p(120 - 4p)$. What is the maximum daily revenue, in dollars?

- A) 15
- B) 60
- C) 900
- D) 1800

22

The function f is defined by $f(x) = a(x - 3)^2 + 5$, where a is a negative constant. Which of the following must be true?

- A) The graph of $y = f(x)$ has a minimum at the point $(3, 5)$.
- B) The maximum value of f is 5.
- C) The graph of $y = f(x)$ has no x -intercepts.
- D) The graph of $y = f(x)$ is symmetric about the line $x = 5$.



Module 2

1

The function f is defined by $f(x) = -(x+6)^2 + 11$. For what value of x does f attain its maximum value?

- A) -11
- B) -6
- C) 6
- D) 11

2

The function g is defined by $g(x) = 2x^2 - 16x + 5$. The graph of $y = g(x)$ in the xy -plane is symmetric about the line $x = c$. What is the value of c ?

3

The polynomial function p is defined by $p(x) = x(x-7)(x+2)$. What are the x -intercepts of the graph of $y = p(x)$ in the xy -plane?

- A) $x = 0$, $x = 7$, and $x = -2$
- B) $x = 0$, $x = -7$, and $x = 2$
- C) $x = 7$ and $x = -2$
- D) $x = 0$, $x = -7$, and $x = -2$

4

What is the remainder when the polynomial $p(x) = x^3 - 2x^2 + 5$ is divided by $x + 1$?

- A) -2
- B) 2
- C) 4
- D) 8

5

The function h is defined by $h(x) = 3(x-2)^2 - 5$. What is the y -coordinate of the y -intercept of the graph of $y = h(x)$ in the xy -plane?

6

The polynomial function q is defined by $q(x) = 4x^4 - x^3 + 2$. Which of the following describes the end behaviour of the graph of $y = q(x)$ in the xy -plane?

- A) As x increases or decreases without bound, $q(x)$ increases without bound.
- B) As x increases or decreases without bound, $q(x)$ decreases without bound.
- C) As x increases without bound, $q(x)$ increases without bound; as x decreases without bound, $q(x)$ decreases without bound.
- D) As x increases without bound, $q(x)$ decreases without bound; as x decreases without bound, $q(x)$ increases without bound.

7

In the xy -plane, which of the following equations defines a parabola with x -intercepts at $(6,0)$ and $(-1,0)$?

- A) $y = (x+6)(x-1)$
- B) $y = (x-6)(x-1)$
- C) $y = (x-6)(x+1)$
- D) $y = (x+6)(x+1)$

8

The function f is defined by $f(x) = x^2 - 10x + 21$. When $f(x)$ is written in the form $(x-h)^2 + k$, where h and k are constants, what is the value of $h + k$?

9

A rocket is launched from a platform. Its height above the ground, in metres, t seconds after launch is given by $h(t) = -5t^2 + 40t + 2$. How many seconds after launch does the rocket reach its greatest height?

- A) 2
- B) 4
- C) 8
- D) 82



10

Which of the following expressions is equivalent to $2x^2 - 12x + 13$ and displays the minimum value of the function $f(x) = 2x^2 - 12x + 13$ as a constant or coefficient?

- A) $2(x - 3)^2 + 5$
- B) $2(x - 6)^2 - 5$
- C) $2(x - 3)^2 - 5$
- D) $2(x + 3)^2 - 5$

11

The polynomial function p is defined by $p(x) = (x - 1)(x + 3)(x + 4)$. What is the value of $p(2)$?

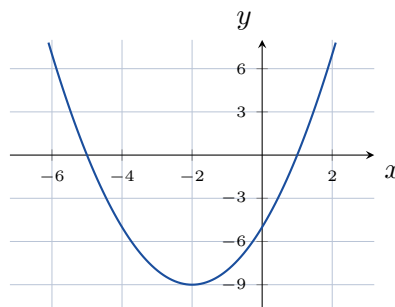
- A) -30
- B) 6
- C) 24
- D) 30

12

The polynomial function p is defined by $p(x) = (x + 1)^3(x - 2)^2$. Which of the following describes the behaviour of the graph of $y = p(x)$ at its x -intercepts?

- A) The graph crosses the x -axis at $x = -1$ and touches the x -axis at $x = 2$.
- B) The graph touches the x -axis at $x = -1$ and crosses the x -axis at $x = 2$.
- C) The graph crosses the x -axis at both $x = -1$ and $x = 2$.
- D) The graph touches the x -axis at both $x = -1$ and $x = 2$.

13



A quadratic function f has the graph shown in the xy -plane above. Which of the following could be an equation of that graph?

- A) $y = (x + 2)^2 - 9$
- B) $y = (x - 2)^2 - 9$
- C) $y = (x + 2)^2 + 9$
- D) $y = -(x + 2)^2 - 9$

14

In the xy -plane, the graph of the quadratic function f crosses the x -axis at $(1, 0)$ and $(9, 0)$, and the minimum value of f is -32 . The function is defined by $f(x) = a(x - 1)(x - 9)$, where a is a constant. What is the value of a ?

15

Which of the following is a factor of the polynomial $p(x) = x^3 + 2x^2 - 11x - 12$?

- A) $x - 2$
- B) $x - 1$
- C) $x + 3$
- D) $x - 3$

16

Which of the following expressions is equivalent to $2x^2 - 2x - 12$ and displays the x -intercepts of the graph of $y = 2x^2 - 2x - 12$ as constants or coefficients?

- A) $(x - 3)(x + 2)$
- B) $2(x - 3)(x + 2)$
- C) $2(x + 3)(x - 2)$
- D) $2(x - 6)(x + 1)$

17

The polynomial p is defined by $p(x) = 3x^3 - 5x^2 + kx + 8$, where k is a constant. If $x - 2$ is a factor of $p(x)$, what is the value of k ?



18

The function f is defined by $f(x) = x^2 + 5x + 3$. What is the minimum value of $f(x)$?

- A) $-\frac{13}{2}$
- B) $-\frac{13}{4}$
- C) $-\frac{5}{2}$
- D) $\frac{13}{4}$

19

In the xy -plane, the graph of $y = x^2 + bx + c$ has its vertex at $(3, -4)$, where b and c are constants. What is the value of c ?

- A) -6
- B) -4
- C) 5
- D) 13

20

A company's daily profit, in dollars, from selling x units of a product is modelled by $P(x) = -2x^2 + 56x - 180$. What is the maximum daily profit, in dollars?

21

The polynomial function p is defined by $p(x) = -(x - 1)^2(x + 3)$. Which of the following statements about the graph of $y = p(x)$ in the xy -plane is true?

- A) The graph touches the x -axis at $x = 1$, and $p(x)$ decreases without bound as x increases without bound.
- B) The graph crosses the x -axis at $x = 1$, and $p(x)$ increases without bound as x increases without bound.
- C) The graph touches the x -axis at $x = -3$, and $p(x)$ decreases without bound as x increases without bound.
- D) The graph touches the x -axis at $x = 1$, and $p(x)$ increases without bound as x increases without bound.

22

The polynomial p is defined by $p(x) = x^3 - 6x^2 + 11x - 6$. If $p(x) = (x - 3)q(x)$ for some polynomial q , which of the following is the expression $q(x)$?

- A) $x^2 - 3x + 2$
- B) $x^2 - 3x - 2$
- C) $x^2 + 3x + 2$
- D) $x^2 - 9x + 2$



Lesson 9: Exponential Functions, Radicals and Rational Expressions

Module 1

1

The function $P(x) = 750(1.06)^x$ gives the population of a town x years after 2015. What was the population of the town in 2015?

- A) 1.06
- B) 750
- C) 795
- D) 45

2

A car purchased for \$24,000 loses 15% of its value each year. Which function gives the value $V(t)$, in dollars, of the car t years after it was purchased?

- A) $V(t) = 24000(0.15)^t$
- B) $V(t) = 24000(1.15)^t$
- C) $V(t) = 24000 - 0.15t$
- D) $V(t) = 24000(0.85)^t$

3

What is the solution to $\sqrt{2x+1} = 7$?

4

What value of x satisfies $\frac{12}{x-2} = 4$?

- A) 1
- B) 3
- C) 5
- D) 7

5

For an exponential function f , $f(0) = 5$, $f(1) = 15$, and $f(2) = 45$. What is the value of $f(4)$?

- A) 75
- B) 135
- C) 405
- D) 1215

6

The function $A(x) = 80(0.5)^x$ gives the number of grams of a sample that remain after x half-lives. How many grams remain after 3 half-lives?

7

A quantity increases by 7% each year. Which of the following is the growth factor for one year?

- A) 0.07
- B) 0.93
- C) 1.07
- D) 7

8

What is the solution to $\sqrt{x+7} = x+1$?

- A) -3
- B) 2
- C) 3
- D) -3 and 2

9

What is the solution to $\frac{1}{x} + \frac{1}{2x} = \frac{3}{4}$?

10

The function $P(t) = 1200(1.03)^t$ models the population of a town t years after 2010. Which of the following is the best interpretation of the number 1.03 in this context?

- A) The population increases by 1.03 people each year.
- B) The population increases by 103% each year.
- C) Each year the population is 1.03 times the population the year before.
- D) The population in 2010 was 1.03 times 1200.



11

A savings account with an initial balance of \$1500 earns 6% annual interest, compounded monthly. Which function gives the balance $B(t)$, in dollars, after t years?

- A) $B(t) = 1500(1.06)^{12t}$
- B) $B(t) = 1500(1.5)^t$
- C) $B(t) = 1500(1.005)^t$
- D) $B(t) = 1500(1.005)^{12t}$

12

What is the solution to $\frac{5}{x+1} = \frac{2}{x-1}$?

- A) $\frac{7}{3}$
- B) $\frac{3}{7}$
- C) $-\frac{7}{3}$
- D) 1

13

A machine purchased for \$30,000 depreciates by 20% of its value each year. What is the value of the machine, in dollars, after 3 years?

- A) 12,000
- B) 15,360
- C) 18,000
- D) 51,840

14

Squaring both sides of $\sqrt{3x+4} = x$ produces an equation with two solutions. Which of the following is an extraneous solution of the original equation?

- A) -1
- B) 1
- C) 4
- D) -4

15

A bacteria population doubles every 5 hours. If there are 300 bacteria at time $t = 0$, how many bacteria are there after 20 hours?

16

The value of a rare coin is \$5000 today and increases by 12% every 4 years. Which function gives the value $V(t)$, in dollars, of the coin t years from today?

- A) $V(t) = 5000(1.12)^{4t}$
- B) $V(t) = 5000(1.12)^{t/4}$
- C) $V(t) = 5000(0.12)^{t/4}$
- D) $V(t) = 5000(1.03)^t$

17

What is the solution to $\frac{x^2}{x-3} = \frac{9}{x-3}$?

- A) -3
- B) 3
- C) -3 and 3
- D) There is no solution.

18

What is the solution to $\sqrt{x+5} - \sqrt{x} = 1$?

19

The value of an investment doubles every 9 years. If the investment is worth \$2500 today, which function gives its value $V(t)$, in dollars, t years from today?

- A) $V(t) = 2500(2)^{9t}$
- B) $V(t) = 2500(9)^{t/2}$
- C) $V(t) = 2500 + 2^{t/9}$
- D) $V(t) = 2500(2)^{t/9}$

20

The expression $\frac{8^{x+1}}{2^x}$ can be written in the form 2^{ax+b} , where a and b are constants. What is the value of $a+b$?

- A) 2
- B) 3
- C) 5
- D) 6

21

What is the solution to $\sqrt{2x+3} = x-6$?



22

The population of a town was 8000 in 2000 and 12,000 in 2010, and the population grows exponentially. Which function gives the population $P(t)$ of the town t years after 2000?

- A) $P(t) = 8000(1.5)^{t/10}$
- B) $P(t) = 8000(1.5)^{10t}$
- C) $P(t) = 8000(0.5)^{t/10}$
- D) $P(t) = 8000 + 400t$





Module 2

1

The function $f(x) = 200(1.15)^x$ increases by $p\%$ for each increase of 1 in the value of x . What is the value of p ?

- A) 0.15
- B) 15
- C) 115
- D) 1.15

2

What is the solution to $\frac{8}{x+3} = 2$?

3

What is the solution to $\sqrt{x-1} + 4 = 9$?

- A) 6
- B) 10
- C) 24
- D) 26

4

The value of a piece of equipment decreases by 35% each year. Which of the following is the decay factor for one year?

- A) 0.35
- B) 1.35
- C) 35
- D) 0.65

5

The function g is defined by $g(x) = 6(3)^x$. What is the value of $g(-1)$?

- A) -18
- B) $\frac{1}{2}$
- C) 2
- D) 18

6

A 400-milligram sample of a radioactive substance has a half-life of 12 years. Which function gives the amount $A(t)$, in milligrams, of the sample remaining after t years?

- A) $A(t) = 400(0.5)^{12t}$
- B) $A(t) = 400(0.5)^{t/12}$
- C) $A(t) = 400(1.5)^{t/12}$
- D) $A(t) = 400 - \frac{t}{12}$

7

The value of an investment, in dollars, is modeled by $V(t) = 800(1.05)^t$, where t is the number of years since the investment was made. What is the value, in dollars, of the investment after 2 years?

8

What is the solution to $\frac{x}{x-1} = \frac{2}{x-1} + 3$?

- A) $\frac{1}{2}$
- B) $-\frac{1}{2}$
- C) 1
- D) 2

9

What is the sum of all solutions to $\sqrt{5x-4} = x$?

10

The function $P(t) = 3000(1.08)^t$ gives the value, in dollars, of an account t years after it was opened. Which function gives the value of the account m months after it was opened?

- A) $P(m) = 3000(1.08)^{12m}$
- B) $P(m) = 3000\left(\frac{1.08}{12}\right)^m$
- C) $P(m) = 250(1.08)^m$
- D) $P(m) = 3000(1.08)^{m/12}$



11

The number of bacteria in a culture triples every 4 hours. There are 90 bacteria in the culture at time $t = 0$. How many bacteria are in the culture after 12 hours?

- A) 270
- B) 810
- C) 1080
- D) 2430

12

What is the positive solution to $\frac{1}{x} + \frac{1}{x+2} = \frac{3}{4}$?

- A) $\frac{2}{3}$
- B) 2
- C) $-\frac{4}{3}$
- D) 4

13

The value of a car t years after it was purchased is modeled by $V(t) = 18000(0.88)^t$. Which of the following is the best interpretation of 0.88 in this model?

- A) The car loses 88% of its value each year.
- B) The car is worth \$0.88 less each year.
- C) Each year the car is worth 0.88 times its value the previous year.
- D) The car was worth \$18,000 times 0.88 when it was purchased.

14

Which of the following is equivalent to $\frac{x^2 - 9}{x + 3}$ for $x \neq -3$?

- A) $x - 3$
- B) $x + 3$
- C) $x^2 - 3$
- D) $\frac{1}{x-3}$

15

What is the solution to $2\sqrt{x} = x - 3$?

- A) 1
- B) 3
- C) 9
- D) 1 and 9

16

The function $N(t) = 1600\left(\frac{1}{2}\right)^{t/8}$ gives the number of grams of a sample remaining t years after the sample was formed. After how many years will 100 grams remain?

- A) 16
- B) 24
- C) 32
- D) 64

17

What is the sum of all solutions to $\sqrt{2x+7} = 1 + \sqrt{x+3}$?

- A) -3
- B) -2
- C) 1
- D) 4

18

A population grows exponentially. The population is 500 at time $t = 0$ and 720 when $t = 2$, where t is measured in years. If the population increases by $p\%$ each year, what is the value of p ?

- A) 20
- B) 22
- C) 44
- D) 144

19

What is the solution to $\frac{3}{x-1} + \frac{2}{x+1} = \frac{10}{x^2-1}$?

20

What value of x satisfies $3^{2x-1} = 27^{x-2}$?

21

The function $P(d) = 8(2)^d$ gives the number of thousands of people living in a city d decades after 1990. Which function gives the number of thousands of people living in the city y years after 1990?

- A) $P(y) = 8(2)^{10y}$
- B) $P(y) = 8(2)^{y/10}$
- C) $P(y) = 8(20)^y$
- D) $P(y) = 80(2)^y$



22

How many solutions does the equation $\frac{1}{x-2} + \frac{1}{x+2} = \frac{4}{x^2-4}$ have?

- A) 0
- B) 1
- C) 2
- D) Infinitely many





Lesson 10: Nonlinear Systems and Function Transformations

Module 1

1

The graph of $y = x^2$ is shifted 5 units up in the xy -plane. Which equation represents the resulting graph?

- A) $y = x^2 + 5$
- B) $y = x^2 - 5$
- C) $y = (x + 5)^2$
- D) $y = (x - 5)^2$

2

The function g is defined by $g(x) = f(x - 3)$. Which of the following describes the graph of $y = g(x)$ compared with the graph of $y = f(x)$?

- A) The graph of f shifted 3 units to the left.
- B) The graph of f shifted 3 units to the right.
- C) The graph of f shifted 3 units up.
- D) The graph of f shifted 3 units down.

3

In the xy -plane, the graphs of $y = 2x$ and $y = x^2 - 3$ intersect at two points. What is the greatest x -coordinate of an intersection point?

4

The graph of $y = f(x)$ is reflected across the x -axis. Which equation represents the reflected graph?

- A) $y = -f(x)$
- B) $y = f(-x)$
- C) $y = -f(-x)$
- D) $y = |f(x)|$

5

The graph of $y = 3f(x)$ is obtained from the graph of $y = f(x)$ by which transformation?

- A) A shift of 3 units up.
- B) A horizontal stretch by a factor of 3.
- C) A vertical stretch by a factor of 3.
- D) A horizontal compression by a factor of 3.

6

The graphs of $y = x^2$ and $y = x + 6$ intersect at two points in the xy -plane. What is the sum of the y -coordinates of those two points?

- A) 1
- B) 5
- C) 9
- D) 13

7

How many points of intersection do the graphs of $y = x^2 + 1$ and $y = x - 2$ have in the xy -plane?

- A) 0
- B) 1
- C) 2
- D) 4

8

$$y = x^2 - 4x + 7$$

$$y = 2x - 1$$

Which ordered pair (x, y) is a solution to the system of equations above?

- A) (1, 1)
- B) (2, 3)
- C) (3, 5)
- D) (4, 9)

9

In the xy -plane, the line $y = 4x + c$ is tangent to the parabola $y = x^2$, where c is a constant. What is the value of c ?

10

The system of equations $x^2 + y^2 = 25$ and $y = x + 1$ has two solutions in the xy -plane. What is the greatest y -coordinate of a solution?

- A) 3
- B) 4
- C) 5
- D) 7



11

The table gives five values of the function f .

x	-1	0	1	2	3
$f(x)$	5	2	6	-1	8

The function g is defined by $g(x) = f(x + 2)$. What is the value of $g(1)$?

- A) 2
- B) 5
- C) 6
- D) 8

12

The graph of $y = x^2$ is transformed to produce the graph of $y = (x + 4)^2 - 3$. Which of the following describes the transformation?

- A) A shift 4 units to the right and 3 units down.
- B) A shift 4 units to the right and 3 units up.
- C) A shift 4 units to the left and 3 units down.
- D) A shift 4 units to the left and 3 units up.

13

The point $(2, 5)$ lies on the graph of $y = f(x)$. The graph of $y = 3f(x)$ passes through the point $(2, k)$. What is the value of k ?

- A) $\frac{5}{3}$
- B) 8
- C) 15
- D) 45

14

The function f is defined by $f(x) = x^2 - 6x$, and $g(x) = f(-x)$. Which expression is equivalent to $g(x)$?

- A) $x^2 - 6x$
- B) $x^2 + 6x$
- C) $-x^2 + 6x$
- D) $-x^2 - 6x$

15

In the xy -plane, the graph of $y = -x^2 + 6x$ intersects the line $y = 5$ at two points. What is the positive difference between the x -coordinates of those two points?

16

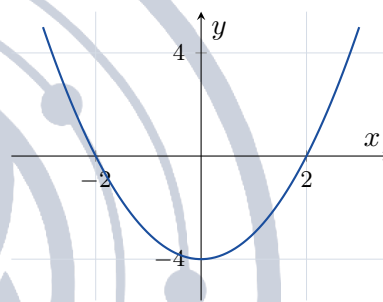
In the xy -plane, the line $y = mx - 4$ is tangent to the graph of $y = x^2$, where m is a positive constant. What is the value of m ?

- A) 2
- B) 4
- C) 8
- D) 16

17

The solutions to the system $y = x^2 - 2x - 3$ and $y = 2x + 2$ are (x_1, y_1) and (x_2, y_2) . What is the value of $y_1 + y_2$?

18



The graph of $y = f(x)$ is shown, where $f(x) = x^2 - 4$. The function g is defined by $g(x) = f(x + 3)$. What is the vertex of the graph of $y = g(x)$?

- A) $(-3, -4)$
- B) $(3, -4)$
- C) $(-3, -1)$
- D) $(0, -1)$

19

The minimum value of the function f is -3 , and it occurs at $x = 1$. The function g is defined by $g(x) = -2f(x)$. What is the maximum value of g ?

- A) -6
- B) -3
- C) 3
- D) 6

20

In the xy -plane, the line $y = 2x + c$ is tangent to the circle $x^2 + y^2 = 45$, where c is a positive constant. What is the value of c ?



21

The graph of $y = f(x)$ is first shifted 2 units to the right and then reflected across the x -axis. Which equation represents the resulting graph?

- A) $y = -f(x + 2)$
- B) $y = f(-x - 2)$
- C) $y = -f(x - 2)$
- D) $y = -f(x) - 2$

22

The graph of $y = f(x)$ has an x -intercept at $x = 6$. The function g is defined by $g(x) = f(2x)$. At what value of x does the graph of $y = g(x)$ have an x -intercept?

- A) 3
- B) 4
- C) 8
- D) 12





Module 2

1

The function h is defined by $h(x) = f(x + 6)$. Which of the following describes the graph of $y = h(x)$ compared with the graph of $y = f(x)$?

- A) The graph of f shifted 6 units to the left.
- B) The graph of f shifted 6 units to the right.
- C) The graph of f shifted 6 units up.
- D) The graph of f shifted 6 units down.

2

The graphs of $y = x^2 + 3x$ and $y = x + 8$ intersect at two points in the xy -plane. What is the greater of the two x -coordinates?

3

In the xy -plane, how many points does the graph of $y = x^2 - 4$ have in common with the line $y = -4$?

- A) 0
- B) 1
- C) 2
- D) 3

4

How does the graph of $y = f(x) - 7$ compare with the graph of $y = f(x)$?

- A) It is shifted 7 units to the left.
- B) It is shifted 7 units to the right.
- C) It is shifted 7 units up.
- D) It is shifted 7 units down.

5

The point $(-3, 8)$ lies on the graph of $y = f(x)$. The graph of $y = f(x) + 5$ passes through the point $(-3, k)$. What is the value of k ?

6

The function f is defined by $f(x) = x^3 - 2x$, and $h(x) = f(-x)$. Which expression is equivalent to $h(x)$?

- A) $x^3 - 2x$
- B) $x^3 + 2x$
- C) $-x^3 + 2x$
- D) $-x^3 - 2x$

7

The graph of $y = \frac{1}{2}f(x)$ is obtained from the graph of $y = f(x)$ by which transformation?

- A) A horizontal compression by a factor of 2.
- B) A horizontal stretch by a factor of 2.
- C) A shift down $\frac{1}{2}$ unit.
- D) A vertical shrink by a factor of $\frac{1}{2}$.

8

In the xy -plane, the line $y = 2x + c$ intersects the parabola $y = x^2 + 6x + 11$ at exactly one point, where c is a constant. What is the value of c ?

9

How many solutions does the system of equations $y = x^2 - 6x + 9$ and $y = x - 3$ have?

- A) 0
- B) 1
- C) 2
- D) 3

10

The table gives four values of the function f .

x	1	3	5	7
$f(x)$	2	6	-1	4

The function g is defined by $g(x) = f(x - 2) + 3$. What is the value of $g(5)$?

- A) 2
- B) 6
- C) 7
- D) 9

11

Which transformation takes the graph of $y = x^2$ to the graph of $y = -(x - 2)^2$?

- A) A reflection across the x -axis and a shift 2 units to the left.
- B) A reflection across the x -axis and a shift 2 units to the right.
- C) A reflection across the y -axis and a shift 2 units to the right.
- D) A reflection across the y -axis and a shift 2 units to the left.



12

The function f is defined by $f(x) = (x + 1)^2$. The graph of $y = f(x)$ is shifted 3 units to the right. Which equation represents the resulting graph?

- A) $y = (x + 4)^2$
- B) $y = (x + 1)^2 + 3$
- C) $y = (x - 2)^2$
- D) $y = (x - 3)^2 + 1$

13

The function f is defined by $f(x) = x^2 - 16$, and g is defined by $g(x) = f\left(\frac{x}{2}\right)$. What is the positive x -intercept of the graph of $y = g(x)$?

14

$$y = x^2 + 5x + 4$$

$$y = 3x + 7$$

The system above has two solutions. What is the sum of the y -coordinates of the two solutions?

- A) -2
- B) 8
- C) 10
- D) 12

15

The function g is defined by $g(x) = -f(-x)$. Which of the following describes the graph of $y = g(x)$ in terms of the graph of $y = f(x)$?

- A) The graph of f reflected across both the x -axis and the y -axis.
- B) The graph of f reflected across the x -axis only.
- C) The graph of f reflected across the y -axis only.
- D) The graph of f reflected across the line $y = x$.

16

In the xy -plane, at how many points does the graph of $x^2 + y^2 = 16$ intersect the graph of $y = x^2 - 4$?

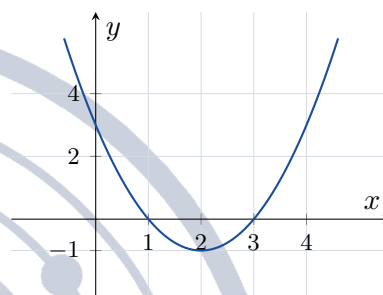
- A) 1
- B) 2
- C) 3
- D) 4

17

In the xy -plane, the line $y = kx$ is tangent to the graph of $y = x^2 - 2x + 4$, where k is a positive constant. What is the value of k ?

- A) 2
- B) 4
- C) 6
- D) 8

18



The graph of $y = f(x)$ is shown, where $f(x) = (x - 2)^2 - 1$. What is the vertex of the graph of $y = f(x + 1) - 2$?

- A) $(3, -3)$
- B) $(1, 1)$
- C) $(3, 1)$
- D) $(1, -3)$

19

The table gives five values of the function f .

x	0	1	2	3	4
$f(x)$	5	-3	8	6	1

The function h is defined by $h(x) = 2f(x + 1)$. What is the value of $h(2)$?

20

Which sequence of transformations takes the graph of $y = x^2$ to the graph of $y = 2(x + 3)^2$?

- A) A vertical stretch by a factor of 2 and a shift 3 units to the right.
- B) A vertical stretch by a factor of 2 and a shift 3 units to the left.
- C) A horizontal stretch by a factor of 2 and a shift 3 units to the left.
- D) A shift 3 units to the left and 2 units up.



21

In the xy -plane, the graph of $x^2 + y^2 = 25$ and the graph of $y = x^2 - 13$ intersect at how many points?

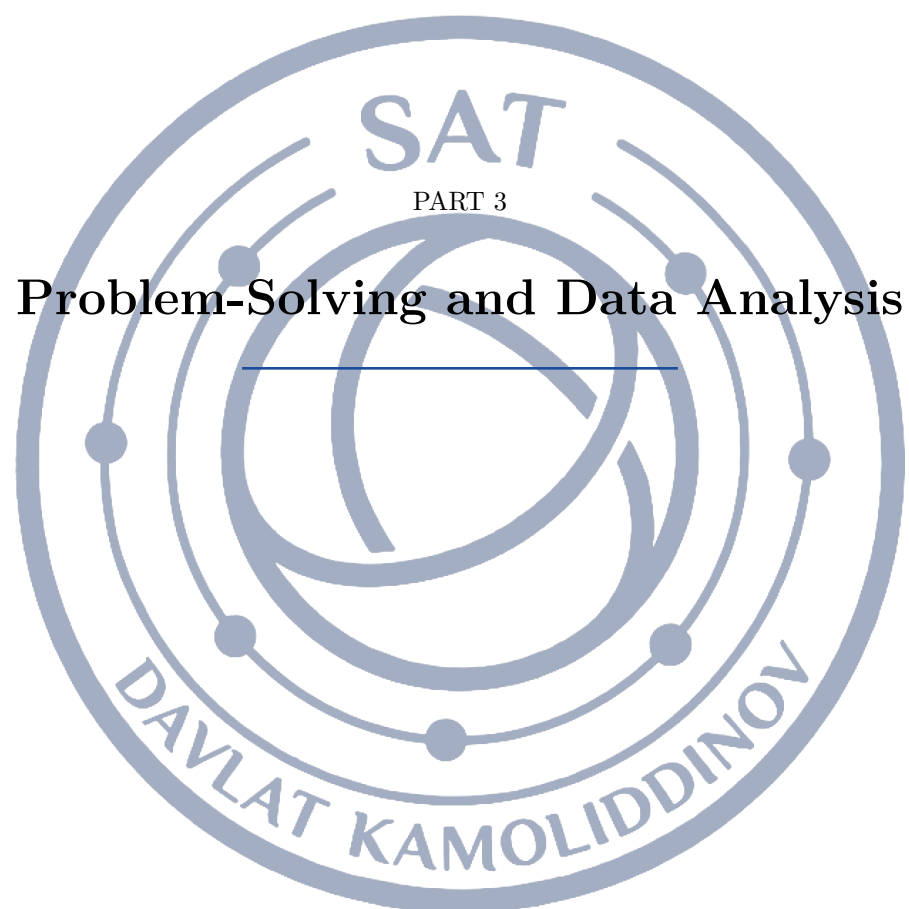
22

$$y = x^2 - 2x + c$$

$$y = 2x - 5$$

In the system of equations above, c is a constant. For what value of c does the system have exactly one real solution?

A) -1 B) 1 C) 5 D) 9 



Problem-Solving and Data Analysis



Lesson 11: Ratios, Rates, Proportional Relationships and Units

Module 1

1

In a school choir, the ratio of boys to girls is 2 : 3. If there are 18 boys in the choir, how many girls are in the choir?

- A) 12
- B) 15
- C) 18
- D) 27

2

A car travels 245 kilometres using 35 litres of fuel. At this rate, how many kilometres does the car travel per litre of fuel?

3

A bag of rice has a mass of 3.5 kilograms. Given that 1 kilogram = 1000 grams, what is the mass of the bag, in grams?

- A) 350
- B) 3500
- C) 35 000
- D) 350 000

4

Two business partners divide a profit of \$60 in the ratio 2 : 3. How many dollars does the partner with the larger share receive?

- A) 12
- B) 24
- C) 30
- D) 36

5

Four identical notebooks cost \$6.00. At this rate, what is the cost, in dollars, of 10 of these notebooks?

- A) 15
- B) 20
- C) 24
- D) 60

6

On a map, 1 centimetre represents 25 kilometres. Two cities are 6 centimetres apart on the map. What is the actual distance, in kilometres, between the two cities?

7

A metal block has a mass of 480 grams and a volume of 60 cubic centimetres. What is the density of the block, in grams per cubic centimetre?

- A) 0.125
- B) 8
- C) 80
- D) 540

8

A bag contains only red marbles and blue marbles. The ratio of red marbles to blue marbles is 3 : 4, and there are 84 marbles in the bag. How many of the marbles are red?

- A) 12
- B) 36
- C) 48
- D) 63

9

A train travels at a constant speed of 72 kilometres per hour. What is the speed of the train, in metres per second?

10

A recipe that uses 3 cups of flour makes 24 cookies. At the same rate, how many cups of flour are needed to make 60 cookies?

- A) 4.5
- B) 7.5
- C) 8
- D) 20



11

A bank exchanges currency at the rate 1 euro = 1.25 dollars. How many euros does a traveller receive in exchange for 400 dollars?

- A) 320
- B) 400
- C) 425
- D) 500

12

A photograph is 4 inches wide and 6 inches long. The photograph is enlarged proportionally so that the new width is 10 inches. What is the new length, in inches?

- A) 2.4
- B) 12
- C) 15
- D) 18

13

A tabletop has an area of 2 square metres. Given that 1 metre = 100 centimetres, what is the area of the tabletop, in square centimetres?

- A) 20
- B) 200
- C) 2000
- D) 20 000

14

Aluminium has a density of 2.7 grams per cubic centimetre. What is the mass, in grams, of a solid aluminium part with a volume of 50 cubic centimetres?

- A) 27
- B) 52.7
- C) 135
- D) 270

15

The quantity y is directly proportional to the quantity x . When $x = 12$, the value of y is 45. What is the value of y when $x = 20$?

- A) 53
- B) 75
- C) 80
- D) 135

16

A bottling machine fills 250 bottles per hour, and each bottle holds 0.75 litres. Given that 1 cubic metre = 1000 litres, how many cubic metres of liquid does the machine bottle in 8 hours?

- A) 0.0015
- B) 0.015
- C) 0.15
- D) 1.5

17

A pastry recipe mixes flour, sugar and butter by mass in the ratio 5 : 2 : 3. One batch of the recipe uses 400 grams of sugar. What is the total mass, in grams, of the three ingredients in that batch?

18

A map is drawn to a scale of 1 : 25 000. Two landmarks are 8 centimetres apart on the map. What is the actual distance between the landmarks, in kilometres?

- A) 0.02
- B) 0.2
- C) 2
- D) 20

19

On a map, 1 centimetre represents 4 kilometres. A lake covers an area of 12 square centimetres on the map. What is the actual area of the lake, in square kilometres?

- A) 48
- B) 96
- C) 192
- D) 768

20

A car uses 8 litres of fuel for every 100 kilometres driven, and fuel costs \$1.60 per litre. What is the cost, in dollars, of the fuel used on a trip of 250 kilometres?

- A) 32
- B) 40
- C) 50
- D) 320



21

A solid metal cube has edges of length 5 centimetres and is made of a metal with a density of 8.4 grams per cubic centimetre. What is the mass of the cube, in kilograms?

22

A printer prints 15 pages per minute, and each page uses one sheet of paper. Paper costs \$6.00 per package of 500 sheets. What is the cost, in dollars, of the paper used if the printer runs continuously for 3 hours?

A) 16.20

B) 27

C) 32.40

D) 54





Module 2

1

A paint mixture is made by combining red paint and blue paint in the ratio 4 : 7. A batch of this mixture uses 28 litres of red paint. How many litres of blue paint does it use?

- A) 16
- B) 25
- C) 35
- D) 49

2

A printer produces 84 pages in 6 minutes. At this constant rate, how many pages does the printer produce per minute?

3

An experiment ran for 5400 seconds. Given that 1 hour = 3600 seconds, how many hours did the experiment run?

- A) 0.9
- B) 1.5
- C) 15
- D) 90

4

A prize of \$96 is divided among three students in the ratio 1 : 2 : 5. How many dollars does the student with the largest share receive?

- A) 12
- B) 24
- C) 60
- D) 80

5

Seven identical textbooks have a total mass of 2.8 kilograms. What is the total mass, in kilograms, of 12 of these textbooks?

- A) 4.8
- B) 5.6
- C) 7.2
- D) 8.4

6

A model car is built to a scale of 1 : 18. The model is 25 centimetres long. What is the length, in centimetres, of the actual car?

7

A rural district has a population density of 250 people per square kilometre and an area of 36 square kilometres. What is the population of the district?

- A) 286
- B) 900
- C) 9000
- D) 90 000

8

In a survey, the ratio of the number of people who prefer tea to the number who prefer coffee is 5 : 7. There are 96 more people who prefer coffee than prefer tea. How many people were surveyed?

- A) 288
- B) 576
- C) 672
- D) 1152

9

A drone flies at a constant speed of 15 metres per second. What is this speed, in kilometres per hour?

10

A storage tank has a volume of 2.5 cubic metres. Given that 1 metre = 100 centimetres, what is the volume of the tank, in cubic centimetres?

- A) 2500
- B) 25 000
- C) 250 000
- D) 2 500 000



11

A triangle has sides of length 6 centimetres, 8 centimetres and 10 centimetres. A second triangle is similar to the first and has a longest side of length 25 centimetres. What is the perimeter, in centimetres, of the second triangle?

- A) 30
- B) 48
- C) 60
- D) 75

12

In a certain country, petrol costs 1.40 euros per litre, and 1 euro is worth 1.10 dollars. What is the cost, in dollars, of 40 litres of petrol?

- A) 44
- B) 50.91
- C) 56
- D) 61.60

13

A map has a scale of 1 : 50 000. Two towns are 12 kilometres apart. How far apart, in centimetres, are the two towns on the map?

14

A 40-litre mixture of water and syrup contains water and syrup in the ratio 7 : 3. How many litres of syrup must be added so that the mixture contains equal amounts of water and syrup?

- A) 16
- B) 20
- C) 24
- D) 28

15

Cooking oil has a density of 0.8 grams per cubic centimetre. What is the volume, in cubic centimetres, of 1.2 kilograms of this oil?

- A) 150
- B) 960
- C) 1500
- D) 1600

16

A rectangular field measures 250 metres by 160 metres. Fertiliser is applied at a rate of 3 kilograms per hectare, where 1 hectare = 10 000 square metres. How many kilograms of fertiliser are needed to cover the field?

- A) 1.2
- B) 12
- C) 120
- D) 12 000

17

A car travels at a constant speed of 90 kilometres per hour. How many metres does the car travel in 12 seconds?

18

Two rectangles are similar, and the ratio of the length of a side of the smaller rectangle to the length of the corresponding side of the larger rectangle is 3 : 5. The area of the smaller rectangle is 27 square centimetres. What is the area, in square centimetres, of the larger rectangle?

- A) 45
- B) 75
- C) 125
- D) 225

19

A solid rectangular block measures 20 centimetres by 15 centimetres by 8 centimetres and has a mass of 6.72 kilograms. What is the density of the block, in grams per cubic centimetre?

20

A car travels 30 miles on each gallon of fuel. Using 1 mile = 1.6 kilometres and 1 gallon = 4 litres, how many kilometres does the car travel on each litre of fuel?

- A) 12
- B) 18.75
- C) 48
- D) 75



21

A floor plan is drawn to a scale of 1 : 200. On the plan, a rectangular room measures 4.5 centimetres by 3 centimetres. What is the actual floor area of the room, in square metres?

- A) 13.5
- B) 27
- C) 54
- D) 108

22

A water tank with a capacity of 4.5 cubic metres is filled by a hose that delivers 15 litres every 10 seconds. Given that 1 cubic metre = 1000 litres, how many minutes does it take to fill the tank?

- A) 30
- B) 50
- C) 75
- D) 300





Lesson 12: Percentages

Module 1

1

What is 30% of 250?

- A) 7.5
- B) 75
- C) 175
- D) 750

2

45 is what percent of 180? (Disregard the percent sign when entering your answer.)

3

A jacket regularly priced at \$80 is on sale for 25% off. What is the sale price of the jacket?

- A) \$20
- B) \$55
- C) \$60
- D) \$75

4

The number of members of a chess club increased from 200 to 250. What was the percent increase in the number of members?

- A) 20%
- B) 25%
- C) 50%
- D) 80%

5

A dinner bill is \$60. A 15% tip is added to the bill. What is the total amount paid?

- A) \$9
- B) \$63
- C) \$69
- D) \$75

6

In a shipment of 150 bicycles, 24 of the bicycles are electric. What percent of the bicycles in the shipment are electric?

- A) 16%
- B) 24%
- C) 76%
- D) 84%

7

On a 40-question test, Maria answered 34 questions correctly. What percent of the questions did Maria answer correctly?

- A) 6%
- B) 15%
- C) 34%
- D) 85%

8

Increasing a quantity by 8% gives the same result as multiplying that quantity by which of the following?

- A) 0.08
- B) 0.8
- C) 0.92
- D) 1.08

9

The price of a laptop was reduced from \$850 to \$680. By what percent was the price of the laptop reduced? (Disregard the percent sign when entering your answer.)

10

40% of what number is 26?

- A) 10.4
- B) 65
- C) 66
- D) 104



11

In a survey of 800 residents, 45% of the residents said they own a car. Of the residents who own a car, 25% said they also own a bicycle. How many of the residents surveyed own both a car and a bicycle?

- A) 90
- B) 200
- C) 360
- D) 560

12

Favorite subject	Number of students
Mathematics	96
Science	84
History	72
English	148
Total	400

The table shows the favorite subject reported by each of the 400 students at a school. What percent of these students reported that their favorite subject is mathematics or science?

- A) 21%
- B) 24%
- C) 45%
- D) 55%

13

A carpenter estimated that a board was 480 centimeters long, but the board was actually 500 centimeters long. The percent error of an estimate is the positive difference between the estimate and the actual value, expressed as a percent of the actual value. What is the percent error of the carpenter's estimate?

- A) 4%
- B) 4.2%
- C) 20%
- D) 25%

14

After a 20% decrease, the monthly rent for an apartment is \$960. What was the monthly rent, in dollars, before the decrease?

- A) \$768
- B) \$1000
- C) \$1152
- D) \$1200

15

The price of a bicycle is \$200. The price is first increased by 10%, and then the increased price is decreased by 10%. What is the final price of the bicycle?

- A) \$160
- B) \$180
- C) \$198
- D) \$200

16

A coat is priced at \$120. A store takes 30% off the price of the coat and then takes an additional 10% off the discounted price. What is the final price of the coat?

- A) \$72
- B) \$75.60
- C) \$78
- D) \$84

17

Including a 6% sales tax, the total cost of a television is \$265. What was the price of the television, in dollars, before tax?

18

The value of an investment increased by 20% during its first year and then decreased by 25% during its second year. At the end of the second year, the value of the investment was what percent of its original value?

- A) 90%
- B) 95%
- C) 100%
- D) 105%

19

A number is increased by 30%, and the result is then decreased by 30%. The final value is 91. What was the original number?



20

60% of a number x is equal to 45% of 80. What is the value of x ?

- A) 36
- B) 45
- C) 48
- D) 60

21

A store increases the price of a lamp by 25%. During a sale, the store then reduces the increased price by 25%. If the sale price of the lamp is \$60, what was the price of the lamp before the increase?

- A) \$48
- B) \$60
- C) \$64
- D) \$80

22

A shop raises the price of every bicycle by 25%. The shop then wants to advertise a discount that returns the price of a bicycle to its value before the increase. What percent discount must the shop advertise? (Disregard the percent sign when entering your answer.)





Module 2

1

What is 65% of 400?

- A) 26
- B) 235
- C) 260
- D) 465

2

18 is what percent of 45? (Disregard the percent sign when entering your answer.)

3

A store sold 240 phones last month, and 35% of those phones were refurbished. How many of the phones sold last month were refurbished?

- A) 35
- B) 84
- C) 156
- D) 205

4

A store's daily sales fell from \$250 on Monday to \$200 on Tuesday. What was the percent decrease in daily sales?

- A) 20%
- B) 25%
- C) 50%
- D) 80%

5

30% of what number is 72?

- A) 21.6
- B) 102
- C) 216
- D) 240

6

Decreasing a quantity by 35% gives the same result as multiplying that quantity by which of the following?

- A) 0.35
- B) 0.65
- C) 1.35
- D) 1.65

7

A student's score on a practice test rose from 60 points to 75 points. What was the percent increase in the student's score?

- A) 15%
- B) 20%
- C) 25%
- D) 80%

8

After a 25% discount, the price of a jacket is \$96. What was the price of the jacket, in dollars, before the discount?

9

The population of a town was 500 at the start of 2021. The population increased by 10% during 2021 and then increased by 20% during 2022. What was the population of the town at the end of 2022?

- A) 620
- B) 650
- C) 660
- D) 680

10

A recipe calls for 250 grams of flour, but a baker used 265 grams. The amount of flour the baker used was what percent greater than the amount the recipe calls for?

- A) 5.7%
- B) 6%
- C) 15%
- D) 106%

11

Ticket type	Number sold
Adult	180
Student	260
Senior	60
Total	500

The table shows the number of tickets of each type sold for a concert. What percent of the tickets sold were not student tickets? (Disregard the percent sign when entering your answer.)



12

96 is what percent of 60?

- A) 36%
- B) 60%
- C) 62.5%
- D) 160%

13

A quantity is decreased by 40%, and the result is then increased by 40%. The final value is what percent of the original value?

- A) 84%
- B) 96%
- C) 100%
- D) 116%

14

A pair of headphones is on sale for \$52 after a 20% discount. What was the price of the headphones before the discount?

- A) \$41.60
- B) \$62.40
- C) \$65
- D) \$72

15

40% of a number is 26 more than 15% of the same number. What is the number?

16

A town's population was 8000 at the start of 2022. During 2022 the population decreased by 15%, and during 2023 it increased by 20%. What was the town's population at the end of 2023?

- A) 8000
- B) 8160
- C) 8400
- D) 8560

17

The price of a share of a stock increased by 20% during January and then decreased by 10% during February. If the price of a share at the end of February was \$216, what was the price of a share at the start of January?

- A) \$194.40
- B) \$200
- C) \$233.28
- D) \$240

18

A store buys a jacket from a supplier and sets a listed price that is 40% greater than what the store paid. The store later sells the jacket at a 25% discount off the listed price. The amount the store received for the jacket is what percent of the amount the store paid for it?

- A) 15%
- B) 95%
- C) 100%
- D) 105%

19

The price of a share of a stock fell by 20%. By what percent must the reduced price rise in order to return the share to its original price? (Disregard the percent sign when entering your answer.)

20

Year	Revenue (thousands of dollars)
2021	250
2022	300
2023	270
2024	324

The table shows a company's annual revenue, in thousands of dollars, for four years. According to the table, the company's revenue in 2024 was what percent greater than its revenue in 2021?

- A) 22.8%
- B) 24%
- C) 29.6%
- D) 74%



21

A printer is sold for \$180 after two successive discounts off its original price: first a discount of 25%, and then a discount of 20% off the reduced price. What was the original price of the printer?

- A) \$240
- B) \$270
- C) \$288
- D) \$300

22

In 2023 a company's revenue was 25% greater than its revenue in 2022. In 2024 the company's revenue was 8% less than its revenue in 2023. If the company's revenue in 2024 was \$115,000, what was its revenue, in dollars, in 2022?





Lesson 13: One-Variable Data: Distributions and Measures of Center and Spread

Module 1

1

The high temperatures, in degrees Fahrenheit, recorded in a town on five days were 61, 70, 72, 78, and 84. What is the mean of these five temperatures?

- A) 71
- B) 72
- C) 73
- D) 74

2

The numbers of books read last month by seven students were 5, 2, 9, 4, 7, 3, and 6. What is the median number of books read?

- A) 4
- B) 5
- C) 6
- D) 9

3

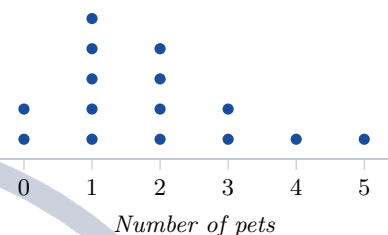
The masses, in grams, of six mineral samples were 12, 19, 7, 15, 22, and 10. What is the range of these masses, in grams?

4

A shoe store recorded the sizes of the eight pairs of shoes it sold in one hour: 8, 9, 7, 9, 10, 8, 9, and 11. What is the mode of these sizes?

- A) 3
- B) 8
- C) 9
- D) 10

5



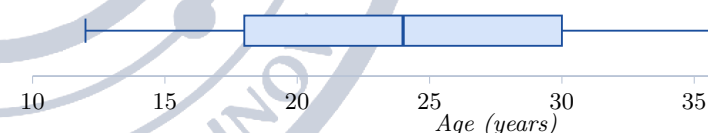
The dot plot shows the number of pets owned by each of 15 families. What is the median number of pets?

- A) 1
- B) 2
- C) 2.5
- D) 3

6

Over eight weeks, a runner ran 14, 18, 12, 20, 16, 22, 19, and 15 miles. What was the mean number of miles the runner ran per week?

7



The box plot summarizes the ages, in years, of the trees in a small orchard. What is the median age of the trees?

- A) 12
- B) 18
- C) 24
- D) 30



8

Number of siblings	Number of students
0	4
1	7
2	5
3	3
4	1

The frequency table shows the number of siblings reported by each of 20 students. What is the mean number of siblings per student?

- A) 1
- B) 1.5
- C) 2
- D) 4

9

A biologist recorded the heights, in centimetres, of ten seedlings: 31, 45, 28, 52, 39, 47, 33, 41, 36, and 50. What is the median height?

- A) 39
- B) 40
- C) 41
- D) 43

10

The mean of five test scores is 84. Four of the scores are 78, 91, 85, and 80. What is the fifth score?

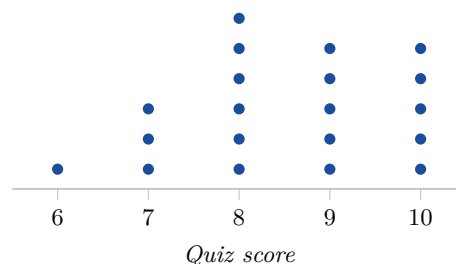
- A) 83.5
- B) 84
- C) 86
- D) 88

11

A data set consists of the nine values 12, 14, 15, 16, 18, 19, 21, 23, and 24. The value 90 is then added to the data set. Which of the following best describes the effect of adding 90?

- A) The mean and the median increase by the same amount.
- B) The median increases by more than the mean does.
- C) The mean increases and the median decreases.
- D) The mean increases by much more than the median does.

12



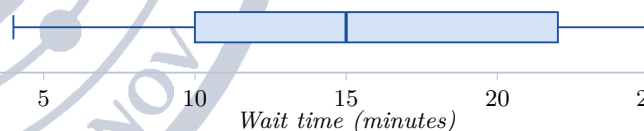
The dot plot shows the scores of 20 students on a 10-point quiz. What is the mean score?

13

Data set X consists of the values 20, 21, 22, 23, and 24. Data set Y consists of the values 12, 17, 22, 27, and 32. Each data set has a mean of 22. Which of the following is true about the standard deviations of the two data sets?

- A) Data set X has the larger standard deviation, because its values are closer to the mean.
- B) The standard deviations are equal, because the two data sets have the same mean.
- C) The standard deviations are equal, because the two data sets have the same number of values.
- D) Data set Y has the larger standard deviation, because its values are farther from the mean.

14



The box plot summarizes the number of minutes that each of 40 commuters waited for a train. What is the interquartile range of the wait times, in minutes?

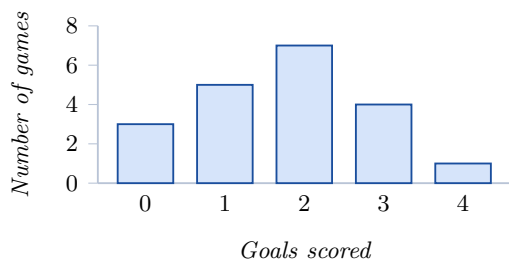
- A) 12
- B) 15
- C) 18
- D) 26

15

The mean of six numbers is 15. When one of the numbers is removed, the mean of the remaining five numbers is 16. What is the value of the number that was removed?



16



The histogram shows the number of goals scored in each of 20 soccer games. What is the mean number of goals per game?

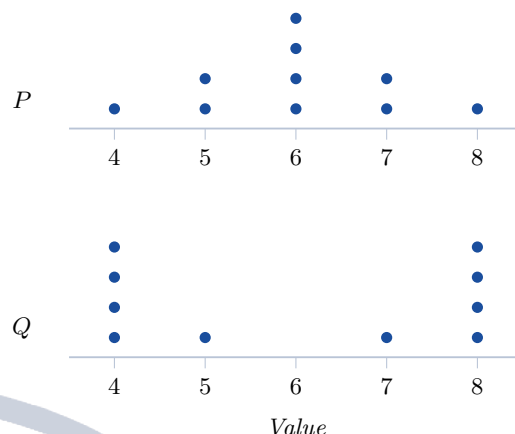
- A) 1.75
- B) 2
- C) 4
- D) 7

17

Value	Frequency
2	5
3	8
4	x
5	3

In the frequency table shown, the value 4 occurs x times. The mean of all the data is 3.4. What is the value of x ?

18



The dot plots show two data sets, each consisting of 10 values with a mean of 6. Which data set has the larger standard deviation, and why?

- A) Data set P , because more of its values are far from the mean.
- B) Data set P , because its values are more tightly clustered.
- C) Neither, because the two data sets have the same mean.
- D) Data set Q , because more of its values are far from the mean.

19

The yearly salaries, in thousands of dollars, of the eight employees at a small company are 32, 35, 36, 38, 40, 42, 45, and 180. The salary of 180 is removed from the data set. Which of the following describes the change in the mean and in the median?

- A) The mean decreases by about 18 and the median decreases by 1.
- B) The mean decreases by 1 and the median decreases by about 18.
- C) The mean and the median each decrease by about 18.
- D) The mean decreases and the median is unchanged.

20

In one class of 24 students, the mean score on a test was 78. In a second class of 16 students, the mean score on the same test was 88. What is the mean score of all 40 students?

**21**

Days absent	Number of students
0	6
1	9
2	8
3	7
4	5

The table shows the number of days absent for each of the 35 students in a class. What is the median number of days absent?

- A) 0
- B) 1
- C) 2
- D) 7

22

The mean of 12 numbers is 7. A thirteenth number is added to the data set, and the mean of all 13 numbers is 8. What is the thirteenth number?

- A) 8
- B) 15
- C) 20
- D) 21





Module 2

1

The prices, in dollars, of eight items sold at a shop are 14, 9, 21, 16, 12, 18, 25, and 11. What is the median price, in dollars?

- A) 14
- B) 15
- C) 15.75
- D) 16

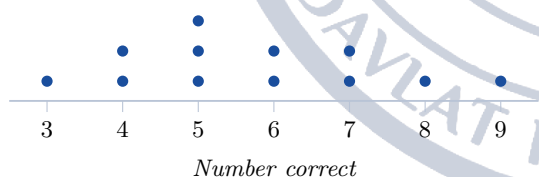
2

Number of children	Number of households
0	6
1	14
2	19
3	8
4	3

The table shows the number of children in each of 50 households in a neighbourhood. What is the mode of the number of children per household?

- A) 2
- B) 3
- C) 14
- D) 19

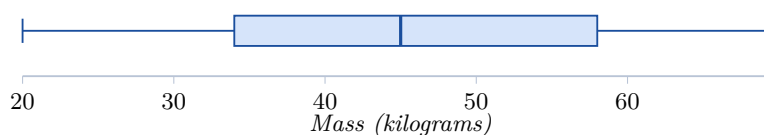
3



The dot plot shows the number of questions answered correctly by each of 12 students on a short test. What is the range of the number of correct answers?

- A) 6
- B) 7
- C) 9
- D) 12

4



The box plot summarizes the masses, in kilograms, of the dogs treated at a clinic in one week. What is the value of the third quartile, Q_3 ?

- A) 34
- B) 45
- C) 46
- D) 58

5

The rainfall, in inches, recorded in a town in each of six months was 2.4, 3.1, 1.8, 2.9, 3.6, and 4.2. What was the mean monthly rainfall, in inches?

6

Data set A consists of the five values 50, 50, 50, 50, and 50. Data set B consists of the five values 46, 48, 50, 52, and 54. Which data set has the greater standard deviation, and why?

- A) Data set A , because all of its values are equal to the mean.
- B) Data set A , because it has the greater mean.
- C) They have the same standard deviation, because they have the same mean.
- D) Data set B , because its values vary while those of data set A do not.

7

Number correct	Number of students
1	2
2	3
3	4
4	6

The table shows the number of correct answers, out of 4, for each of 15 students on a short quiz. What is the median number of correct answers?

- A) 3
- B) 4
- C) 6
- D) 15



8

Number of cars	Number of families
0	5
1	8
2	6
3	4
4	2

The table shows the number of cars owned by each of 25 families. What is the mean number of cars per family?

- A) 1
- B) 1.6
- C) 2
- D) 5

9

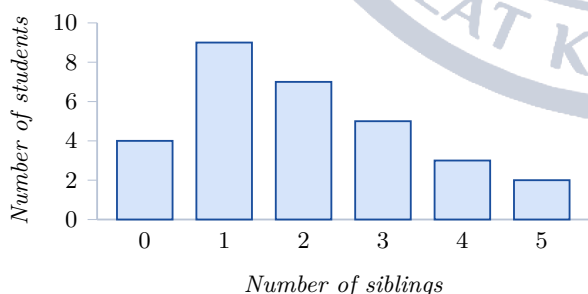
A data set consists of the six values 4, 9, 11, x , 17, and 20, listed in increasing order. The median of the data set is 13. What is the value of x ?

10

A data set of 10 numbers has a mean of 46. One value of 23 is replaced by 63, and no other value is changed. What is the mean of the new data set?

- A) 46
- B) 48
- C) 50
- D) 86

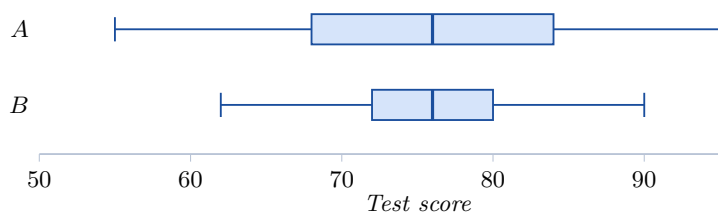
11



The histogram shows the number of siblings of each of 30 students. What is the median number of siblings?

- A) 1
- B) 2
- C) 2.5
- D) 5

12



The box plots summarize the test scores of the students in class A and class B. Which of the following statements is true?

- A) The two classes have the same median, and class A has the larger interquartile range.
- B) Class B has the larger median and the larger interquartile range.
- C) The two classes have the same median, and class B has the larger range.
- D) Class A has the larger median, and the two classes have the same range.

13

The 18 students in a class have a mean height of 62 inches. Two new students, of heights 70 inches and 74 inches, join the class. What is the mean height, in inches, of the 20 students?

- A) 62.4
- B) 63
- C) 64
- D) 67

14

The numbers of customers a cashier served during each of eight hours were 3, 7, 8, 12, 13, 15, 18, and 25. What is the interquartile range of these numbers?

15

A data set consists of the seven values 4, 6, 7, 9, 10, 12, and 40. The value 40 is removed from the data set. Which of the following describes the change in the mean and in the median?

- A) The median decreases by more than the mean does.
- B) The mean is unchanged and the median decreases.
- C) Both the mean and the median increase.
- D) The mean decreases by more than the median does.



16

The mean of seven numbers is 24. Two of the numbers, 15 and 28, are removed from the data set. What is the mean of the remaining five numbers?

17

Two data sets each consist of 20 values with a mean of 50. In data set R , 16 of the values are between 48 and 52. In data set S , 16 of the values are less than 40 or greater than 60. Which of the following statements is true?

- A) Data set R has the larger standard deviation, because more of its values are near 50.
- B) The standard deviations are equal, because the means are equal.
- C) The standard deviations cannot be compared without calculating each one.
- D) Data set S has the larger standard deviation, because more of its values are far from 50.

18

The mean of nine numbers is 15. Each of the nine numbers is increased by 4, and each result is then doubled. What is the mean of the nine new numbers?

- A) 30
- B) 34
- C) 38
- D) 46

19

Score	Number of students
70	6
75	9
80	11
85	9
90	5

The table shows the scores of the 40 students who took a quiz. What is the median score?

- A) 75
- B) 79.75
- C) 80
- D) 85

20

A data set consists of 15 values, each between 40 and 60. A new value of 200 is then added to the data set. Compared with the original data set, which of the following must be true of the new data set?

- A) Both the mean and the standard deviation are larger.
- B) The mean is larger and the standard deviation is smaller.
- C) The mean is unchanged and the standard deviation is larger.
- D) The median increases by more than the mean does.

21

A class of 30 students took a test. The 12 students in group A had a mean score of 85, and the class as a whole had a mean score of 79. What was the mean score of the other 18 students?

22

A data set of 20 values has a mean of 42. When two of the values are removed, the mean of the remaining 18 values is 40. What is the mean of the two values that were removed?

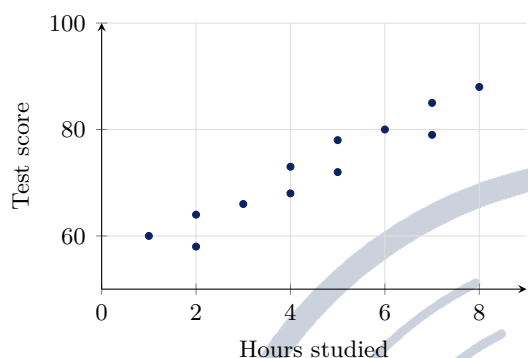
- A) 41
- B) 44
- C) 46
- D) 60



Lesson 14: Two-Variable Data: Models and Scatterplots

Module 1

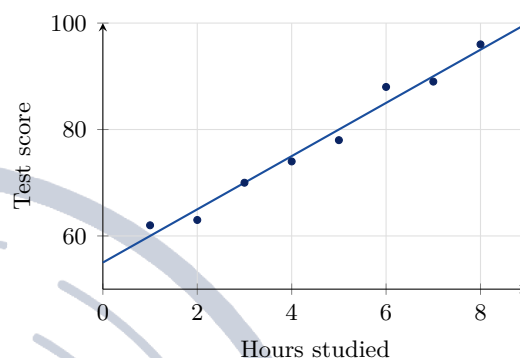
1



The scatterplot shows the number of hours each of 12 students spent studying for an exam and the score that student earned. Which of the following best describes the association between hours studied and score?

- A) The association is positive and approximately linear.
- B) The association is negative and approximately linear.
- C) The association is positive and exponential, with the score doubling each hour.
- D) There is no association between hours studied and score.

2



The scatterplot shows test scores against hours studied for 8 students, along with the line of best fit. According to the line of best fit, which of the following is closest to the predicted score of a student who studied for 5 hours?

- A) 70
- B) 75
- C) 80
- D) 85

3

A line of best fit passes through the points (2, 11) and (6, 27). What is the slope of this line of best fit?

4

A gym recorded its membership each month after opening. For the scatterplot of x , the number of months since the gym opened, and y , the number of members, the line of best fit is $\hat{y} = 24x + 150$. Which of the following is the best interpretation of the slope of this line?

- A) The predicted number of members increases by 24 each month.
- B) The predicted number of members increases by 150 each month.
- C) The gym had 24 members when it opened.
- D) It takes 24 months for the gym to gain one member.



5

For a scatterplot of x , the number of weeks since a seedling was potted, and y , the height of the seedling in centimeters, the line of best fit is $\hat{y} = 1.8x + 4$. Which of the following is the best interpretation of the y -intercept of this line?

- A) The predicted height increases by 4 centimeters each week.
- B) The predicted height of the seedling when it was potted was 1.8 centimeters.
- C) The seedling reaches a height of 4 centimeters after 1.8 weeks.
- D) The predicted height of the seedling when it was potted was 4 centimeters.

6

For a group of cafés, the line of best fit relating x , the number of tables, to y , the number of chairs, is $\hat{y} = 6x + 13$. How many chairs does this model predict for a café with 12 tables?

7

x	0	1	2	3
y	200	400	800	1600

The table shows four values of a data set. Which of the following best describes a model for these data?

- A) An exponential model, because the values increase by a constant factor.
- B) A linear model, because the values increase by a constant amount.
- C) A linear model, because the values increase by a constant factor.
- D) An exponential model, because the values increase by a constant amount.

8

For a scatterplot of the daily high temperature and the number of gallons of ice cream a shop sells, the line of best fit predicts 58 gallons on a certain day. The shop actually sold 64 gallons that day. What is the residual for that day?

- A) -6
- B) 6
- C) 58
- D) 122

9

A line of best fit predicts that a repair will cost \$212, but the repair actually costs \$197. Which of the following correctly describes the model at this data point?

- A) The model underpredicts the cost by \$15, so the residual is 15 dollars.
- B) The model overpredicts the cost by \$15, so the residual is 15 dollars.
- C) The model underpredicts the cost by \$15, so the residual is -15 dollars.
- D) The model overpredicts the cost by \$15, so the residual is -15 dollars.

10

For a scatterplot of x , the number of years since 2010, and y , the number of landline telephone subscriptions per 100 households in a county, the line of best fit is $\hat{y} = -3.4x + 61$. Which of the following is the best interpretation of the slope of this line?

- A) The predicted number of subscriptions per 100 households increases by 3.4 each year.
- B) The predicted number of subscriptions per 100 households decreases by 61 each year.
- C) In 2010 there were 3.4 subscriptions per 100 households.
- D) The predicted number of subscriptions per 100 households decreases by 3.4 each year.

11

The line of best fit for the height of a certain species of tree is $\hat{y} = 4.5x + 12$, where x is the number of years since the tree was planted and $\hat{y}$ is the predicted height in feet. What height, in feet, does the model predict for a tree 16 years after planting?

12

A taxi company plotted x , the length of a ride in miles, against y , the fare in dollars, and found the line of best fit $\hat{y} = 2.4x + 3.5$. Which of the following is the best interpretation of the number 3.5 in this model?

- A) The predicted fare for a ride of 0 miles is \$3.50.
- B) The predicted fare increases by \$3.50 for each additional mile.
- C) The predicted fare for a ride of 0 miles is \$2.40.
- D) The predicted length of a ride costing \$0 is 3.5 miles.



13

x	1	2	3	4	5	6
Actual y	28	33	42	47	60	63

The table shows six data points from a study. The line of best fit for these data is $\hat{y} = 7x + 20$. What is the residual for the data point with $x = 5$?

14

t	0	1	2	3
P	80	120	180	270

The table shows the value of P at four times t . Which of the following describes the type of model that fits the data and the constant that governs it?

- A) Linear, with a constant increase of 40.
- B) Linear, with a constant increase of 1.5.
- C) Exponential, with a constant multiplier of 1.5.
- D) Exponential, with a constant multiplier of 40.

15

A town's population was modeled by $\hat{P} = 1.6t + 34$, where t is the number of years since 2000 and $\hat{P}$ is the predicted population in thousands. Which of the following is the best interpretation of the slope of this model?

- A) The population is predicted to increase by 1.6 people each year.
- B) The population is predicted to increase by 34000 people each year.
- C) The population in 2000 was predicted to be 1600 people.
- D) The population is predicted to increase by 1600 people each year.

16

A reading study produced the line of best fit $\hat{y} = 13x + 45$, where x is the number of minutes spent reading and $\hat{y}$ is the predicted number of words recalled. According to this model, how many more words are predicted for a student who reads for 18 minutes than for a student who reads for 11 minutes?

- A) 7
- B) 13
- C) 91
- D) 324

17

A museum uses a line of best fit to predict daily attendance. On one day the model predicted 138 visitors and the residual for that day was -24 . How many visitors actually came that day?

- A) 24
- B) 114
- C) 138
- D) 162

18

A tutoring company plotted x , the number of practice tests a student completed, against y , that student's score improvement in points, and found the line of best fit $\hat{y} = 17x + 5$. Which of the following is the best interpretation of the slope of this line?

- A) Each additional practice test causes a score improvement of exactly 17 points.
- B) On average, each additional point of score improvement is associated with 17 more practice tests.
- C) A student who takes no practice tests is predicted to improve by 17 points.
- D) On average, each additional practice test is associated with a predicted increase of 17 points in score improvement.

19

The line of best fit relating x , the number of weeks of training, to $\hat{y}$, the predicted number of push-ups an athlete can complete, is $\hat{y} = 2.5x + 18$. For what value of x does this model predict 68 push-ups?

20

Day	1	2	3	4	5
Actual	12	24	29	42	48
Predicted	13	22	31	40	49

The table shows the actual number of bicycles a shop rented and the number predicted by the line of best fit on each of five days. On how many of these days does the model overpredict the number of rentals?

- A) 1
- B) 2
- C) 3
- D) 4

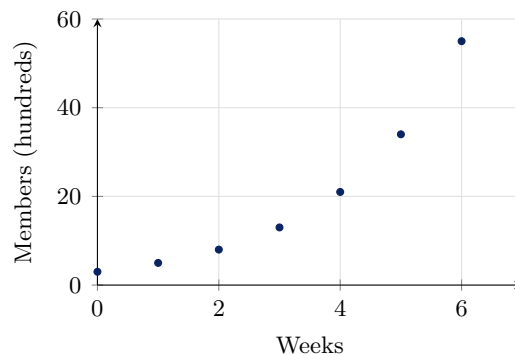


21

A biologist restocked a pond and then counted the fish each month. For the scatterplot of x , the number of months since restocking, and y , the number of fish, the line of best fit is $\hat{y} = 34x + 210$. Which of the following statements about this model is true?

- A) The predicted number of fish increases by 210 each month, and the pond contained 34 fish at restocking.
- B) The predicted number of fish increases by 34 each month, and the pond contained 210 fish at restocking.
- C) The predicted number of fish increases by 34 each month, and the pond contained 34 fish at restocking.
- D) The predicted number of fish increases by 210 each month, and the pond contained 210 fish at restocking.

22



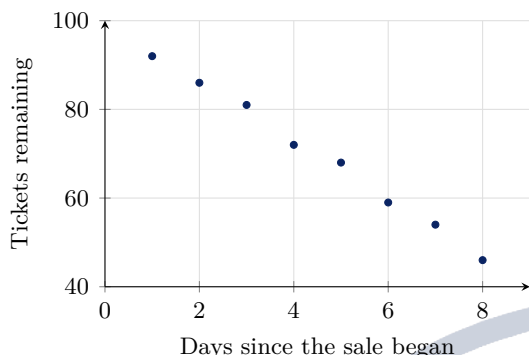
The scatterplot shows the membership of a new club, in hundreds, over seven weeks. Which of the following best describes a model for these data?

- A) A linear model, because the points fall close to a straight line.
- B) An exponential decay model, because the values decrease at a decreasing rate.
- C) An exponential growth model, because the values increase at an increasing rate.
- D) A linear model, because the values increase by a constant amount for each additional week.



Module 2

1



The scatterplot shows the number of tickets remaining for a concert on each of eight days after a sale began. Which of the following best describes the association shown?

- A) The association is negative and approximately linear.
- B) The association is positive and approximately linear.
- C) The association is negative and exponential, with the tickets halving each day.
- D) There is no association between the day and the number of tickets remaining.

2

A candle's height was measured as it burned. For the scatterplot of x , the number of minutes since the candle was lit, and y , the height of the candle in centimeters, the line of best fit is $\hat{y} = 15 - 0.2x$. Which of the following is the best interpretation of the number 15 in this model?

- A) The candle's predicted height decreases by 15 centimeters per minute.
- B) The candle's predicted height when it was lit was 0.2 centimeters.
- C) The candle's predicted height when it was lit was 15 centimeters.
- D) The candle is predicted to burn out after 15 minutes.

3

In a study of 30 precincts, the line of best fit relating x , the number of registered voters in hundreds, to $\hat{y}$, the predicted number of ballots cast in hundreds, is $\hat{y} = 0.8x + 12$. How many hundreds of ballots does the model predict for a precinct with 35 hundred registered voters?

4

For a group of phone customers, the line of best fit relating x , the number of text messages sent in a month, to y , that month's bill in dollars, is $\hat{y} = 0.05x + 30$. Which of the following is the best interpretation of the slope of this line?

- A) The predicted bill increases by \$30 for each additional text message.
- B) The predicted bill increases by \$0.05 for each additional text message.
- C) The predicted bill is \$0.05 when no text messages are sent.
- D) For each additional dollar in the bill, 0.05 more text messages are predicted.

5

A line of best fit passes through the points (4, 30) and (14, 5). What is the slope of this line of best fit?

6

Which of the following situations is best modeled by an exponential function?

- A) A car's value decreases by 15% of its current value each year.
- B) A pool is drained at a constant rate of 12 gallons per minute.
- C) A worker earns 18 dollars for each hour worked.
- D) A tank gains 5 liters of water every 2 minutes.

7

In a scatterplot with a line of best fit, one data point lies below the line. Which of the following must be true about that data point?

- A) The residual for that point is positive, and the model underpredicts the value.
- B) The residual for that point is negative, and the model underpredicts the value.
- C) The residual for that point is negative, and the model overpredicts the value.
- D) The residual for that point is positive, and the model overpredicts the value.

8

The line of best fit for a data set is $\hat{y} = 5.5x + 30$. One data point in the set has $x = 8$ and an actual y -value of 70. What is the residual for that data point?



9

The population of a town is modeled by $P(t) = 4800(0.85)^t$, where t is the number of years since 2015. Which of the following is the best interpretation of this model?

- A) The population decreases by 85% each year from an initial value of 4800.
- B) The population decreases by 15 people each year from an initial value of 4800.
- C) The population increases by 15% each year from an initial value of 4800.
- D) The population decreases by 15% each year from an initial value of 4800.

10

A researcher collected data for x between 1 and 12 months and found the line of best fit $\hat{y} = 45x + 120$. She then used the model to predict $\hat{y}$ when $x = 90$. Which of the following is the best reason to doubt that prediction?

- A) The slope of the line of best fit is positive, so the model cannot be used to make predictions.
- B) A line of best fit can only be used for values of x that appear exactly in the data set.
- C) The y -intercept of 120 is too large for the model to be accurate.
- D) The value $x = 90$ is far outside the range of the data used to build the model, so the pattern may not continue.

11

The line of best fit for a set of data is $\hat{y} = -0.6x + 45$. For what value of x does this model predict a y -value of 15?

12

For a scatterplot of x , the number of years since 1990, and y , the average price of a movie ticket in dollars, the line of best fit is $\hat{y} = 0.19x + 4.2$. Which of the following is the best interpretation of the slope of this line?

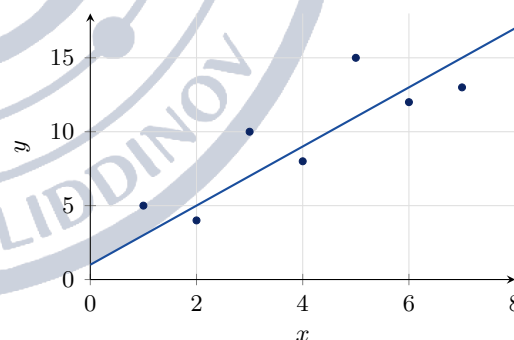
- A) The predicted price of a ticket increased by about 19 cents per year.
- B) The predicted price of a ticket increased by about 19 dollars per year.
- C) The predicted price of a ticket increased by about 4.20 dollars per year.
- D) The predicted price of a ticket in 1990 was about 19 cents.

13

A machine purchased for \$25000 loses value according to the model $V(t) = 25000(0.8)^t$, where t is the number of years since purchase. What value does this model predict after 3 years?

- A) \$10000
- B) \$12800
- C) \$15000
- D) \$20000

14



The scatterplot shows seven data points together with a line of best fit. For how many of these data points is the residual positive?



15

On one afternoon a weather station recorded the temperature every hour. For the scatterplot of x , the number of hours after noon, and y , the temperature in degrees Fahrenheit, the line of best fit is $\hat{y} = 2.3x + 68$. Which of the following is the best interpretation of the number 68 in this model?

- A) The temperature increased by 68 degrees Fahrenheit each hour.
- B) The predicted temperature at noon was 2.3 degrees Fahrenheit.
- C) It took 68 hours for the temperature to rise 2.3 degrees Fahrenheit.
- D) The predicted temperature at noon was 68 degrees Fahrenheit.

16

t	0	1	2	3
N	6	18	54	162

The table shows the number N of cells in a culture t hours after an experiment began. Which of the following equations best models these data?

- A) $N = 3(6)^t$
- B) $N = 6t + 3$
- C) $N = 6(3)^t$
- D) $N = 12t + 6$

17

Day	1	2	3	4
Actual visitors	46	48	71	68
Predicted visitors	40	55	62	80

The table shows the actual number of visitors to a park and the number predicted by a line of best fit on each of four days. Which of the following correctly identifies the day on which the model overpredicts by the greatest amount?

- A) On day 3 the model overpredicts by 9 visitors.
- B) On day 4 the model overpredicts by 12 visitors.
- C) On day 2 the model overpredicts by 7 visitors.
- D) On day 1 the model overpredicts by 6 visitors.

18

A study of high school students produced the line of best fit $\hat{y} = 8.4 - 0.006x$, where x is the number of minutes per day a student spends on social media and $\hat{y}$ is the predicted number of hours of sleep that student gets per night. Which of the following is the best interpretation of the number -0.006 in this model?

- A) For each additional minute per day on social media, the model predicts 8.4 fewer hours of sleep per night.
- B) For each additional hour of sleep per night, the model predicts 0.006 fewer minutes per day on social media.
- C) A student who spends no time on social media is predicted to sleep 0.006 hours per night.
- D) For each additional minute per day on social media, the model predicts 0.006 fewer hours of sleep per night.

19

For a set of data the line of best fit is $\hat{y} = 7.2x + 15$. According to this model, how much greater is the predicted value of y when $x = 25$ than the predicted value of y when $x = 10$?

20

A linear model is fitted to a data set, and the residuals, listed in order of increasing x , are 6, 2, -3 , -5 , -2 , 3, and 8. Which of the following conclusions is best supported by these residuals?

- A) The linear model is appropriate, because the residuals sum to a small number.
- B) The linear model is appropriate, because every residual is less than 10 in absolute value.
- C) A nonlinear model would fit better, because the residuals move from positive to negative and back in a U-shaped pattern.
- D) A nonlinear model would fit better, because every residual is positive.

21

For a data set the line of best fit is $\hat{y} = 3.4x + 12$. One data point in the set has $x = 15$, and its residual is -7 . What is the actual y -value of that data point?

- A) -7
- B) 56
- C) 63
- D) 70



22

For 40 employees at a firm, the line of best fit relating x , the number of years of experience, to y , the annual salary in thousands of dollars, is $\hat{y} = 2.8x + 41$. Which of the following gives the best interpretation of both the slope and the y -intercept of this model?

- A) Each additional year of experience is associated with a predicted increase of \$2800 in salary, and an employee with no experience is predicted to earn \$41000.
- B) Each additional year of experience is associated with a predicted increase of \$41000 in salary, and an employee with no experience is predicted to earn \$2800.
- C) Each additional year of experience is associated with a predicted increase of \$2.80 in salary, and an employee with no experience is predicted to earn \$41.
- D) Each additional \$2800 of salary is associated with one more year of experience, and an employee earning no salary is predicted to have 41 years of experience.



Lesson 15: Probability and Conditional Probability

Module 1

1

A bag contains 5 red marbles, 7 blue marbles, and 8 green marbles. If one marble is selected at random from the bag, what is the probability that it is red?

- A) $\frac{1}{4}$
- B) $\frac{1}{3}$
- C) $\frac{2}{5}$
- D) $\frac{3}{4}$

2

A basket contains only 12 lemon candies and 18 cherry candies. If one candy is selected at random, what is the probability that it is *not* a lemon candy?

- A) $\frac{2}{5}$
- B) $\frac{3}{5}$
- C) $\frac{2}{3}$
- D) $\frac{3}{4}$

3

An integer is selected at random from the integers 1 through 20, inclusive. What is the probability that the integer selected is a multiple of 5?

- A) $\frac{1}{20}$
- B) $\frac{1}{10}$
- C) $\frac{1}{5}$
- D) $\frac{1}{4}$

4

A class contains 25 students, 15 of whom are girls. If one student is selected at random from the class, what is the probability that the student is a boy?

5

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. If one of these 100 students is selected at random, what is the probability that the student is in the robotics club?

- A) $\frac{2}{5}$
- B) $\frac{1}{2}$
- C) $\frac{13}{25}$
- D) $\frac{13}{20}$

6

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. Given that a randomly selected member is a junior, what is the probability that the member is in the debate club?

- A) $\frac{9}{25}$
- B) $\frac{2}{5}$
- C) $\frac{3}{5}$
- D) $\frac{18}{25}$



7

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. If one of these 100 students is selected at random, what is the probability that the student is a junior in the robotics club?

- A) $\frac{6}{25}$
 B) $\frac{2}{5}$
 C) $\frac{12}{25}$
 D) $\frac{21}{25}$

8

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. A member of the robotics club is selected at random. What is the probability that this member is a senior?

9

A charity sells 250 raffle tickets and will draw one winning ticket at random. Maria buys 10 of the tickets. What is the probability that Maria's tickets do *not* include the winning ticket?

- A) $\frac{1}{25}$
 B) $\frac{1}{24}$
 C) $\frac{23}{25}$
 D) $\frac{24}{25}$

10

	Bus	Walk	Car	Total
Grade 9	45	30	15	90
Grade 10	35	20	15	70
Total	80	50	30	160

The table shows how 160 students at a school travel to school, by grade. If one of the 160 students is selected at random, what is the probability that the student walks to school?

- A) $\frac{3}{16}$
 B) $\frac{5}{16}$
 C) $\frac{1}{3}$
 D) $\frac{2}{5}$

11

	Bus	Walk	Car	Total
Grade 9	45	30	15	90
Grade 10	35	20	15	70
Total	80	50	30	160

The table shows how 160 students at a school travel to school, by grade. Given that a randomly selected student is in grade 10, what is the probability that the student takes the bus?

- A) $\frac{7}{32}$
 B) $\frac{2}{7}$
 C) $\frac{7}{16}$
 D) $\frac{1}{2}$

12

	Bus	Walk	Car	Total
Grade 9	45	30	15	90
Grade 10	35	20	15	70
Total	80	50	30	160

The table shows how 160 students at a school travel to school, by grade. A student is selected at random from those who take the bus. What is the probability that the student is in grade 9?



13

	Bus	Walk	Car	Total
Grade 9	45	30	15	90
Grade 10	35	20	15	70
Total	80	50	30	160

The table shows how 160 students at a school travel to school, by grade. If one of the 160 students is selected at random, what is the probability that the student does *not* travel by car?

- A) $\frac{3}{16}$
 B) $\frac{3}{14}$
 C) $\frac{5}{16}$
 D) $\frac{13}{16}$

14

A jar contains only red marbles and green marbles. The probability that a marble selected at random is red is $\frac{3}{8}$, and the jar contains 15 red marbles. How many green marbles are in the jar?

- A) 24
 B) 25
 C) 40
 D) 64

15

	Bus	Walk	Car	Total
Grade 9	45	30	15	90
Grade 10	35	20	15	70
Total	80	50	30	160

The table shows how 160 students at a school travel to school, by grade. If one of the 160 students is selected at random, what is the probability that the student is in grade 10 and walks to school?

- A) $\frac{1}{8}$
 B) $\frac{2}{7}$
 C) $\frac{5}{16}$
 D) $\frac{2}{5}$

16

A bag contains 8 red marbles and 12 blue marbles. How many additional red marbles must be added to the bag so that the probability of selecting a red marble at random becomes $\frac{1}{2}$?

17

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. A member of the debate club is selected at random. What is the probability that this member is a junior?

- A) $\frac{18}{25}$
 B) $\frac{3}{5}$
 C) $\frac{9}{25}$
 D) $\frac{7}{25}$

18

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. If one of these 100 students is selected at random, what is the probability that the student is in the debate club or is a senior?

- A) $\frac{7}{50}$
 B) $\frac{31}{50}$
 C) $\frac{19}{25}$
 D) $\frac{9}{10}$

19

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. How many additional students, all of whom join the robotics club, must enrol so that the probability that a randomly selected member of the two clubs is in the robotics club becomes $\frac{3}{5}$?

- A) 10
 B) 15
 C) 20
 D) 25



20

A store's display holds 30 shirts, 6 of which are blue. How many additional blue shirts must be added to the display so that the probability that a randomly selected shirt from the display is blue becomes $\frac{1}{4}$?

21

	Bus	Walk	Car	Total
Grade 9	45	30	15	90
Grade 10	35	20	15	70
Total	80	50	30	160

The table shows how 160 students at a school travel to school, by grade. Given that a randomly selected student is in grade 9, what is the probability that the student does *not* take the bus?

- A) $\frac{9}{32}$
- B) $\frac{1}{3}$
- C) $\frac{1}{2}$
- D) $\frac{9}{16}$

22

At a gathering of 60 people, 35 like tea, 30 like coffee, and 12 like both tea and coffee. If one of the 60 people is selected at random, what is the probability that the person likes neither tea nor coffee?

- A) $\frac{1}{12}$
- B) $\frac{7}{60}$
- C) $\frac{1}{5}$
- D) $\frac{53}{60}$



Module 2

1

A spinner is divided into 8 sections of equal area, 3 of which are shaded. If the spinner is spun once and lands on a random section, what is the probability that it lands on a shaded section?

- A) $\frac{1}{8}$
 B) $\frac{3}{8}$
 C) $\frac{3}{5}$
 D) $\frac{5}{8}$

2

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. If one of the 400 residents is selected at random, what is the probability that the resident supports the proposal?

- A) $\frac{1}{2}$
 B) $\frac{3}{5}$
 C) $\frac{2}{3}$
 D) $\frac{3}{4}$

3

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. Given that a randomly selected resident is under 40, what is the probability that the resident supports the proposal?

- A) $\frac{3}{10}$
 B) $\frac{1}{2}$
 C) $\frac{3}{5}$
 D) $\frac{3}{4}$

4

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. If one of the 400 residents is selected at random, what is the probability that the resident is 40 or older and opposes the proposal?

5

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. A resident who supports the proposal is selected at random. What is the probability that this resident is under 40?

- A) $\frac{4}{5}$
 B) $\frac{3}{4}$
 C) $\frac{3}{5}$
 D) $\frac{3}{10}$

6

Each of 12 cards is labelled with a different integer from 1 through 12. If one card is selected at random, what is the probability that the number on the card is prime?

- A) $\frac{1}{4}$
 B) $\frac{1}{3}$
 C) $\frac{5}{12}$
 D) $\frac{1}{2}$



7

	Passed	Defective	Total
Machine A	285	15	300
Machine B	165	35	200
Total	450	50	500

A factory inspected the 500 items produced in one shift by its two machines. The table shows the results. If one of the 500 items is selected at random, what is the probability that it is defective?

- A) $\frac{1}{10}$
 B) $\frac{7}{40}$
 C) $\frac{3}{10}$
 D) $\frac{9}{10}$

8

	Passed	Defective	Total
Machine A	285	15	300
Machine B	165	35	200
Total	450	50	500

A factory inspected the 500 items produced in one shift by its two machines, with the results shown. An item produced by machine B is selected at random. What is the probability that the item is defective?

9

	Passed	Defective	Total
Machine A	285	15	300
Machine B	165	35	200
Total	450	50	500

A factory inspected the 500 items produced in one shift by its two machines, with the results shown. A defective item is selected at random. What is the probability that it was produced by machine A?

- A) $\frac{3}{10}$
 B) $\frac{1}{10}$
 C) $\frac{1}{20}$
 D) $\frac{3}{100}$

10

	Passed	Defective	Total
Machine A	285	15	300
Machine B	165	35	200
Total	450	50	500

A factory inspected the 500 items produced in one shift by its two machines, with the results shown. Given that a randomly selected item passed inspection, what is the probability that it was produced by machine B?

- A) $\frac{33}{100}$
 B) $\frac{11}{30}$
 C) $\frac{19}{30}$
 D) $\frac{33}{40}$

11

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. A resident who opposes the proposal is selected at random. What is the probability that this resident is 40 or older?

- A) $\frac{2}{5}$
 B) $\frac{3}{5}$
 C) $\frac{2}{3}$
 D) $\frac{4}{5}$

12

A bag contains 60 marbles. The probability that a marble selected at random from the bag is green is $\frac{1}{5}$. How many of the marbles are *not* green?



13

	Passed	Defective	Total
Machine A	285	15	300
Machine B	165	35	200
Total	450	50	500

A factory inspected the 500 items produced in one shift by its two machines, with the results shown. An item produced by machine A is selected at random. What is the probability that the item passed inspection?

- A) $\frac{11}{30}$
 B) $\frac{57}{100}$
 C) $\frac{9}{10}$
 D) $\frac{19}{20}$

14

An integer is selected at random from the integers 1 through 30, inclusive. What is the probability that the integer selected is *not* a multiple of 3?

- A) $\frac{1}{3}$
 B) $\frac{1}{2}$
 C) $\frac{2}{3}$
 D) $\frac{3}{4}$

15

	Yes	No	Total
Male		30	70
Female	20	30	50
Total		60	120

The incomplete table shows the results of a survey of 120 people who were asked whether they support a change to the bus timetable. Given that a randomly selected respondent answered Yes, what is the probability that the respondent is female?

- A) $\frac{1}{6}$
 B) $\frac{1}{3}$
 C) $\frac{2}{5}$
 D) $\frac{2}{3}$

16

	Passed	Defective	Total
Machine A	285	15	300
Machine B	165	35	200
Total	450	50	500

A factory inspected the 500 items produced in one shift by its two machines, with the results shown. The next day the factory produces some additional items, all of which are defective, and adds them to this group. How many additional defective items would make the probability that a randomly selected item from the enlarged group is defective equal to $\frac{1}{6}$?

17

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. How many additional residents under 40, all of whom support the proposal, would have to be surveyed so that $\frac{5}{9}$ of all the residents surveyed support the proposal?

- A) 40
 B) 50
 C) 60
 D) 100

18

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. Of the residents who do *not* support the proposal, what fraction are under 40?

- A) $\frac{4}{5}$
 B) $\frac{1}{4}$
 C) $\frac{1}{5}$
 D) $\frac{1}{10}$



19

A bag contains 24 marbles, 6 of which are red. How many red marbles must be added to the bag so that the probability of selecting a red marble at random is $\frac{1}{2}$?

20

In a school of 120 students, 70 play a sport, 45 play a musical instrument, and 25 do both. If one of the 120 students is selected at random, what is the probability that the student does neither?

- A) $\frac{3}{4}$
- B) $\frac{3}{8}$
- C) $\frac{1}{4}$
- D) $\frac{5}{24}$

21

A distributor receives lamps from two plants: 60% come from plant X and the rest from plant Y. Of the lamps from plant X, 5% are defective, and of the lamps from plant Y, 10% are defective. If one lamp is selected at random from the distributor's stock, what is the probability that it is defective?

- A) $\frac{3}{100}$
- B) $\frac{7}{100}$
- C) $\frac{3}{40}$
- D) $\frac{3}{20}$

22

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. How many additional residents aged 40 or older, all of whom oppose the proposal, would have to be surveyed so that the probability that a randomly selected respondent aged 40 or older opposes the proposal is $\frac{3}{4}$?



Lesson 16: Inference, Margin of Error and Evaluating Statistical Claims

Module 1

1

A researcher selected a random sample of 200 of the 1,500 students at a high school and found that 60 of the students in the sample walk to school. Based on this sample, which of the following is the best estimate of the total number of students at the school who walk to school?

- A) 60
- B) 300
- C) 450
- D) 750

2

A poll of a random sample of voters estimated that 42% of all voters in a city support a new transit plan, with an associated margin of error of 3%. What is the least percent of all voters that is consistent with this estimate? (Ignore the percent sign when entering your answer.)

3

A city library has 2,000 registered members. A librarian surveyed 150 members selected at random from the list of all registered members. Which of the following is the population that this survey is designed to describe?

- A) The 150 members who were surveyed.
- B) All 2,000 registered members of the library.
- C) All residents of the city.
- D) All people who visited the library on the day of the survey.

4

A researcher will use a random sample to estimate the mean number of hours per week that adults in a town exercise. If the researcher increases the size of the random sample but keeps the confidence level the same, what is the most likely effect on the margin of error?

- A) It will decrease.
- B) It will increase.
- C) It will stay exactly the same.
- D) It will become negative.

5

In a random sample of 80 of the 1,200 households in a town, 28 households own a pet. Based on this sample, which of the following is the best estimate of the number of households in the town that own a pet?

- A) 336
- B) 420
- C) 480
- D) 560

6

Based on a random sample, a biologist estimates that the mean mass of a certain species of frog is between 12.4 grams and 13.6 grams. What is the margin of error associated with this estimate?

- A) 0.6
- B) 1.2
- C) 12.4
- D) 13.0



7

A store manager wants to estimate the mean number of minutes that all of the store's customers wait in line. Which of the following sampling methods is most likely to produce a biased estimate?

- A) Selecting a random sample of customers from a list of all purchases made during the week.
- B) Selecting a random sample of customers from each day of the week.
- C) Selecting only customers who shop during the single busiest hour of the week.
- D) Selecting a random sample of customers throughout the entire week.

8

A survey of a random sample of the students at a college found that the mean number of hours per week that the students in the sample study is 15.2 hours, with an associated margin of error of 1.3 hours. Which of the following is the most appropriate conclusion?

- A) Every student at the college studies between 13.9 and 16.5 hours per week.
- B) Exactly 95 percent of the students at the college study between 13.9 and 16.5 hours per week.
- C) The mean number of hours per week that all students at the college study is exactly 15.2 hours.
- D) It is plausible that the mean number of hours per week that all students at the college study is between 13.9 and 16.5 hours.

9

A random sample of 250 adults in a city was asked how many minutes they commute to work each day. The mean for the sample was 27.5 minutes, with an associated margin of error of 1.2 minutes. What is the greatest number of minutes that is consistent with this estimate?

10

A swim club at one gym has 400 members. All 400 names were listed, 50 members were selected at random, and each was asked about a proposed change to the pool schedule. Of those surveyed, 34 supported the change. Which of the following conclusions is best supported by these results?

- A) Most members of the swim club at this gym likely support the change.
- B) Most people who swim in the city likely support the change.
- C) The proposed schedule change causes members to swim more often.
- D) Exactly 34 of the 400 members of the swim club support the change.

11

A random sample of 320 of the 12,000 registered voters in a county was surveyed, and 208 of those surveyed said they plan to vote in an upcoming election. Based on this sample, which of the following is the best estimate of the number of registered voters in the county who plan to vote?

- A) 6 000
- B) 7 800
- C) 8 400
- D) 9 600

12

Two polls were conducted to estimate the percent of a city's residents who support a bond measure. Both polls used a random sample of residents and the same confidence level, but Poll A surveyed 400 residents and Poll B surveyed 1,600 residents. Which of the following statements about the margins of error is most likely true?

- A) Poll A has the smaller margin of error, because it surveyed fewer residents.
- B) The two polls must have the same margin of error, because they used the same confidence level.
- C) The margins of error cannot be compared without knowing the results of the two polls.
- D) Poll B has the smaller margin of error, because it surveyed more residents.



13

Based on a random sample of the trees in a forest, a plausible range for the proportion of all trees in the forest that are oak trees is 0.38 to 0.46. What is the proportion of oak trees estimated from the sample?

14

A researcher recruited 200 volunteers who experience frequent headaches and randomly assigned each volunteer to take either a new medication or a placebo. After eight weeks, the volunteers taking the new medication reported significantly fewer headaches than those taking the placebo. Which of the following conclusions is best supported?

- A) The new medication caused a reduction in headaches for the volunteers in the study.
- B) The new medication would reduce headaches for all adults who experience frequent headaches.
- C) Adults who take medication are more likely to experience frequent headaches.
- D) No conclusion about cause can be drawn, because the volunteers were not selected at random.

15

Based on a random sample of a population of fish, a 95% confidence interval for the mean weight of the fish in the population is 2.1 kilograms to 2.7 kilograms. Which of the following statements is best supported by this interval?

- A) Ninety-five percent of the fish in the population weigh between 2.1 kilograms and 2.7 kilograms.
- B) Ninety-five percent of the fish in the sample weigh between 2.1 kilograms and 2.7 kilograms.
- C) It is plausible that the mean weight of all the fish in the population is 2.5 kilograms.
- D) The mean weight of all the fish in the population must be 2.4 kilograms.

16

A nutritionist selected a random sample of 250 students from the 3,000 students at one university and recorded each student's breakfast habits and reported energy level. Students in the sample who ate breakfast reported higher energy levels than students who did not. Which of the following conclusions is best supported by this study?

- A) Eating breakfast causes higher energy levels in students at this university.
- B) Eating breakfast causes higher energy levels in university students throughout the country.
- C) Students at this university who want more energy must eat breakfast.
- D) Among the students at this university, eating breakfast is associated with higher reported energy levels.

17

A random sample of 500 of the 6,400 employees at a company was surveyed. Based on the sample, the proportion of all employees at the company who bike to work is estimated to be 0.36, with an associated margin of error of 0.04. What is the greatest number of employees at the company who bike to work that is consistent with this estimate?

18

From one random sample of batteries, a statistician computes both a 90% confidence interval and a 99% confidence interval for the mean lifetime of the batteries produced at a factory. Which of the following correctly compares the two intervals?

- A) The 99 percent interval is wider, because a higher confidence level requires a larger margin of error.
- B) The 90 percent interval is wider, because a lower confidence level requires a larger margin of error.
- C) The two intervals have the same width, because they are computed from the same sample.
- D) The 99 percent interval is narrower, because a higher confidence level is more precise.



19

A magazine printed a survey in one issue and asked readers to mail in their responses. Of the 3,000 readers who responded, 82% said they were dissatisfied with a new city policy. Which of the following is the best reason that this result may be a poor estimate of the percent of all city residents who are dissatisfied with the policy?

- A) The sample of 3,000 is too small for a city-wide estimate.
- B) A percent cannot be estimated unless a mean is also reported.
- C) The survey asked about only one policy instead of several.
- D) The readers who chose to respond were not randomly selected, so people with strong opinions are likely overrepresented.

20

Based on a random sample of the bags of chips produced at a factory, a 95% confidence interval for the mean mass, in grams, of all bags produced at the factory is 142.9 to 148.3. What is the margin of error for this estimate?

21

Based on a random sample of the students at a school, a 95% confidence interval for the mean number of text messages sent per day by the students at the school is 48 to 62. Which of the following best describes what the confidence level of 95% means?

- A) There is a 95 percent chance that the mean for all students at the school is between 48 and 62.
- B) If this sampling procedure were repeated many times, about 95 percent of the intervals produced would contain the mean for all students at the school.
- C) About 95 percent of the students at the school send between 48 and 62 text messages per day.
- D) About 95 percent of the students in the sample send between 48 and 62 text messages per day.

22

A survey of a random sample of adults in a country found that the adults who own a dog walk more minutes per week, on average, than the adults who do not own a dog. A reporter concluded that owning a dog causes adults to walk more. Which of the following changes to the study would be needed to support the reporter's conclusion?

- A) Surveying a larger random sample of adults in the country.
- B) Reporting a margin of error along with each estimate.
- C) Randomly assigning adults to own a dog or not to own a dog and then comparing the two groups.
- D) Surveying only adults who already walk regularly.



Module 2

1

A random sample of 150 of the 900 books in a school library was examined, and 45 of the books in the sample were published before 2010. Based on this sample, which of the following is the best estimate of the number of books in the library that were published before 2010?

- A) 135
- B) 225
- C) 270
- D) 300

2

Based on a random sample of its residents, a city estimates that the proportion of all residents who recycle is 0.58, with an associated margin of error of 0.03. What is the least proportion of all residents who recycle that is consistent with this estimate?

3

Two independent random samples were taken from the same population to estimate the mean daily rainfall, in millimeters, using the same confidence level. Study A used 100 days and Study B used 900 days. The margin of error reported by Study B was smaller than the one reported by Study A. Which of the following best explains why?

- A) A larger random sample gives more information about the population, which reduces the margin of error.
- B) A larger random sample always produces a larger mean, which reduces the margin of error.
- C) Study B must have used a higher confidence level than Study A.
- D) Study B measured rainfall on more days, so its measurements must be more accurate individually.

4

A random sample of 60 employees at one branch of a company was surveyed about the branch's parking policy. To which of the following groups can the results of the survey be generalized?

- A) All employees at that branch of the company.
- B) All employees at every branch of the company.
- C) All people who work in the city where the branch is located.
- D) Only the 60 employees who were surveyed.

5

In a random sample of 300 of the 5,500 students at a university, 24 students said that they own a car. Based on this sample, what is the best estimate of the number of students at the university who own a car?

6

Based on a random sample of the students at a school, the mean number of hours of sleep per night for all students at the school is estimated to be 7.4 hours, with an associated margin of error of 0.5 hours. Which of the following is a plausible value for the mean number of hours of sleep per night for all students at the school?

- A) 6.8
- B) 7.1
- C) 8.0
- D) 8.4

7

A researcher wants to estimate the mean number of hours per week that all residents of a town spend exercising. Which of the following samples is most likely to produce an estimate that is too high?

- A) A random sample of residents chosen from the town census list.
- B) A random sample of residents chosen from a list of all households in the town.
- C) A random sample of residents chosen from a list of all registered voters in the town.
- D) A sample of residents chosen as they leave a fitness center on a weekday evening.



8

Researchers followed a random sample of 1,200 adults for one year and recorded how much coffee each adult drank and how many hours each adult slept. Adults who drank more coffee reported fewer hours of sleep. The researchers did not assign any adult to drink a particular amount of coffee. Which of the following conclusions is best supported?

- A) Drinking coffee causes adults to sleep fewer hours.
- B) Sleeping fewer hours causes adults to drink more coffee.
- C) There is an association between coffee consumption and hours of sleep among adults, but the study cannot establish a cause.
- D) The study shows that coffee consumption and hours of sleep are unrelated.

9

Based on a random sample of the packages shipped by a company, a plausible range for the mean weight, in pounds, of all packages shipped by the company is 21.4 to 24.8. What is the mean weight, in pounds, estimated from the sample?

10

A researcher randomly selected 100 of the 2,500 members of a professional organization and asked each selected member whether they attended the organization's annual conference. Which of the following is an appropriate use of the results?

- A) Estimating the proportion of all professionals in this field who attended the conference.
- B) Concluding that attending the conference causes members to be more satisfied with their jobs.
- C) Determining the exact number of members of the organization who attended the conference.
- D) Estimating the proportion of all 2,500 members of the organization who attended the conference.

11

A pollster used a random sample of voters to compute a 90% confidence interval for the percent of all voters who support a measure. Using the same sample, the pollster then computes a 99% confidence interval instead. Compared with the first interval, the second interval will be described by which of the following?

- A) It is wider, and the estimate at its center is the same.
- B) It is narrower, and the estimate at its center is the same.
- C) It is wider, and the estimate at its center is larger.
- D) It is unchanged, because the sample did not change.

12

In a random sample of 240 of the 8,000 trees in a park, 84 of the trees are maple trees. Based on this sample, which of the following is the best estimate of the number of maple trees in the park?

- A) 1 960
- B) 2 400
- C) 2 600
- D) 2 800

13

Based on a random sample of a store's customers, a 95% confidence interval for the mean number of minutes per visit that all of the store's customers spend in the store is 28.6 to 31.2. Which of the following values of the mean for all customers is NOT plausible based on this interval?

- A) 28.9
- B) 30.0
- C) 31.1
- D) 32.4



14

A school emailed a survey to all 1,000 of its students, and 120 students responded. Of the students who responded, 90% said they were satisfied with the cafeteria. Which of the following is the best criticism of using this result to estimate the percent of all students at the school who are satisfied with the cafeteria?

- A) The sample of 120 students is a random sample, so the estimate applies only to those 120 students.
- B) The students who chose to respond may differ systematically from the students who did not respond.
- C) The survey should have been emailed to fewer students so that more would respond.
- D) A percent cannot be computed from only 120 responses.

15

Based on a random sample of the households in a town, the mean number of hours per day that a television is on in the households of the town is estimated to be 3.6 hours, with an associated margin of error of 0.45 hours. What is the greatest number of hours that is consistent with this estimate?

16

A city randomly selected 500 of its residents and surveyed them about how they commute and how much stress they experience. Residents who reported commuting by bicycle also reported lower stress levels. Which of the following is the strongest conclusion justified by this survey?

- A) Commuting by bicycle lowers stress for the residents of the city.
- B) Lower stress causes residents of the city to commute by bicycle.
- C) Because the sample was randomly selected, the association between bicycle commuting and lower stress can be generalized to the city's residents, but no cause-and-effect relationship is established.
- D) Because the sample was randomly selected, the survey establishes that bicycle commuting causes lower stress for the city's residents.

17

Study 1 surveyed a random sample of 800 adults selected from a national list and found an association between hours of screen time and reported sleep quality. Study 2 recruited 200 volunteers and randomly assigned each volunteer to one of two screen-time schedules, then compared the groups' reported sleep quality. Which of the following correctly describes what the two studies can support?

- A) Study 1 can establish cause and effect for adults nationally, and Study 2 cannot establish cause and effect at all.
- B) Study 2 can establish cause and effect for its volunteers, and Study 1 can generalize an association to adults nationally.
- C) Both studies can establish cause and effect for adults nationally.
- D) Neither study can support any conclusion about anyone other than the people who took part.

18

A random sample of 400 of the 4,500 students at a college was surveyed. Based on the sample, the proportion of all students at the college who live on campus is estimated to be 0.42, with an associated margin of error of 0.03. What is the greatest number of students at the college who live on campus that is consistent with this estimate?

19

A survey of a random sample of 400 people produced a margin of error of 4 percentage points at a 95% confidence level. The margin of error is inversely proportional to the square root of the sample size. Approximately how many people would need to be surveyed, at the same confidence level, to reduce the margin of error to 2 percentage points?

- A) 800
- B) 1 200
- C) 1 600
- D) 3 200



20

A polling organization reports, “Based on our random sample, we are 95% confident that between 46% and 52% of all voters support the measure.” Which of the following is the most accurate interpretation of this report?

- A) There is a 95 percent probability that the percent of all voters who support the measure is between 46 and 52.
- B) Between 46 percent and 52 percent of the voters in the sample support the measure.
- C) The interval was produced by a method that, used repeatedly, captures the percent of all voters who support the measure about 95 percent of the time.
- D) If 95 more random samples were taken, every one of them would give a percent between 46 and 52.

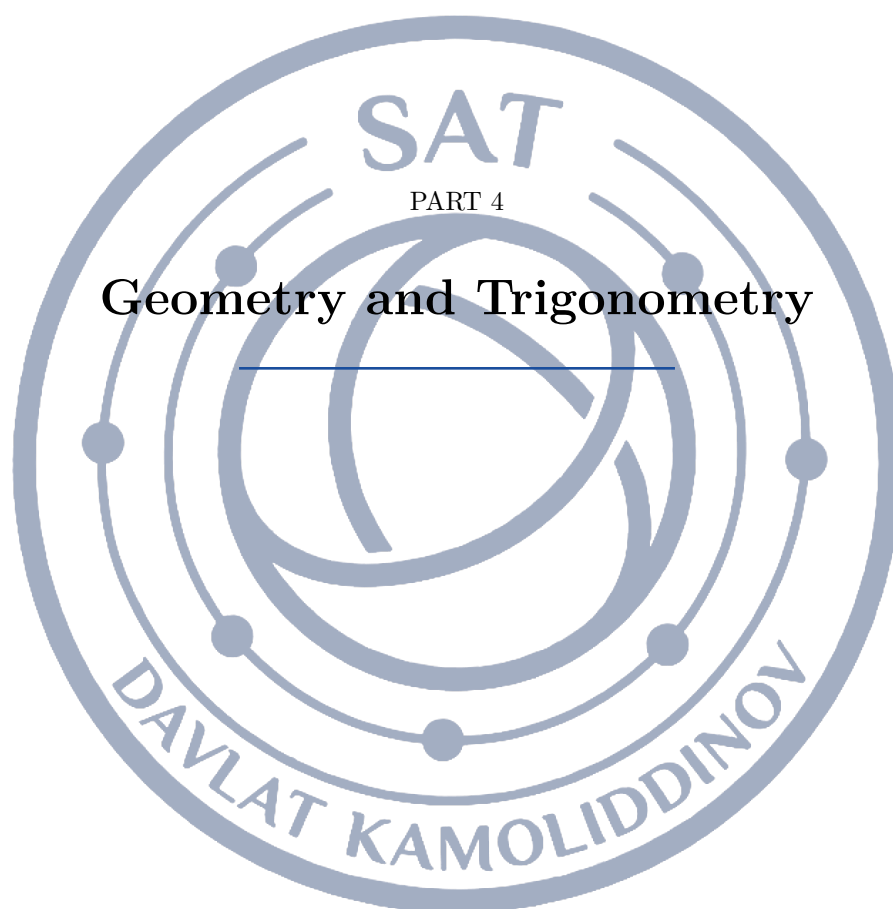
21

A study using a random sample of 900 adults reported a margin of error of 2.4 percentage points. The margin of error is inversely proportional to the square root of the sample size. How many adults must be included in a random sample, at the same confidence level, for the margin of error to be 0.8 percentage points?

22

Researchers recruited 300 volunteers who responded to an online advertisement and randomly assigned each volunteer to one of two exercise programs. After twelve weeks, the volunteers in Program A had significantly greater improvement in endurance than those in Program B. Which of the following conclusions is best supported?

- A) Program A causes greater improvement in endurance than Program B for all adults.
- B) Program A caused greater improvement in endurance for the volunteers in this study, but the result may not extend to all adults.
- C) No conclusion about cause can be drawn, because the volunteers were not randomly selected.
- D) Adults who want to improve their endurance are more likely to choose Program A.





Lesson 17: Area and Volume

Module 1

1

A rectangular tabletop is 12 inches long and 7 inches wide. What is the area, in square inches, of the tabletop?

- A) 19
- B) 38
- C) 42
- D) 84

2

A triangle has a base of length 10 centimeters and a height of 6 centimeters. What is the area, in square centimeters, of the triangle?

- A) 16
- B) 30
- C) 32
- D) 60

3

A circle has a radius of 5 inches. What is the area, in square inches, of the circle?

- A) 5π
- B) 10π
- C) 25π
- D) 100π

4

A rectangular box has a length of 9 inches, a width of 5 inches, and a height of 4 inches. What is the volume, in cubic inches, of the box?

- A) 7π
- B) 14π
- C) 28π
- D) 49π

6

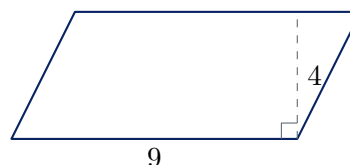


Figure not drawn to scale.

In the parallelogram shown, the base has length 9 and the height is 4. What is the area of the parallelogram?

- A) 13
- B) 18
- C) 26
- D) 36

7

A cube has edges of length 4 centimeters. What is the surface area, in square centimeters, of the cube?

8

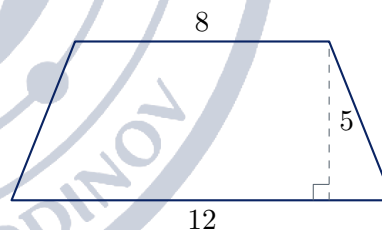


Figure not drawn to scale.

The trapezoid shown has parallel sides of length 8 and 12 and a height of 5. What is the area of the trapezoid?

- A) 40
- B) 50
- C) 60
- D) 100



9

A cylindrical can has a radius of 3 inches and a height of 10 inches. What is the volume, in cubic inches, of the can?

- A) 30π
- B) 60π
- C) 90π
- D) 360π

10

A right circular cone has a base radius of 6 centimeters and a height of 10 centimeters. What is the volume, in cubic centimeters, of the cone?

- A) 120π
- B) 180π
- C) 360π
- D) 480π

11

A triangle has an area of 84 square units and a base of length 24 units. What is the length of the height drawn to that base?

12

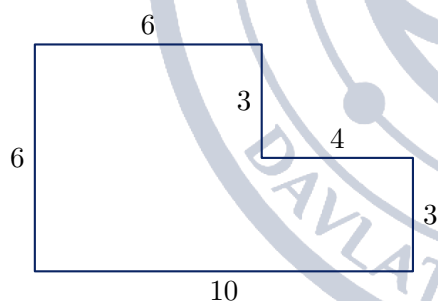


Figure not drawn to scale.

In the figure shown, every angle is a right angle. What is the area of the region?

- A) 36
- B) 48
- C) 60
- D) 72

13

Rectangle B is formed by multiplying both the length and the width of rectangle A by 3. The area of rectangle B is how many times the area of rectangle A ?

- A) 3
- B) 6
- C) 9
- D) 27

14

A sphere has a radius of 3 inches. What is the volume, in cubic inches, of the sphere?

- A) 27π
- B) 36π
- C) 108π
- D) 288π

15

A right circular cylinder has a volume of 100π cubic centimeters and a height of 4 centimeters. What is the radius, in centimeters, of the cylinder's base?

16

The radius of a right circular cone is multiplied by 2 and its height is multiplied by 3. The volume of the new cone is how many times the volume of the original cone?

- A) 6
- B) 12
- C) 24
- D) 36



17

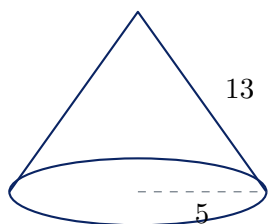


Figure not drawn to scale.

A right circular cone has a base radius of 5 and a slant height of 13, as shown. What is the volume of the cone?

- A) 100π
- B) $\frac{325\pi}{3}$
- C) 300π
- D) 325π

18

A rectangular prism with a square base has a volume of 500 cubic inches and a height of 5 inches. What is the surface area, in square inches, of the prism?

- A) 200
- B) 300
- C) 400
- D) 500

19

A sculptor builds a scale model of a statue. Every length on the statue is 4 times the corresponding length on the model. The model has a total surface area of 25 square inches. What is the total surface area, in square inches, of the statue?

20

A cylindrical tank with a base radius of 4 feet and a height of 9 feet is filled with water to $\frac{2}{3}$ of its capacity. What is the volume, in cubic feet, of the water in the tank?

- A) 48π
- B) 96π
- C) 144π
- D) 384π

21

A trapezoid has an area of 60 square units and a height of 6 units. One of its parallel sides is 3 times as long as the other. What is the length of the longer parallel side?

- A) 5
- B) 10
- C) 15
- D) 20

22

The volume of a sphere is 27 times the volume of a smaller sphere. The surface area of the larger sphere is how many times the surface area of the smaller sphere?

- A) 3
- B) 9
- C) 18
- D) 27



Module 2

1

A square has an area of 144 square meters. What is the perimeter, in meters, of the square?

- A) 12
- B) 24
- C) 48
- D) 576

2

A circle has an area of 49π square centimeters. What is the circumference, in centimeters, of the circle?

- A) 7π
- B) 14π
- C) 49π
- D) 98π

3

A rectangular box has a volume of 240 cubic inches. The base of the box measures 8 inches by 6 inches. What is the height, in inches, of the box?

- A) 5
- B) 10
- C) 30
- D) 40

4

A cube has a volume of 125 cubic centimeters. What is the surface area, in square centimeters, of the cube?

5

A circular pizza has a diameter of 16 inches. What is the area, in square inches, of the pizza?

- A) 16π
- B) 32π
- C) 64π
- D) 256π

6

Each edge of a cube is multiplied by 4 to produce a larger cube. The volume of the larger cube is how many times the volume of the original cube?

- A) 4
- B) 12
- C) 16
- D) 64

7

A tabletop is shaped like a trapezoid whose parallel sides measure 6 feet and 10 feet, with a distance of 4 feet between them. What is the area, in square feet, of the tabletop?

- A) 20
- B) 24
- C) 32
- D) 64

8

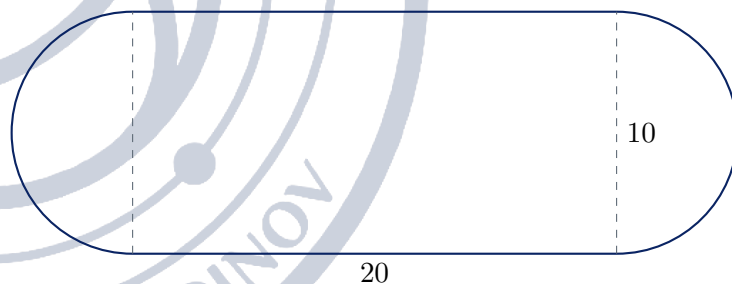


Figure not drawn to scale.

The region shown consists of a rectangle that is 20 meters long and 10 meters wide, with a semicircle attached to each of its two shorter sides. What is the area, in square meters, of the region?

- A) $200 + 25\pi$
- B) $200 + 50\pi$
- C) $400 + 25\pi$
- D) $200 + 100\pi$

9

A right circular cone has a volume of 96π cubic inches and a height of 8 inches. What is the radius, in inches, of the cone's base?



10

A closed cylindrical container has a base radius of 5 centimeters and a height of 6 centimeters. What is the total surface area, in square centimeters, of the container?

- A) 60π
- B) 85π
- C) 110π
- D) 260π

11

A rectangle has a perimeter of 34 inches, and one of its sides has length 12 inches. What is the area, in square inches, of the rectangle?

- A) 5
- B) 17
- C) 22
- D) 60

12

A rectangular photograph with an area of 48 square inches is enlarged so that each of its dimensions is multiplied by 2.5. What is the area, in square inches, of the enlarged photograph?

13

A solid consists of a right circular cylinder of radius 3 inches and height 10 inches with a hemisphere of radius 3 inches attached to its top. What is the volume, in cubic inches, of the solid?

- A) 90π
- B) 108π
- C) 126π
- D) 234π

14

A closed rectangular box measures 3 inches by 4 inches by 5 inches. What is the minimum number of square inches of paper needed to cover the outside of the box exactly?

- A) 24
- B) 47
- C) 60
- D) 94

15

A rectangular garden measures 15 feet by 8 feet. A walkway of uniform width 2 feet is built around the outside of the garden. What is the area, in square feet, of the walkway alone?

- A) 46
- B) 92
- C) 108
- D) 228

16

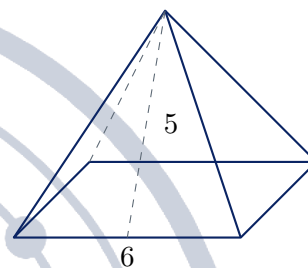


Figure not drawn to scale.

A pyramid has a square base with edges of length 6 centimeters. The slant height, measured from the apex to the midpoint of a base edge, is 5 centimeters. What is the volume, in cubic centimeters, of the pyramid?

- A) 48
- B) 60
- C) 144
- D) 180

17

Cylinder A and cylinder B are similar, and every length on cylinder B is 3 times the corresponding length on cylinder A . The volume of cylinder A is 20 cubic centimeters. What is the volume, in cubic centimeters, of cylinder B ?

18

A sphere has a volume of 288π cubic units. What is the surface area, in square units, of the sphere?

- A) 36π
- B) 144π
- C) 288π
- D) 576π



19

Cylinder P has radius r and height h . Cylinder Q has a radius half as long as cylinder P 's and a height twice as long as cylinder P 's. The volume of cylinder Q is what fraction of the volume of cylinder P ?

- A) $\frac{1}{2}$
- B) 1
- C) 2
- D) 4

21

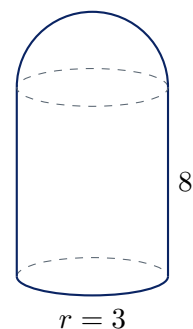


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20

A rectangular prism has a square base, and its height is 4 times the length of an edge of the base. The volume of the prism is 500 cubic centimeters. What is the length, in centimeters, of an edge of the base?

A solid is formed by attaching a hemisphere of radius 3 inches to the top of a right circular cylinder of radius 3 inches and height 8 inches, as shown. The bottom of the cylinder is a flat circular disk. What is the total surface area, in square inches, of the solid?

- A) 48π
- B) 57π
- C) 66π
- D) 75π

22

Two similar right circular cones have volumes of 27 cubic centimeters and 125 cubic centimeters. The base radius of the smaller cone is 6 centimeters. What is the base radius, in centimeters, of the larger cone?



Lesson 18: Lines, Angles and Triangles

Module 1

1

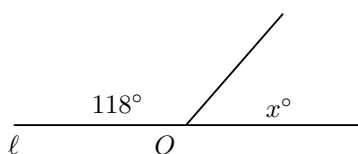


Figure not drawn to scale.

In the figure, ray OA meets line ℓ at point O , forming the two marked angles. What is the value of x ?

- A) 28
- B) 62
- C) 118
- D) 152

2

Angle A and angle B are complementary. The measure of angle A is 27° . What is the measure, in degrees, of angle B ?

3

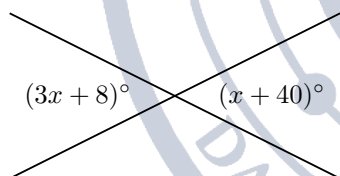


Figure not drawn to scale.

In the figure, two lines intersect and the two marked angles are vertical angles. What is the value of x ?

- A) 8
- B) 16
- C) 20
- D) 33

4

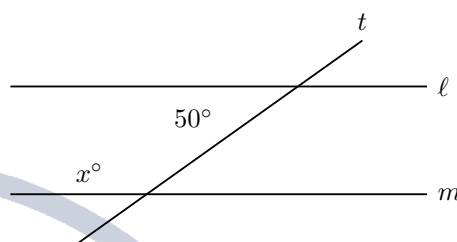


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In the figure, line ℓ is parallel to line m and both are crossed by transversal t . The two marked angles are interior angles on the same side of t . What is the value of x ?

- A) 40
- B) 50
- C) 65
- D) 130

5

Two of the interior angles of a triangle measure 47° and 68° . What is the measure, in degrees, of the third interior angle?

- A) 65
- B) 112
- C) 115
- D) 133

6

In isosceles triangle ABC , $AB = AC$ and the measure of $\angle A$ is 40° . What is the measure, in degrees, of $\angle B$?



7

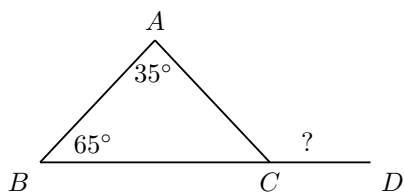


Figure not drawn to scale.

In the figure, side $\overline{BC}$ of triangle ABC is extended through C to point D . If $\angle A$ measures 35° and $\angle B$ measures 65° , what is the measure, in degrees, of $\angle ACD$?

- A) 80
- B) 100
- C) 115
- D) 145

8

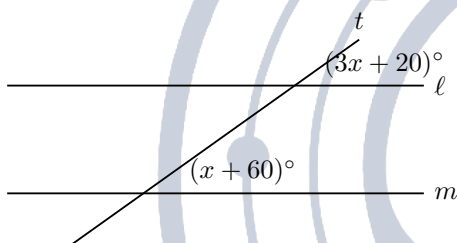


Figure not drawn to scale.

In the figure, $\ell \parallel m$ and the two marked angles are corresponding angles. What is the measure, in degrees, of each marked angle?

- A) 20
- B) 25
- C) 80
- D) 100

9

Two sides of a triangle have lengths 7 and 12. Which of the following could be the length of the third side?

- A) 4
- B) 5
- C) 17
- D) 19

10

The lengths of two sides of a triangle are 9 and 5. What is the greatest possible integer length of the third side?

11

Triangle ABC is similar to triangle DEF , where A corresponds to D , B corresponds to E , and C corresponds to F . If $AB = 8$, $BC = 12$, and $DE = 6$, what is the length of $\overline{EF}$?

- A) 9
- B) 10
- C) 16
- D) 18

12

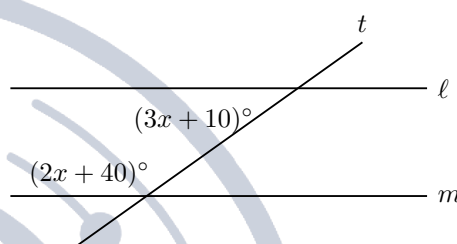


Figure not drawn to scale.

In the figure, $\ell \parallel m$ and the two marked angles are interior angles on the same side of the transversal. What is the value of x ?

- A) 26
- B) 30
- C) 88
- D) 92

13

In triangle ABC , side $\overline{AB}$ is extended beyond B to point D . The measure of $\angle CBD$ is $(5x)^\circ$, the measure of $\angle A$ is 40° , and the measure of $\angle C$ is $(3x + 10)^\circ$. What is the value of x ?

14

In triangle ABC , $AB = AC$. The measure of $\angle A$ is $(x + 20)^\circ$ and the measure of $\angle B$ is $(2x)^\circ$. What is the value of x ?

- A) 20
- B) 32
- C) 52
- D) 64



15

Triangle ABC is similar to triangle DEF , and the ratio of AB to DE is 3 to 5. The area of triangle ABC is 18. What is the area of triangle DEF ?

- A) 10.8
- B) 30
- C) 50
- D) 90

16

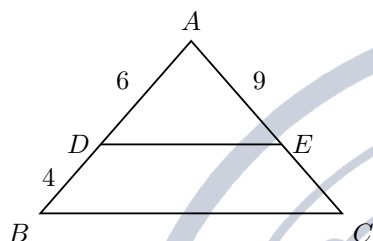


Figure not drawn to scale.

In the figure, D lies on $\overline{AB}$, E lies on $\overline{AC}$, and $\overline{DE} \parallel \overline{BC}$. If $AD = 6$, $DB = 4$, and $AE = 9$, what is the length of $\overline{EC}$?

- A) 4
- B) 6
- C) 13.5
- D) 15

17

In triangle PQR , point S lies on $\overline{PQ}$ and point T lies on $\overline{PR}$ so that $\overline{ST} \parallel \overline{QR}$. If $PS = 8$, $SQ = 12$, and $ST = 10$, what is the length of $\overline{QR}$?

18

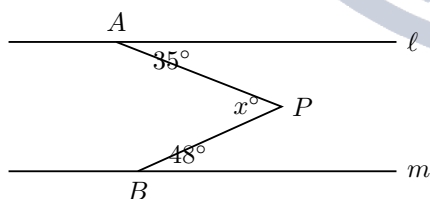


Figure not drawn to scale.

In the figure, $\ell \parallel m$, point A is on ℓ , point B is on m , and point P lies between the two lines. Segment $\overline{PA}$ makes a 35° angle with ℓ and segment $\overline{PB}$ makes a 48° angle with m , as shown. What is the value of x , the measure in degrees of $\angle APB$?

- A) 13
- B) 48
- C) 83
- D) 97

19

The side lengths of a triangle are 5, 11, and x , where x is an integer. What is the greatest possible perimeter of the triangle?

- A) 21
- B) 23
- C) 31
- D) 32

20

Triangle ABC is similar to triangle DEF , and the ratio of the perimeter of triangle ABC to the perimeter of triangle DEF is 2 to 3. If the area of triangle ABC is 24, what is the area of triangle DEF ?

21

At the same time of day, a person who is 6 feet tall casts a shadow 9 feet long, and a nearby flagpole casts a shadow 24 feet long. How tall, in feet, is the flagpole?

- A) 16
- B) 21
- C) 27
- D) 36

22

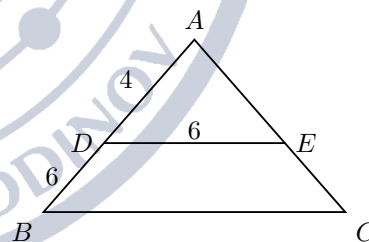


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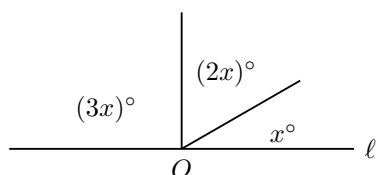
In the figure, D lies on $\overline{AB}$ and E lies on $\overline{AC}$, and $\angle ADE$ is congruent to $\angle ABC$. If $AD = 4$, $DB = 6$, and $DE = 6$, what is the length of $\overline{BC}$?

- A) 2.4
- B) 9
- C) 12
- D) 15



Module 2

1



In the figure, two rays drawn from point O on line ℓ divide the straight angle above the line into three angles with measures x° , $(2x)^\circ$, and $(3x)^\circ$. What is the value of x ?

- A) 20
- B) 30
- C) 45
- D) 60

2

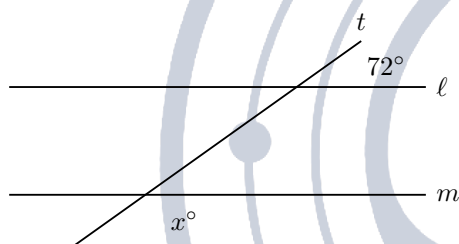


Figure not drawn to scale.

In the figure, line ℓ is parallel to line m and both are crossed by transversal t . What is the value of x ?

- A) 18
- B) 72
- C) 108
- D) 162

3

The measures of the three interior angles of a triangle are in the ratio 2 : 3 : 4. What is the measure, in degrees, of the largest angle?

4

In isosceles triangle ABC , $AB = AC$ and the measure of $\angle B$ is 37° . What is the measure, in degrees, of $\angle A$?

- A) 37
- B) 53
- C) 74
- D) 106

5

In triangle ABC , side $\overline{AB}$ is extended through B to point D , and $\angle CBD$ measures 130° . If $\angle A$ measures 55° , what is the measure, in degrees, of $\angle C$?

- A) 50
- B) 75
- C) 105
- D) 125

6

Triangle RST is similar to triangle XYZ , where R corresponds to X , S corresponds to Y , and T corresponds to Z . If $RS = 12$, $ST = 15$, and $XY = 8$, what is the length of $\overline{YZ}$?

- A) 8
- B) 10
- C) 11
- D) 22.5

7

Two similar triangles have corresponding side lengths in the ratio 2 to 5. The area of the smaller triangle is 12. What is the area of the larger triangle?

8

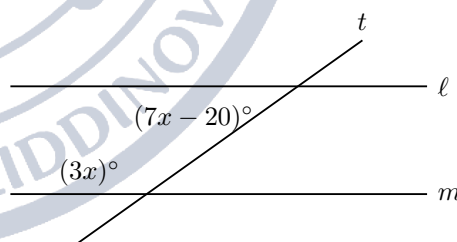


Figure not drawn to scale.

In the figure, $\ell \parallel m$ and the two marked angles are interior angles on the same side of the transversal. What is the measure, in degrees, of the larger of the two marked angles?

- A) 20
- B) 40
- C) 60
- D) 120

9

The measures of the three interior angles of a triangle are $(x + 10)^\circ$, $(2x - 20)^\circ$, and $(3x + 10)^\circ$. What is the measure, in degrees, of the largest angle?



10

Two sides of a triangle have lengths 8 and 15. If the length of the third side is an integer, how many different values are possible for that length?

- A) 13
- B) 14
- C) 15
- D) 16

11

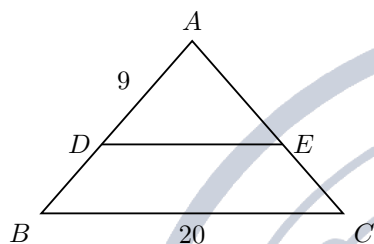


Figure not drawn to scale.

In the figure, D lies on $\overline{AB}$, E lies on $\overline{AC}$, and $\overline{DE} \parallel \overline{BC}$. If $AD = 9$, $AB = 15$, and $BC = 20$, what is the length of $\overline{DE}$?

- A) 12
- B) 14
- C) 20
- D) 30

12

In triangle ABC , $AB = AC$. Side $\overline{BC}$ is extended through C to point D , and $\angle ACD$ measures 118° . What is the measure, in degrees, of $\angle A$?

- A) 56
- B) 62
- C) 118
- D) 124

13

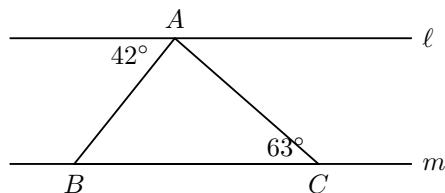


Figure not drawn to scale.

In the figure, $\ell \parallel m$, vertex A of triangle ABC lies on ℓ , and B and C lie on m . Side $\overline{AB}$ makes a 42° angle with line ℓ as shown, and $\angle ACB$ measures 63° . What is the measure, in degrees, of $\angle BAC$?

- A) 75
- B) 105
- C) 117
- D) 138

14

Triangle ABC is similar to triangle PQR , where A corresponds to P and B corresponds to Q . The area of triangle ABC is 27 and the area of triangle PQR is 48. If $AB = 9$, what is the length of $\overline{PQ}$?

- A) 6.75
- B) 9
- C) 12
- D) 16

15

Triangle ABC is congruent to triangle XYZ , where A corresponds to X , B corresponds to Y , and C corresponds to Z . If $AB = 3x - 4$, $XY = 20$, $BC = 2y + 1$, and $YZ = 15$, what is the value of $x + y$?

- A) 7
- B) 8
- C) 15
- D) 16



16

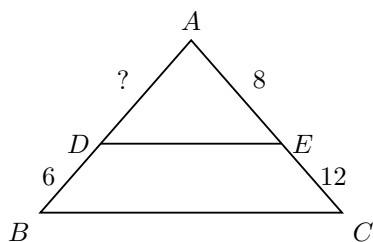


Figure not drawn to scale.

In the figure, D lies on $\overline{AB}$, E lies on $\overline{AC}$, and $\overline{DE} \parallel \overline{BC}$. If $DB = 6$, $AE = 8$, and $EC = 12$, what is the length of $\overline{AD}$?

17

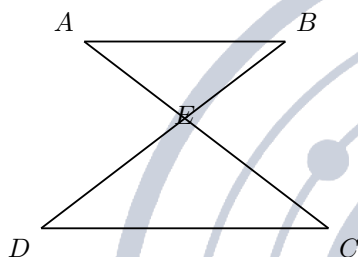


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In the figure, $\overline{AC}$ and $\overline{BD}$ intersect at point E , and $\overline{AB} \parallel \overline{CD}$. If $AE = 4$, $EC = 10$, and $BE = 6$, what is the length of $\overline{ED}$?

- A) 2.4
- B) 6
- C) 10
- D) 15

18

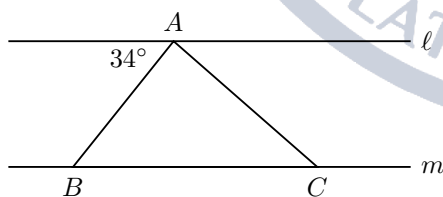


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In the figure, $\ell \parallel m$, point A is on ℓ , and points B and C are on m , where $AB = AC$. Side $\overline{AB}$ makes a 34° angle with line ℓ as shown. What is the measure, in degrees, of $\angle BAC$?

- A) 56
- B) 68
- C) 112
- D) 146

19

Two similar triangular plots of land have corresponding side lengths in the ratio 1 to 1.5. The area of the smaller plot is 40 square meters. What is the area, in square meters, of the larger plot?

- A) 60
- B) 90
- C) 120
- D) 160

20

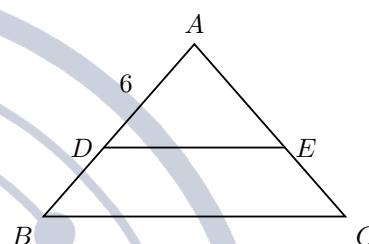


Figure not drawn to scale.

In triangle ABC , point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE} \parallel \overline{BC}$. The area of triangle ADE is 16 and the area of quadrilateral $DBCE$ is 48. If $AD = 6$, what is the length of $\overline{AB}$?

21

An isosceles triangle has two sides of lengths 5 and 12. What is the perimeter of the triangle?

- A) 17
- B) 22
- C) 24
- D) 29

22

In triangle ABC , point D lies on $\overline{BC}$ so that $\angle BAD$ is congruent to $\angle BCA$. If $AB = 8$ and $BD = 4$, what is the length of $\overline{BC}$?

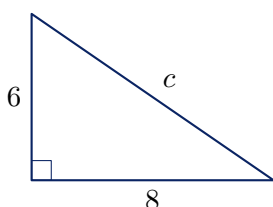
- A) 4
- B) 8
- C) 12
- D) 16



Lesson 19: Right Triangles and Trigonometry

Module 1

1



4

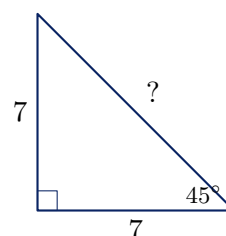


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In the right triangle shown, the two legs have lengths 6 and 8. What is the length of the hypotenuse?

Figure not drawn to scale.

In a right triangle, both acute angles measure 45° and each leg has length 7. What is the length of the hypotenuse?

- A) $\sqrt{28}$
- B) 10
- C) 14
- D) 48

- A) $\frac{7}{2}\sqrt{2}$
- B) $7\sqrt{2}$
- C) $7\sqrt{3}$
- D) 14

2

A right triangle has a hypotenuse of length 13 and one leg of length 5. What is the length of the other leg?

5

In a right triangle, the hypotenuse has length 10 and one of the acute angles measures 30° . What is the length of the side opposite the 30° angle?

- A) 8
- B) 12
- C) $\sqrt{194}$
- D) 18

- A) 5
- B) $5\sqrt{2}$
- C) $5\sqrt{3}$
- D) $10\sqrt{3}$

3

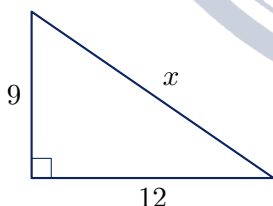


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The legs of a right triangle have lengths 9 and 12. What is the length of the hypotenuse of the triangle?



6

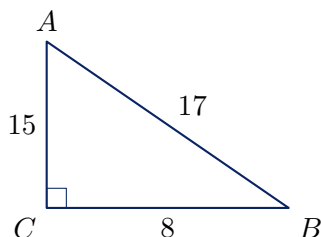


Figure not drawn to scale.

In the right triangle shown, angle A is an acute angle. What is the value of $\sin A$?

- A) $\frac{8}{17}$
- B) $\frac{8}{15}$
- C) $\frac{15}{17}$
- D) $\frac{17}{8}$

7

If $\sin(x^\circ) = \frac{3}{5}$, what is the value of $\cos((90 - x)^\circ)$?

- A) $\frac{3}{5}$
- B) $\frac{3}{4}$
- C) $\frac{4}{5}$
- D) $\frac{5}{3}$

8

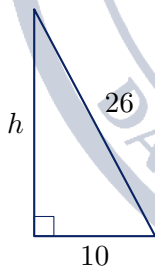


Figure not drawn to scale.

A 26-foot ladder leans against a vertical wall, with the base of the ladder 10 feet from the wall on level ground. How many feet up the wall does the top of the ladder reach?

9

In right triangle ABC , angle C is a right angle, $\tan A = \frac{3}{4}$, and $AC = 20$. What is the length of $\overline{BC}$?

- A) 12
- B) 15
- C) 25
- D) $\frac{80}{3}$

10

In a right triangle, one acute angle measures 60° and the side opposite the 30° angle has length 6. What is the length of the side opposite the 60° angle?

- A) $3\sqrt{3}$
- B) $6\sqrt{2}$
- C) $6\sqrt{3}$
- D) 12

11

Right triangles ABC and DEF have right angles at C and F , and angle A is congruent to angle D . If $AC = 6$, $BC = 8$, and $DF = 9$, what is the length of $\overline{EF}$?

- A) 10
- B) 11
- C) 12
- D) 15

12

In right triangle PQR , angle R is a right angle and $\cos P = \frac{7}{25}$. What is the value of $\sin Q$?

- A) $\frac{7}{25}$
- B) $\frac{7}{24}$
- C) $\frac{24}{25}$
- D) $\frac{25}{7}$

13

A vertical pole stands on level ground. The angle of elevation of the sun is 45° , and the pole's shadow is 18 feet long. What is the height, in feet, of the pole?

- A) $9\sqrt{2}$
- B) $9\sqrt{3}$
- C) 18
- D) $18\sqrt{2}$

14

The legs of a right triangle have lengths 8 and 15. What is the perimeter of the triangle?

- A) 17
- B) 23
- C) $\sqrt{161} + 23$
- D) 40



15

In right triangle ABC , angle C is a right angle and $\sin A = \frac{3}{5}$. What is the value of $\cos A$?

16

In right triangle ABC , angle B is a right angle, $AB = 9$, and $BC = 12$. Point D lies on $\overline{AC}$ so that $\overline{BD}$ is perpendicular to $\overline{AC}$. What is the length of $\overline{BD}$?

- A) $\frac{20}{3}$
- B) $\frac{36}{5}$
- C) $\frac{15}{2}$
- D) $\frac{48}{5}$

17

The acute angles of a right triangle measure a° and b° . If $\sin(a^\circ) = \frac{5}{13}$, what is the value of $\cos(b^\circ) \cdot \tan(a^\circ)$?

- A) $\frac{25}{169}$
- B) $\frac{25}{156}$
- C) $\frac{60}{169}$
- D) $\frac{144}{169}$

18

In a right triangle, one acute angle measures $(3x+10)^\circ$ and $\sin((3x+10)^\circ) = \cos((x+20)^\circ)$. What is the value of x ?

19

An equilateral triangle has side length 12. What is the length of an altitude of the triangle?

- A) 6
- B) $6\sqrt{2}$
- C) $6\sqrt{3}$
- D) $12\sqrt{3}$

20

In a right triangle, θ is an acute angle and $\tan \theta = \frac{20}{21}$. What is the value of $\sin \theta$?

- A) $\frac{20}{29}$
- B) $\frac{21}{29}$
- C) $\frac{20}{21}$
- D) $\frac{29}{20}$

21

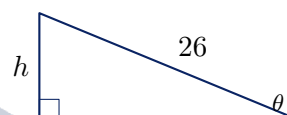


Figure not drawn to scale.

A straight ramp 26 feet long runs from level ground to a loading dock. The ramp makes an angle θ with the ground, where $\sin \theta = \frac{5}{13}$. How many feet above the ground is the top of the ramp?

22

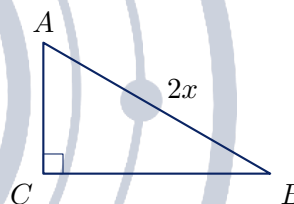


Figure not drawn to scale.

In right triangle ABC , angle C is a right angle and the length of $\overline{AB}$ is twice the length of $\overline{BC}$. What is the value of $\tan A$?

- A) $\frac{1}{2}$
- B) $\frac{1}{3}\sqrt{3}$
- C) $\frac{1}{2}\sqrt{3}$
- D) $\sqrt{3}$



Module 2

1

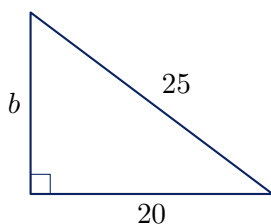


Figure not drawn to scale.

In the right triangle shown, one leg has length 20 and the hypotenuse has length 25. What is the length of the other leg?

- A) 5
- B) 15
- C) $\sqrt{1025}$
- D) 45

2

The hypotenuse of a 45° - 45° - 90° triangle has length $8\sqrt{2}$. What is the length of each leg?

- A) 4
- B) $\frac{8}{3}\sqrt{3}$
- C) $4\sqrt{2}$
- D) 8

3

In right triangle ABC , angle B is a right angle, $AB = 24$, and $BC = 7$. What is the value of $\tan A$?

4

Angles A and B are the two acute angles of a right triangle. If $\tan A = \frac{5}{12}$, what is the value of $\tan B$?

- A) $\frac{5}{13}$
- B) $\frac{5}{12}$
- C) $\frac{13}{12}$
- D) $\frac{12}{5}$

5

In a 30° - 60° - 90° triangle, the side opposite the 60° angle has length $9\sqrt{3}$. What is the length of the hypotenuse?

- A) $\frac{9}{2}\sqrt{3}$
- B) 9
- C) $9\sqrt{3}$
- D) 18

6

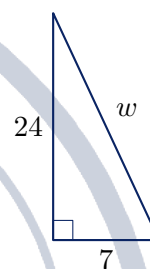


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A support wire runs from the top of a vertical 24-foot pole to a point on level ground 7 feet from the base of the pole. What is the length, in feet, of the wire?

- A) 17
- B) $\sqrt{527}$
- C) 25
- D) 31

7

In right triangle DEF , angle F is a right angle and $\sin D = \frac{8}{17}$. What is the value of $\cos E$?



8

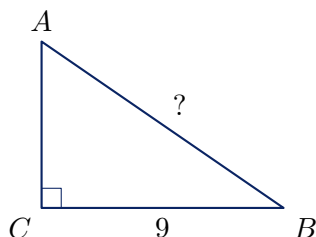


Figure not drawn to scale.

In right triangle ABC , angle C is a right angle, $BC = 9$, and $\sin A = \frac{3}{5}$. What is the length of $\overline{AB}$?

- A) $\frac{27}{5}$
- B) $\frac{45}{4}$
- C) 12
- D) 15

9

In a right triangle, one acute angle measures $(4x - 6)^\circ$ and $\cos((4x - 6)^\circ) = \sin((x + 16)^\circ)$. What is the value of x ?

10

Right triangles ABC and XYZ are similar, with right angles at B and Y and with angle A corresponding to angle X . If $AB = 8$, $BC = 6$, and $XY = 20$, what is the length of $\overline{XZ}$?

- A) 10
- B) 15
- C) $\sqrt{436}$
- D) 25

11

A vertical flagpole 40 feet tall casts a shadow $40\sqrt{3}$ feet long on level ground. What is the measure, in degrees, of the angle of elevation of the sun?

- A) 30
- B) 45
- C) 60
- D) 75

12

The legs of a right triangle have lengths 10 and 24. What is the perimeter of the triangle?

13

In right triangle ABC , angle B is a right angle and $\cos A = \frac{5}{13}$. What is the value of $\tan A$?

- A) $\frac{5}{12}$
- B) $\frac{12}{13}$
- C) $\frac{12}{5}$
- D) $\frac{13}{5}$

14

A square has a diagonal of length 10. What is the area of the square?

- A) 25
- B) 50
- C) $50\sqrt{2}$
- D) 100

15

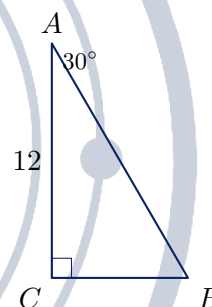


Figure not drawn to scale.

In right triangle ABC , angle C is a right angle, angle A measures 30° , and $AC = 12$. What is the length of $\overline{BC}$?

- A) 6
- B) $4\sqrt{3}$
- C) $6\sqrt{3}$
- D) $12\sqrt{3}$

16

In right triangle ABC , angle C is a right angle and $\sin A = \frac{2}{3}$. What is the value of $\cos A$?

- A) $\frac{1}{3}$
- B) $\frac{2}{3}$
- C) $\frac{1}{3}\sqrt{5}$
- D) $\frac{1}{3}\sqrt{13}$



17

In right triangle ABC , angle C is a right angle and $\tan A = \frac{3}{4}$. The perimeter of triangle ABC is 36. What is the length of $\overline{AB}$?

18

The acute angles of a right triangle measure a° and b° . If $\cos(a^\circ) = \frac{4}{5}$, what is the value of $\frac{\sin(b^\circ)}{\cos(b^\circ)}$?

- A) $\frac{3}{4}$
- B) $\frac{4}{5}$
- C) $\frac{4}{3}$
- D) $\frac{5}{3}$

19

In right triangle ABC the right angle is at C . Point D lies on $\overline{AB}$ so that $\overline{CD}$ is perpendicular to $\overline{AB}$. If $AD = 4$ and $DB = 9$, what is the length of $\overline{CD}$?

- A) 6
- B) $\frac{13}{2}$
- C) $\sqrt{97}$
- D) 13

20

In right triangle ABC , angle C is a right angle, $AB = 20$, and $\sin B = \cos B$. What is the area of triangle ABC ?

21

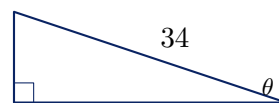


Figure not drawn to scale.

A straight ramp 34 feet long rises from level ground at an angle θ , where $\sin \theta = \frac{8}{17}$. What is the horizontal distance, in feet, covered by the ramp?

- A) 16
- B) 18
- C) 30
- D) 32

22

In right triangle ABC , angle C is a right angle and the measure of angle A is twice the measure of angle B . What is the value of $\frac{AC}{AB}$?

- A) $\frac{1}{2}$
- B) $\frac{1}{2}\sqrt{2}$
- C) $\frac{1}{2}\sqrt{3}$
- D) $\frac{2}{3}\sqrt{3}$



Lesson 20: Circles

Module 1

1

A circle has a radius of 7 inches. What is the circumference of this circle, in inches?

- A) 7π
- B) 14π
- C) 28π
- D) 49π

2

A circle has a diameter of 10 centimeters. What is the area of this circle, in square centimeters?

- A) 5π
- B) 10π
- C) 25π
- D) 100π

3

In the xy -plane, the equation of a circle is $(x - 3)^2 + (y + 5)^2 = 49$. What is the radius of this circle?

4

An angle has a measure of 60° . What is the measure of this angle, in radians?

- A) $\frac{\pi}{6}$
- B) $\frac{\pi}{4}$
- C) $\frac{\pi}{3}$
- D) $\frac{\pi}{2}$

5

In the xy -plane, a circle has the equation $(x + 4)^2 + (y - 1)^2 = 36$. What are the coordinates of the center of this circle?

- A) $(-4, 1)$
- B) $(-4, -1)$
- C) $(4, -1)$
- D) $(4, 1)$

6

The area of a circle is 36π square meters. What is the radius of this circle, in meters?

- A) 6
- B) 18
- C) 36
- D) 72

7

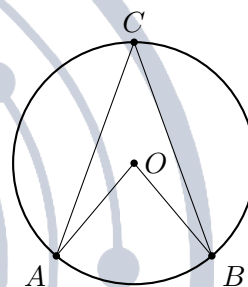


Figure not drawn to scale.

In the circle shown, O is the center and the central angle AOB measures 80° . Point C lies on the major arc AB . What is the measure of the inscribed angle ACB ?

- A) 20°
- B) 40°
- C) 80°
- D) 160°

8

In a circle with radius 9, a central angle measures 40° . What is the length of the arc intercepted by this central angle?

- A) 2π
- B) 4π
- C) 9π
- D) 18π



9

In a circle with radius 6, a sector has a central angle of 60° . What is the area of this sector?

- A) 2π
- B) 6π
- C) 12π
- D) 36π

10

An angle has a measure of $\frac{5\pi}{6}$ radians. What is the measure of this angle, in degrees?

11

In the xy -plane, a circle has center $(0, 3)$ and radius 5. Which of the following is an equation of this circle?

- A) $x^2 + (y + 3)^2 = 25$
- B) $x^2 + (y - 3)^2 = 5$
- C) $(x - 3)^2 + y^2 = 25$
- D) $x^2 + (y - 3)^2 = 25$

12

In the xy -plane, the graph of $(x + 2)^2 + (y - 6)^2 = 100$ is a circle with center (h, k) and radius r . What is the value of $h + k + r$?

13

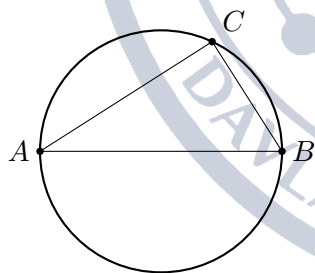


Figure not drawn to scale.

In the figure, $\overline{AB}$ is a diameter of the circle and point C lies on the circle. If the measure of angle CAB is 35° , what is the measure of angle ABC ?

- A) 35°
- B) 45°
- C) 55°
- D) 90°

14

A line is tangent to a circle with center O at point P . The radius of the circle is 5, and Q is a point on the tangent line such that $OQ = 13$. What is the length of $\overline{PQ}$?

- A) 5
- B) 8
- C) 12
- D) 18

15

A circle has a radius of 10. A central angle of $\frac{3\pi}{4}$ radians intercepts an arc of this circle. What is the length of that arc?

- A) $\frac{15\pi}{8}$
- B) $\frac{15\pi}{4}$
- C) $\frac{15\pi}{2}$
- D) 30π

16

In the xy -plane, the equation $x^2 + y^2 - 6x + 8y - 11 = 0$ represents a circle. What is the radius of this circle?

- A) $\sqrt{11}$
- B) $\sqrt{14}$
- C) 6
- D) 36

17

In the xy -plane, the graph of $x^2 + y^2 + 10x - 4y + 13 = 0$ is a circle. What is the radius of this circle?

18

In the xy -plane, the equation $x^2 + y^2 - 12x + 2y + 12 = 0$ represents a circle. What are the coordinates of the center of this circle?

- A) $(-6, 1)$
- B) $(-6, -1)$
- C) $(6, 1)$
- D) $(6, -1)$



19

A sector of a circle has a central angle of 120° and an area of 27π . What is the radius of the circle?

- A) 3
- B) 9
- C) 27
- D) 81

20

In a circle with radius 12, an inscribed angle of 30° intercepts arc AB . What is the length of arc AB ?

- A) π
- B) 2π
- C) 3π
- D) 4π

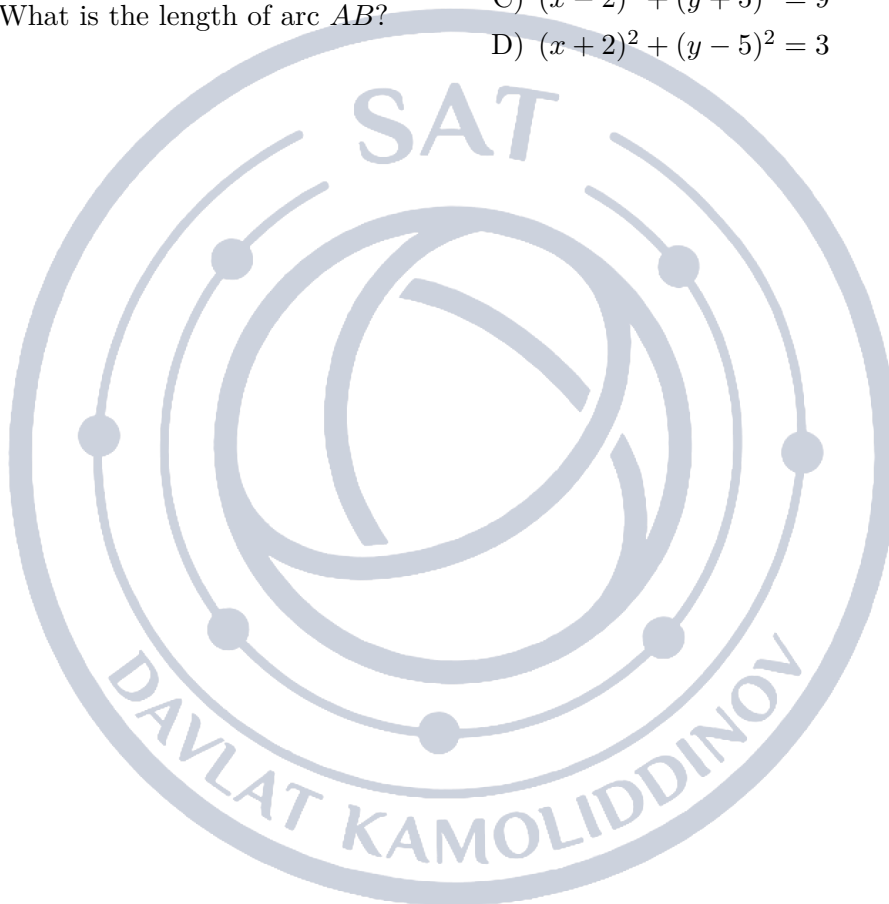
21

In the xy -plane, a circle has center $(4, -3)$ and passes through the point $(9, 9)$. What is the radius of this circle?

22

Which of the following equations is equivalent to $x^2 + y^2 + 4x - 10y + 20 = 0$?

- A) $(x + 2)^2 + (y - 5)^2 = 9$
- B) $(x + 2)^2 + (y - 5)^2 = 29$
- C) $(x - 2)^2 + (y + 5)^2 = 9$
- D) $(x + 2)^2 + (y - 5)^2 = 3$





Module 2

1

The circumference of a circle is 36π centimeters. What is the diameter of this circle, in centimeters?

- A) 6
- B) 18
- C) 36
- D) 72

2

The area of a circle is 49π square inches. What is the circumference of this circle, in inches?

- A) 7π
- B) 14π
- C) 28π
- D) 98π

3

An angle has a measure of $\frac{3\pi}{4}$ radians. What is the measure of this angle, in degrees?

4

In the xy -plane, the equation of a circle is $(x - 7)^2 + (y + 2)^2 = 45$. What is the radius of this circle?

- A) $3\sqrt{5}$
- B) $5\sqrt{3}$
- C) $9\sqrt{5}$
- D) 45

5

A sector of a circle with radius 8 has a central angle of 45° . What is the area of this sector?

- A) 2π
- B) 4π
- C) 8π
- D) 16π

6

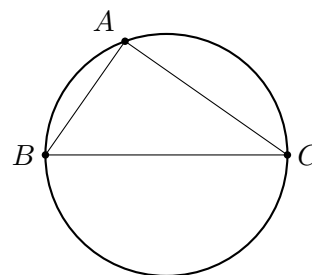


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In the figure, $\overline{BC}$ is a diameter of the circle and point A lies on the circle. If the measure of angle ABC is 28° , what is the measure of angle BCA ?

- A) 28°
- B) 62°
- C) 90°
- D) 152°

7

An angle has a measure of 210° . What is the measure of this angle, in radians?

- A) $\frac{7\pi}{12}$
- B) $\frac{5\pi}{6}$
- C) $\frac{6\pi}{7}$
- D) $\frac{7\pi}{6}$

8

In the xy -plane, a circle has center $(2, -2)$ and passes through the point $(6, 1)$. What is the radius of this circle?

9

A sector of a circle with radius 12 has an area of 24π . What is the measure, in degrees, of the central angle of this sector?

- A) 30
- B) 45
- C) 60
- D) 120

10

In a circle with radius 15, an arc has length 10π . What is the measure, in degrees, of the central angle that intercepts this arc?



11

In the xy -plane, the points $(1, 2)$ and $(7, 10)$ are the endpoints of a diameter of a circle. Which of the following is an equation of this circle?

- A) $(x - 4)^2 + (y - 6)^2 = 25$
- B) $(x - 4)^2 + (y - 6)^2 = 100$
- C) $(x + 4)^2 + (y + 6)^2 = 25$
- D) $(x - 1)^2 + (y - 2)^2 = 100$

12

A central angle of $\frac{2\pi}{3}$ radians intercepts an arc of length 12π . What is the radius of the circle?

- A) 6
- B) 9
- C) 12
- D) 18

13

In the xy -plane, the graph of $x^2 + y^2 - 8x + 6y - 11 = 0$ is a circle. What is the radius of this circle?

14

In a circle, a central angle and an inscribed angle intercept the same arc. The inscribed angle measures $(3x)^\circ$ and the central angle measures $(x + 40)^\circ$. What is the value of x ?

- A) 8
- B) 20
- C) 40
- D) 80

15

The circumference of a circle is 24π . What is the area of a sector of this circle with a central angle of 30° ?

- A) 2π
- B) 6π
- C) 12π
- D) 144π

16

A sector of a circle with radius 12 has an arc length of 8π . What is the area of this sector?

- A) 4π
- B) 24π
- C) 48π
- D) 96π

17

Which of the following equations is equivalent to $x^2 + y^2 + 6x - 14y + 22 = 0$?

- A) $(x + 3)^2 + (y - 7)^2 = 36$
- B) $(x - 3)^2 + (y + 7)^2 = 36$
- C) $(x + 3)^2 + (y - 7)^2 = 6$
- D) $(x + 3)^2 + (y - 7)^2 = 58$

18

In the xy -plane, the equation $2x^2 + 2y^2 - 12x + 4y - 6 = 0$ represents a circle. What is the radius of this circle?

- A) $\sqrt{7}$
- B) $\sqrt{13}$
- C) $\sqrt{26}$
- D) 13

19

In the xy -plane, the graph of $x^2 + y^2 - 10x + 4y + 20 = 0$ is a circle with center (h, k) and radius r . What is the value of $h + k + r$?

20

In the xy -plane, a circle has center $(3, 4)$ and is tangent to the x -axis. Which of the following is an equation of this circle?

- A) $(x - 3)^2 + (y - 4)^2 = 9$
- B) $(x + 3)^2 + (y + 4)^2 = 16$
- C) $(x - 3)^2 + (y - 4)^2 = 25$
- D) $(x - 3)^2 + (y - 4)^2 = 16$

21

In the xy -plane, the graph of $x^2 + y^2 - 4x - 2y - 20 = 0$ is a circle. The area of this circle is $k\pi$. What is the value of k ?

22

In the xy -plane, the equation $x^2 + y^2 - 2x + 8y = 8$ represents a circle. What is the circumference of this circle?

- A) 5π
- B) 10π
- C) 25π
- D) 50π

Answer Key

Topic 01 — Linear Equations in One Variable

Module 1

1 A	2 7	3 B	4 C	5 B	6 10	7 C	8 C
9 A	10 3	11 A	12 D	13 B	14 4	15 B	16 C
17 D	18 12	19 A	20 D	21 41	22 B		

Module 2

1 B	2 30	3 C	4 C	5 D	6 C	7 7	8 C
9 B	10 A	11 14	12 A	13 D	14 B	15 A	16 4
17 C	18 D	19 A	20 D	21 20	22 B		

Topic 02 — Linear Functions

Module 1

1 C	2 C	3 4	4 B	5 D	6 1	7 A	8 B
9 29	10 A	11 A	12 B	13 7	14 A	15 C	16 B
17 5	18 D	19 C	20 16	21 C	22 D		

Module 2

1 D	2 -2	3 B	4 D	5 D	6 B	7 A	8 B
9 4	10 C	11 A	12 6	13 C	14 B	15 C	16 A
17 1	18 D	19 C	20 32	21 C	22 B		

Topic 03 — Linear Equations in Two Variables

Module 1

1 A	2 B	3 19	4 C	5 D	6 3	7 B	8 A
9 C	10 C	11 A	12 12	13 D	14 B	15 15	16 D
17 A	18 2	19 C	20 B	21 A	22 11		

Module 2

1 B	2 -4	3 C	4 D	5 A	6 B	7 C	8 D
9 $-\frac{2}{3}$	10 A	11 C	12 B	13 D	14 C	15 6	16 A
17 -1	18 D	19 B	20 C	21 $\frac{9}{4}$	22 D		

Topic 04 — Systems of Two Linear Equations in Two Variables

**Module 1**

1 A	2 C	3 7	4 C	5 D	6 B	7 10	8 B
9 5	10 A	11 C	12 D	13 B	14 C	15 6	16 A
17 B	18 2	19 D	20 D	21 1	22 B		

Module 2

1 C	2 B	3 16	4 C	5 D	6 C	7 D	8 B
9 3	10 C	11 C	12 11	13 C	14 A	15 2	16 A
17 B	18 C	19 B	20 6	21 D	22 D		

Topic 05 — Linear Inequalities in One or Two Variables**Module 1**

1 B	2 C	3 7	4 A	5 D	6 B	7 C	8 B
9 19	10 C	11 D	12 3	13 B	14 A	15 4	16 C
17 B	18 B	19 A	20 2	21 C	22 A		

Module 2

1 B	2 3	3 D	4 D	5 A	6 6	7 A	8 C
9 43	10 C	11 D	12 B	13 A	14 C	15 5	16 D
17 A	18 16	19 C	20 34	21 C	22 A		

Topic 06 — Equivalent Expressions**Module 1**

1 A	2 B	3 1	4 D	5 A	6 C	7 D	8 D
9 C	10 4	11 D	12 A	13 8	14 C	15 C	16 A
17 D	18 2	19 C	20 A	21 8	22 B		

Module 2

1 A	2 C	3 5	4 C	5 D	6 9	7 B	8 B
9 5	10 A	11 B	12 2	13 B	14 A	15 C	16 B
17 D	18 3	19 B	20 24	21 D	22 C		

Topic 07 — Quadratic Equations**Module 1**

1 B	2 C	3 5	4 D	5 A	6 B	7 A	8 5
9 D	10 C	11 3	12 B	13 A	14 B	15 12	16 C
17 A	18 9	19 B	20 C	21 B	22 D		

Module 2

1 D	2 3	3 A	4 B	5 A	6 5	7 A	8 C
9 -15	10 C	11 B	12 2	13 B	14 D	15 C	16 C
17 C	18 4	19 45	20 C	21 A	22 C		

Topic 08 — Quadratic and Polynomial Functions

**Module 1**

1 C 2 3 3 D 4 A 5 D 6 5 7 C 8 D
 9 2 10 D 11 B 12 D 13 B 14 C 15 B 16 B
 17 -3 18 D 19 C 20 2 21 C 22 B

Module 2

1 B 2 4 3 A 4 B 5 7 6 A 7 C 8 1
 9 B 10 C 11 D 12 A 13 A 14 2 15 D 16 B
 17 -6 18 B 19 C 20 212 21 A 22 A

Topic 09 — Exponential Functions, Radicals and Rational Expressions**Module 1**

1 B 2 D 3 24 4 C 5 C 6 10 7 C 8 B
 9 2 10 C 11 D 12 A 13 B 14 A 15 4800 16 B
 17 A 18 4 19 D 20 C 21 11 22 A

Module 2

1 B 2 1 3 D 4 D 5 C 6 B 7 882 8 A
 9 5 10 D 11 D 12 B 13 C 14 A 15 C 16 C
 17 B 18 A 19 $9/5$ 20 5 21 B 22 A

Topic 10 — Nonlinear Systems and Function Transformations**Module 1**

1 A 2 B 3 3 4 A 5 C 6 D 7 A 8 B
 9 -4 10 B 11 D 12 C 13 C 14 B 15 4 16 B
 17 12 18 A 19 D 20 15 21 C 22 A

Module 2

1 A 2 2 3 B 4 D 5 13 6 C 7 D 8 7
 9 C 10 D 11 B 12 C 13 8 14 B 15 A 16 C
 17 A 18 D 19 12 20 B 21 4 22 A

Topic 11 — Ratios, Rates, Proportional Relationships and Units**Module 1**

1 D 2 7 3 B 4 D 5 A 6 150 7 B 8 B
 9 20 10 B 11 A 12 C 13 D 14 C 15 B 16 D
 17 2000 18 C 19 C 20 A 21 1.05 22 C

Module 2

1 D 2 14 3 B 4 C 5 A 6 450 7 C 8 B
 9 54 10 D 11 C 12 D 13 24 14 A 15 C 16 B
 17 300 18 B 19 2.8 20 A 21 C 22 B

Topic 12 — Percentages

**Module 1**

1 B 2 25 3 C 4 B 5 C 6 A 7 D 8 D
 9 20 10 B 11 A 12 C 13 A 14 D 15 C 16 B
 17 250 18 A 19 100 20 D 21 C 22 20

Module 2

1 C 2 40 3 B 4 A 5 D 6 B 7 C 8 128
 9 C 10 B 11 48 12 D 13 A 14 C 15 104 16 B
 17 B 18 D 19 25 20 C 21 D 22 100000

Topic 13 — One-Variable Data: Distributions and Measures of Center and Spread**Module 1**

1 C 2 B 3 15 4 C 5 B 6 17 7 C 8 B
 9 B 10 C 11 D 12 8.5 13 D 14 A 15 10 16 A
 17 9 18 D 19 A 20 82 21 C 22 C

Module 2

1 B 2 A 3 A 4 D 5 3 6 D 7 A 8 B
 9 15 10 C 11 B 12 A 13 B 14 9 15 D 16 25
 17 D 18 C 19 C 20 A 21 75 22 D

Topic 14 — Two-Variable Data: Models and Scatterplots**Module 1**

1 A 2 C 3 4 4 A 5 D 6 85 7 A 8 B
 9 D 10 D 11 84 12 A 13 5 14 C 15 D 16 C
 17 B 18 D 19 20 20 C 21 B 22 C

Module 2

1 A 2 C 3 40 4 B 5 -2.5 6 A 7 C 8 -4
 9 D 10 D 11 50 12 A 13 B 14 3 15 D 16 C
 17 B 18 D 19 108 20 C 21 B 22 A

Topic 15 — Probability and Conditional Probability**Module 1**

1 A 2 B 3 C 4 $\frac{2}{5}$ 5 B 6 C 7 A 8 $\frac{13}{25}$
 9 D 10 B 11 D 12 $\frac{9}{16}$ 13 D 14 B 15 A 16 4
 17 A 18 C 19 D 20 2 21 C 22 B

Module 2

1 B 2 A 3 D 4 $\frac{2}{5}$ 5 C 6 C 7 A 8 $\frac{7}{40}$
 9 A 10 B 11 D 12 48 13 D 14 C 15 B 16 40
 17 B 18 C 19 12 20 C 21 B 22 80

Topic 16 — Inference, Margin of Error and Evaluating Statistical Claims



Module 1

1 C 2 39 3 B 4 A 5 B 6 A 7 C 8 D
 9 28.7 10 A 11 B 12 D 13 0.42 14 A 15 C 16 D
 17 2560 18 A 19 D 20 2.7 21 B 22 C

Module 2

1 C 2 0.55 3 A 4 A 5 440 6 B 7 D 8 C
 9 23.1 10 D 11 A 12 D 13 D 14 B 15 4.05 16 C
 17 B 18 2025 19 C 20 C 21 8100 22 B

Topic 17 — Area and Volume

Module 1

1 D 2 B 3 C 4 180 5 B 6 D 7 96 8 B
 9 C 10 A 11 7 12 B 13 C 14 B 15 5 16 B
 17 A 18 C 19 400 20 B 21 C 22 B

Module 2

1 C 2 B 3 A 4 150 5 C 6 D 7 C 8 A
 9 6 10 C 11 D 12 300 13 B 14 D 15 C 16 A
 17 540 18 B 19 A 20 5 21 D 22 10

Topic 18 — Lines, Angles and Triangles

Module 1

1 B 2 63 3 B 4 D 5 A 6 70 7 B 8 C
 9 C 10 13 11 A 12 A 13 25 14 B 15 C 16 B
 17 25 18 C 19 C 20 54 21 A 22 D

Module 2

1 B 2 C 3 80 4 D 5 B 6 B 7 75 8 D
 9 100 10 C 11 A 12 A 13 A 14 C 15 C 16 4
 17 D 18 C 19 B 20 12 21 D 22 D

Topic 19 — Right Triangles and Trigonometry

Module 1

1 B 2 B 3 15 4 B 5 A 6 A 7 A 8 24
 9 B 10 C 11 C 12 A 13 C 14 D 15 $\frac{4}{5}$ 16 B
 17 B 18 15 19 C 20 A 21 10 22 B

Module 2

1 B 2 D 3 $\frac{7}{24}$ 4 D 5 D 6 C 7 $\frac{8}{17}$ 8 D
 9 16 10 D 11 A 12 60 13 C 14 B 15 B 16 C
 17 15 18 C 19 A 20 100 21 C 22 A

Topic 20 — Circles



Module 1

1 B	2 C	3 7	4 C	5 A	6 A	7 B	8 A
9 B	10 150	11 D	12 14	13 C	14 C	15 C	16 C
17 4	18 D	19 B	20 D	21 13	22 A		

Module 2

1 C	2 B	3 135	4 A	5 C	6 B	7 D	8 5
9 C	10 120	11 A	12 D	13 6	14 A	15 C	16 C
17 A	18 B	19 6	20 D	21 25	22 B		

